

Each sheet is now built when it is generated, mostly from data. The formula and each raw material's own data (constituent CAS numbers and percentages, flash point, boiling point, specific gravity, VOC, solubility and physical state) go through the hazard engine. The engine also uses the federal hazard records, your CAS determinations, trade-secret declarations and mixture calculations. The resulting hazard codes then go through the transport classifier and the regulatory lists you maintain: TSCA, SARA 313, HAP, SNUR, Prop 65, RCRA and the carcinogen lists. The final hazard codes choose the translated sentences in Sections 2 and 4 to 14. The supplier block, logo and disclaimer come from Admin Settings, or from the manufacturer record on private-label sheets. Recommended Use comes from the product or its family, and the version and effective date are stamped at publish. A per-product, per-language text edit can replace a single line. The regulated tables and statuses cannot be edited. The same publish gates run on every publish path and in every language.

Before this work, only about a third of a sheet came from data. Most paragraphs were canned: one first-aid, fire, handling and stability text for every product, and Section 14 copied in English from the first ingredient that had a DOT entry. OSHA and TSCA status always gave the same answer, exact percentages were printed, and ES/FR/DE sheets still contained English. Flammability came from the lowest raw-material flash point and from ingredient codes, so the sections could disagree. Now all sections share one Flammable Liquids result, and nearly every paragraph follows the product's own hazards. Fixed text is limited to regulatory wording, format labels and last-resort fallbacks.

Key points

- Six things block publishing, on every path and in every language: a blank emergency phone (company, or the manufacturer's on private label), a blank private-label manufacturer name, missing federal hazard data above the threshold (default 1%), a trade-secret ingredient on the Prop 65 list, a Section 14 result of 'Not determined' or an incomplete transport edit, and an inactive finished good. Preview warnings never block.
- Enter a flash point on every liquid raw material. The product flash point is the weight-averaged flash point of the raws that have one, and raws without one are left out. Enter a high flash point for water and other non-flammable liquids, or the average will run low. If no raw has a flash point, the product is treated as not flammable. Boiling points set Category 1 vs 2 and Packing Group I.
- Keep constituent CAS numbers and min/max percentages complete on every raw material. They drive classification, the concentration bands in Sections 3, 8, 11, 12 and 15, the TSCA check and RCRA. Gaps show as warnings, and TSCA then reads 'not verified'.
- Maintain CAS determinations (preferred names, TSCA status, metal and element flags) and the regulatory lists (TSCA, SARA 313 with its PBT and special-concern flags, HAP, SNUR, Prop 65, RCRA waste codes, carcinogens). An edit to them re-queues the affected sheets.
- Maintain each product family's Recommended Use and Restrictions text, plus its active and UV/LED flags, in each language. Also maintain the Admin disclaimer and each manufacturer's disclaimer per language. A blank language prints the translated standard text, never English.
- Text edits are per product and per language and are saved only when they differ from the automatic text. Composition, exposure-limit and regulatory tables, OSHA/TSCA status and the RCRA line cannot be edited. Section 9 temperature edits change only what is displayed.
- Some text stays fixed by regulation: GHS hazard and precautionary wording, the prescribed concentration bands and 0.1% cut-off, the flammability, transport and RCRA thresholds, DOT shipping names, Prop 65 safe-harbor wording and the Appendix D headings.

How to read this document

The pages that follow are laid out like a generated Safety Data Sheet: the same sixteen sections in the same order, under the same banners, with the same field labels. Where the sheet would print a value, this document shows where that value comes from.

Each entry has a coloured tag naming the kind of source, a plain description (when several sources compete, they are listed as A > B > C, and the first one that has a value wins), an example of what prints, **You maintain** (the information a person keeps current) and **If missing** (exactly what prints, or what happens, when that information is blank). The small grey line is

the technical reference for the system team.

FORMULA / RAW MATERIALS	Calculated from the product's formula and the data entered on its raw materials.	HAZARD ENGINE	GHS classification computed from the formula and the competent person determinations.
REGULATORY LIST	Looked up in a maintained regulatory list (Prop 65, HAP, SARA 313, TSCA, RCRA, carcinogens).	PRODUCT FAMILY	Comes from the product's family, assigned by rules or inherited from its raw materials.
TRANSPORT RULES	Derived by the DOT rules from flash point, boiling point and hazard classes.	COMPANY / MANUFACTURER	Company settings, or the private label manufacturer on private label sheets.
ADMIN SETTING	An administrator setting that applies to every sheet.	PRODUCT OVERRIDE	Text typed for one product and language in the SDS editor; replaces the automatic value.
PUBLISH RECORD	Stamped when a version is published (version number, effective date).	TEXT CHOSEN BY HAZARD	Standard wording, selected and combined according to the product's hazards.
FIXED TEXT (TRANSLATED)	Standard wording, the same on every sheet, in the sheet's language.	FIXED TEXT	The same on every sheet (labels, regulatory citations).

At a glance

Fields mapped in Sections 1-16	281
From your product data	128 (46%)
Standard wording chosen by the product's hazards	76 (27%)
Admin settings and per-product edits	20 (7%)
Fixed text (layout, labels, regulatory phrases)	57 (20%)
Conditions that block publishing	8
Languages	English, Spanish, French, German (same sources; labels and standard wording translated)

SECTION 1: IDENTIFICATION

Section 1 combines the product's code and description from the CMS import (alias sheets: the alias's base code), use text from a per-product override, the product's own text or its active product family (resale sheets: a standard resale sentence), and the supplier block and logo from Admin > Settings or the private-label manufacturer record. All of it is set in fixed, translated labels and page furniture.

Product Identifier:

FORMULA / RAW MATERIALS The product code and description from the CMS import. Alias sheets use the alias's base code (before the first hyphen) and its description; resale sheets use the RM base code and supplier product name; private label uses custom > shared alias > product.

On a sheet: Product Identifier: BK1080 — Process Black Offset Ink

You maintain: Product code and description (CMS import / Finished goods screen); supplier product name on resale raw materials; alias code and description (CMS import; the Aliases screen is read-only); private-label custom code/description when used

If missing: A blank description, or one equal to the code, prints the code alone ("BK1080"). A resale RM with no supplier name prints its base code alone. The line is never omitted.

SDSGenerator::productIdentifier :849-857; section1 :1108; resale :463-469; createAliasVariant :818 + AliasResolver::stripPack; PrivateLabelPublisher::resolveIdentity :51

Recommended Use:

PRODUCT FAMILY Order: per-product override > the product's own text > its active family's text in the sheet language > the translated standard sentence. Resale sheets skip the family and use the resale sentence; their override is stored per raw material.

On a sheet: Recommended Use: UV-curable offset ink for sheetfed presses.

You maintain: Recommended Use text in every language on each active Product Family, plus family rules (Recompute now after an import or a rule change). Leave the product's own Recommended Use blank unless it is product-specific.

If missing: "Printing ink for commercial and industrial applications." (resale: "Raw material for industrial formulation."), translated. A family with no text in the sheet language also gives this sentence. Never blank.

resolveUseText :1134: override(1,recommended_use) > fg.recommended_use > active family recommended_use_json[lang] (not resale) > section1.recommended_use[_resale]

Restrictions on Use:

PRODUCT FAMILY The same order as Recommended Use: per-product override > product's own text > active family text in the sheet language > standard sentence. Resale sheets skip the family.

On a sheet: Restrictions on Use: For professional/industrial use only. Not for household consumer use.

You maintain: Restrictions text in every language on each active Product Family. Leave the product's own Restrictions blank unless it is product-specific.

If missing: "For professional/industrial use only. Not for household consumer use." (the same for resale), translated. A family with no text in the sheet language also gives this sentence. Never blank.

resolveUseText(..,'restrictions'): override(1,restrictions) > fg.restrictions_on_use > active family restrictions_json[lang] (not resale) > section1.restrictions[_resale]

Manufacturer / Supplier Information:

FIXED TEXT (TRANSLATED) A fixed translated sub-heading above the supplier block.

You maintain: Nothing

If missing: Always printed, even when every supplier line below it is blank

labels.manufacturer_info via getLabels() :4767 -> meta.labels; PDFService::renderSection1 :383-384; preview.php:67

Company:

COMPANY / MANUFACTURER The company name from Admin > Settings. On a private-label sheet, the manufacturer's name.

On a sheet: Company: Your Ink Company, Inc.

You maintain: Company Name in Admin > Settings > Manufacturer Information; Name on each private-label manufacturer (required)

If missing: A saved blank omits the line, and the FG and resale previews and SDS Review warn; publishing is not blocked. A never-saved key prints the config.php placeholder with a warning. A blank private-label name blocks publishing.

settings company.name (config.php only if never saved) -> buildManufacturerBlock :897; PL manufacturers.name via Manufacturer::toCompanyInfo; PDFService:388

Address:

COMPANY / MANUFACTURER Street, city, state and ZIP, and country from Admin > Settings (or the private-label manufacturer), joined with commas and skipping blanks.

On a sheet: Address: 123 Industrial Blvd, Suite 100, Anytown, OH 44000, US

You maintain: Address, City, State, ZIP, Country in Admin > Settings; the same fields on each manufacturer

If missing: Blank parts are skipped; a part never saved in Admin > Settings prints the config.php placeholder. If every part is blank, the line is omitted. No warning.

formatManufacturerAddress (SDSGenerator:869-881): company.address/city/state/zip/country; PL manufacturers.address/city/state/zip/country; PDFService:389

Phone:	COMPANY / MANUFACTURER The company phone from Admin > Settings, or the private-label manufacturer's phone. <i>On a sheet: Phone: (555) 123-4567</i> You maintain: Phone in Admin > Settings; phone on each manufacturer If missing: A saved blank omits the line; a never-saved key prints the config.php placeholder. Either way the FG and resale previews and SDS Review warn (private label: its preview). Publishing is not blocked. company.phone (config if never saved) / manufacturers.phone -> buildManufacturerBlock :899; PDFService:390; SDSReadinessService::companySupplierWarnings :526
Email:	COMPANY / MANUFACTURER The company email from Admin > Settings, or the private-label manufacturer's email. <i>On a sheet: Email: sds@yourinkcompany.com</i> You maintain: Email in Admin > Settings; email on each manufacturer If missing: A saved blank omits the line; a never-saved key prints the config.php placeholder. No warning. company.email (config if never saved) / manufacturers.email -> buildManufacturerBlock :900; PDFService:392; preview.php:59
Website:	COMPANY / MANUFACTURER The company website from Admin > Settings, or the private-label manufacturer's website. <i>On a sheet: Website: https://www.yourinkcompany.com</i> You maintain: Website in Admin > Settings; website on each manufacturer If missing: A saved blank omits the line; a never-saved key prints the config.php placeholder. No warning. company.website (config if never saved) / manufacturers.website -> buildManufacturerBlock :901; PDFService:393; preview.php:60
Emergency:	COMPANY / MANUFACTURER The Emergency Phone saved in Admin > Settings, with no config fallback; it prints last in the block. A private-label sheet prints only the manufacturer's own emergency number. <i>On a sheet: Emergency: CHEMTREC: (800) 424-9300</i> You maintain: Emergency Phone in Admin > Settings (required); emergency phone on every private-label manufacturer (required) If missing: The line is omitted, every publish path for that sheet refuses, and the FG, resale and private-label previews show "Publishing blocked: ...". settings company.emergency_phone only (getCompanySettings :4567-4575) -> section1 :1116; PL trim(manufacturers.emergency_phone) :931; PDFService:394
Preview Warnings box (operator only):	FIXED TEXT English operator messages above the preview: Section 1 blocks and warnings (emergency phone, supplier name and phone, inactive product), plus generator warnings from other sections. Never on the customer PDF. <i>On a sheet: Publishing blocked: the company emergency phone number is blank or has never been saved.</i> You maintain: Nothing (clear the underlying data) If missing: No box when there are no warnings preview-pdf.php:32-41; SDSController::preview :78-91, previewResale :142-149; createManufacturerVariant :932-945; SDSGenerator :273-337

SECTION 2: HAZARD(S) IDENTIFICATION

Section 2 is built from the hazard classification of the formula. The inputs are ingredient hazard records (PubChem, now attributed per class), competent-person determinations, trade-secret entries, mixture calculations, a Flammable Liquids class derived from the product flash point and boiling point, the optional per-product hazard override and the carcinogen list. All wording comes from fixed GHS phrase and label tables in the sheet language. Two per-product text overrides can change it: Other Hazards and the Section 8 PPE fields. The flammability and physical-state warnings appear only in the preview Warnings box. The missing-data publish block applies on every publish path.

Not a hazardous substance or mixture.:	TEXT CHOSEN BY HAZARD A fixed sentence. It prints only when, after every hazard step, the product has no signal word, pictogram, hazard class or H-statement. P-statements alone do not count. <i>On a sheet: Not a hazardous substance or mixture.</i> You maintain: Nothing for the wording. Whether it shows depends on the hazard data listed under Signal word. If missing: Snapshots stored without is_classified are judged on their own stored signal word, pictograms, classes and H-statements, so the sentence never appears above printed hazards. section2.not_classified (en.php:16); SDSGenerator::isClassified :1257-1263, section2IsClassified :1275-1281; PDFService.php:412-415; preview.php:73-75
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Signal word (DANGER / WARNING):	HAZARD ENGINE The most severe signal word of the classes that trigger: the GHS default for ingredient, mixture-rule and flash-point classes, and the word entered on determinations and trade-secret entries. Precedence: replace-mode hazard override > most severe of (computed, additive override, carcinogen-list rule) > powder carbon-black/TiO2 rule (sets Warning only when nothing else set a word). <i>On a sheet: DANGER (red) or WARNING (orange)</i> You maintain: Hazard data per ingredient CAS (Raw materials, hazard refresh) or a determination (CAS determinations). On Raw materials: trade-secret hazards, flash point, boiling point and physical state. Hazard override and physical state (Finished goods). Carcinogen list (Regulatory lists). If missing: The signal word line is omitted. Carbon black or TiO2 adds nothing (no WARNING) when any raw material is not Solid or Powder (a blank state counts). A Warning left only by acute toxicity Category 5 is removed. HazardEngine::classify :669-1215 (contributionForRow; parseDeterminationStructure :3033), applyFinishedGoodOverride :2045-2053; SDSGenerator :5014-5100, :5517-5522; PDF :417-425
Pictograms:	FIXED TEXT (TRANSLATED) A fixed label in the sheet language, shown only when pictograms are present. <i>On a sheet: Pictograms:</i> You maintain: Nothing If missing: Order of fallback: label stored with the sheet > sheet-language translation > the key. labels.pictograms (getLabels SDSGenerator.php:4774; en.php:578); PDFService::renderPictogramRow :633-690; preview.php:84
GHS pictogram images (GHS01-GHS09):	HAZARD ENGINE The pictograms of each class that triggers (no longer the substance's whole PubChem list), of determinations, trade-secret entries, mixture rules, the flash-point Flammable Liquids class (GHS02) and the hazard override. GHS07 is dropped when GHS06 or GHS05 is present, and the rest are ordered GHS01, 05, 06, 02, 04, 03, 08, 07, 09; GHS08 added by the carcinogen-list or powder carbon-black rule prints last. <i>On a sheet: GHS02 flame + GHS07 exclamation mark diamonds</i> You maintain: The same hazard data as the signal word. Optional replacement artwork on Admin > Pictograms. If missing: No pictograms: the block is omitted. In the PDF a code with no artwork is skipped, and if no code has artwork, text such as 'GHS02 (Flame), ...' prints instead. The preview shows the code as a badge. contributionForRow :367; applyFlammableLiquidsFromFlashPoint :2849; applyPictogramPrecedence :2971-2990; SDSGenerator :5089, :5513; PictogramHelper::getPngPath :61-79
Pictogram caption under each image:	FIXED TEXT (TRANSLATED) The fixed GHS pictogram name for each code, in the sheet language. <i>On a sheet: Flame / Exclamation Mark</i> You maintain: Nothing If missing: Falls back to the English name, then to the code itself, e.g. 'GHS02'. GHSStatements::pictogramName :345-358 (ghs_{lang}.pictogram_names > PICTOGRAM_NAMES > code); PDFService.php:679-684; preview.php:92, 97
GHS Classification:	FIXED TEXT (TRANSLATED) A fixed heading, shown only when at least one hazard class exists. <i>On a sheet: GHS Classification:</i> You maintain: Nothing If missing: Falls back to the sheet-language translation, then the key. labels.ghs_classification (en.php:579); PDFService.php:478; preview.php:104
Physical Hazards: / Health Hazards: / Environmental Hazards:	FIXED TEXT (TRANSLATED) Three fixed headings, all printed whenever any class exists. Each class is placed by its English class name, which is now the standard GHS name for ingredient, mixture, flash-point and carcinogen-rule classes and for determination classes picked from the list (legacy free-text determination classes keep their typed name); unknown names go under Health Hazards. <i>On a sheet: Physical Hazards:</i> You maintain: Nothing for the headings. A class typed into the hazard override must use the standard GHS name, spelled exactly, to land in the right group. If missing: Falls back to the sheet-language translation. An unrecognised class name (e.g. 'Unclassified', a free-text determination class or a non-standard override class) is listed under Health Hazards. labels.physical_hazards/health_hazards/environmental_hazards (en.php:580-582); HazardEngine::groupByHazardType :3583-3596, HAZARD_GROUP_ORDER :3337; PDF :480-505
None (under an empty hazard group):	FIXED TEXT (TRANSLATED) A fixed word in the sheet language, printed under any group that has no classes. <i>On a sheet: None</i> You maintain: Nothing If missing: Falls back to the sheet-language translation, then the key. labels.none (en.php:743); PDFService.php:503-504; preview.php:150-151

Hazard class lines (H-code — Class (Category)):

HAZARD ENGINE One line per standard hazard class (most severe category kept), from triggered ingredient records, determinations, trade-secret entries, mixture calculations (summation/ATE/aquatic), the flash-point class and the hazard override, plus carcinogen-list and powder carbon-black lines added afterwards. Each line carries the GHS default H-code for its class and category, and lines sort by that code within each group.

On a sheet: H319 — Serious Eye Damage/Eye Irritation (Category 2A): Causes serious eye irritation

You maintain: Ingredient hazard data and determinations (Raw materials, CAS determinations); constituent CAS and %; carcinogen list (Regulatory lists); hazard override classes (Finished goods).

If missing: No classes: the block is omitted. Unrecognised class or category: no H-code prefix (its codes print under Hazard Statements). Override class without its H-code: code prefix, no phrase. Acute toxicity Category 5 never prints. A carcinogen-list line can print as a second Carcinogenicity line.

classify :669-1215; HazardRowNormalizer::expandRows :301, entryFor; h_codes stamped :1134-1146; consolidateHazardClasses :3458; dropUnadoptedAcuteCategories :2496; PDF :437-535

Flammable Liquids class line:

HAZARD ENGINE Derived from the product flash point (wt%-weighted average over raws that have one; a '>' value counts as its number and makes the product '> n', so exactly 60 or 93 °C moves up a band) and the lowest raw boiling point: below 23 °C Cat 1 (boiling point ≤35 °C) or Cat 2, 23-60 °C Cat 3, above 60 to 93 °C Cat 4. Ingredient H224-H227 never classify the mixture, and an override Flammable Liquids class or H224-H227 replaces the derived line (override > flash point).

On a sheet: H226 — Flammable Liquids (Category 3): Flammable liquid and vapour

You maintain: On every raw material (Raw materials): the flash point in °C (with the '>' flag) and the initial boiling point. Enter a high flash point for water and similar raws. Physical state goes on the finished good, or else on the raw materials.

If missing: No raw material with a flash point, a product flash point above 93 °C, or a product that is Solid, Powder or Paste: no Flammable Liquids line. Flash point below 23 °C with no boiling point: Category 2 (H225), with a preview warning.

FormulaCalcService::weightedFlashPoint :690, lowest BP :549-555;
HazardEngine::flammableLiquidCategory :2762, applyFlammableLiquidsFromFlashPoint :2849;
applyFinishedGoodOverride :1950-2001

Unclassified (Health Hazards line):

HAZARD ENGINE A stored hazard row named 'Unclassified' whose H-codes match no GHS class. At 1% or more it prints under Health Hazards with no H-code, its codes go to the Hazard Statements list, and its stored P-codes, pictograms and signal word are used unfiltered.

On a sheet: Unclassified

You maintain: Fix the ingredient's hazard data with a determination (CAS determinations), or refresh PubChem.

If missing: Not printed when its H-codes map to GHS classes (those classes print instead) or when the CAS is below 1%.

HazardRowNormalizer::rowsForStorage :488, expandRows :301-352, contributionForRow :373-376 (no class); HazardEngine::getCutoff 1% default :1371-1378; groupByHazardType

H-statement phrase (text after the colon):

REGULATORY LIST The standard GHS wording for the line's H-code, in the sheet language. The wording on the source record is always replaced.

On a sheet: Causes serious eye irritation

You maintain: Nothing (fixed phrase table).

If missing: Code missing from the sheet-language table: English wording. Missing from both, or the line's code not among the product's H-statements: the line ends at '(Category n)'.

GHSStatements::hText :309-322 (ghs_{lang}.h_statements > H_STATEMENTS), resolveHStatements :410; translateHazardResult :447

Hazard Statements: (H-codes without a class line):

HAZARD ENGINE Lists each H-statement that no class line above carries, worded from the fixed GHS table. Examples: hazard override H-codes entered without a class, and codes of an unrecognised class. Codes that OSHA does not adopt (H303, H313, H333, H316, H305) are never listed.

On a sheet: Hazard Statements: H319: Causes serious eye irritation

You maintain: Nothing beyond the hazard data and the hazard override H-codes (Finished goods).

If missing: When every H-code is on a class line, the heading and list are omitted. A code not in the phrase table prints alone.

labels.hazard_statements (en.php:584); GHSStatements::uncoveredHStatements :551-583, NOT_ADOPTED_H_CODES :496; PDFService.php:538-558; preview.php:176-183

Precautionary Statements:

FIXED TEXT (TRANSLATED) A fixed heading, shown only when P-statements exist.

On a sheet: Precautionary Statements:

You maintain: Nothing

If missing: Falls back to the sheet-language translation.

labels.precautionary_statements (en.php:583); PDFService.php:565; preview.php:186

Precautionary statement lines:	HAZARD ENGINE Each triggered ingredient class gives its GHS default P-codes plus the record's own codes for that class; determinations and trade-secret entries give the P-codes entered on them; mixture rules, the flash-point class and the override add theirs, all sorted by code. Withdrawn P281 becomes P280, and the carcinogen-list and powder carbon-black rules append their class defaults at the end. <i>On a sheet: P210: Keep away from heat, hot surfaces, sparks, open flames and other ignition sources.</i> You maintain: The same hazard data as the class lines; P-codes on determinations and trade-secret entries; P-codes in the hazard override if used (Finished goods). If missing: No P-codes: the block is omitted. A code not in the phrase table prints alone. contributionForRow :367; parseDeterminationStructure :3044-3047, :3250-3252; classify :1151-1161 (replaceWithdrawnPCodes); appendClassDefaultPStatements SDSGenerator :5363; PDF :561-584
Recommended Personal Protective Equipment (PPE):	FIXED TEXT (TRANSLATED) A fixed heading, shown only when at least one PPE field is driven by the product's H-codes. <i>On a sheet: Recommended Personal Protective Equipment (PPE):</i> You maintain: Nothing If missing: Falls back to the sheet-language translation. labels.ppe_recommendations (en.php:585); gate resolvePPE hazard_driven SDSGenerator.php:1290-1294; PDFService.php:587-596; preview.php:193-198
PPE pictogram images (eye, hand, respiratory, skin):	TEXT CHOSEN BY HAZARD One icon for each PPE field whose level is set by the H-codes: for example, H319 gives goggles, H315 gloves and H335 a respirator. P-codes play no part, and H-codes that Section 2 does not list (e.g. the non-adopted H316) still count. <i>On a sheet: Blue eye-protection and glove icons</i> You maintain: Ingredient hazard data that yields H-codes. Optional artwork on Admin > Pictograms. If missing: No hazard-driven field: no icons. Missing artwork: the PDF drops that icon and its caption; the preview keeps the caption without an image. HazardEngine::derivePPE :1279-1358 via SDSGenerator::resolvePPE :2483-2508; PictogramHelper PPE-*.png; PDFService::renderPPEPictogramRow :694
PPE icon captions:	FIXED TEXT (TRANSLATED) A fixed caption for each active PPE icon, in the sheet language. <i>On a sheet: Wear Eye Protection / Wear Gloves</i> You maintain: Nothing If missing: Falls back to the sheet-language translation. labels.ppe_wear_eye/_gloves/_respiratory/_skin (en.php:587-590); PDFService::renderPPEPictogramRow; preview.php:200-218
Respiratory: / Hand Protection: / Eye Protection: / Skin/Body Protection:	FIXED TEXT (TRANSLATED) Fixed sub-labels, printed before each hazard-driven PPE sentence. <i>On a sheet: Hand Protection:</i> You maintain: Nothing If missing: The line is skipped when its sentence is empty. labels.respiratory/hand_protection/eye_protection/skin_body (en.php:629-633); PDFService.php:599-608; preview.php:223-235
PPE recommendation sentences:	TEXT CHOSEN BY HAZARD Section 8 per-product override (if not blank) > translated sentence for the H-code level, plus the UV acrylate sentence on UV products when the field is hazard-driven or H317 is present. Printed in Section 2 only when the level is hazard-driven, and always the same text as Section 8. <i>On a sheet: Chemical-resistant gloves (nitrile or neoprene recommended) meeting 29 CFR 1910.138.</i> You maintain: Ingredient hazard data (H-codes). Optional Section 8 PPE text per product and language (SDS editor). UV/LED flag on the product family (Product families) and the uv_acrylate_rule_pack setting. If missing: Baseline level (general/none): not printed in Section 2 (Section 8 prints it). A Section 8 override replaces the whole sentence, including the UV text. resolvePPE SDSGenerator.php:2483-2508: text_overrides(8, field) > section8.ppe.<field>.<tier> + section8.uv_<field> (UVAcrylateRulePack::getPpeSupplement)
Other Hazards:	FIXED TEXT (TRANSLATED) A fixed label, printed with the Other Hazards text whenever that text is not blank. <i>On a sheet: Other Hazards:</i> You maintain: Nothing If missing: Omitted together with the text when the text is blank (not reachable now that blank rows are ignored). labels.other_hazards (en.php:586); PDFService.php:615-619; preview.php:239-242

Other Hazards text:	PRODUCT OVERRIDE An override typed in the SDS editor, per language and per product (per raw material on resale sheets), wins over the translated default 'None known.' <i>On a sheet: None known.</i> You maintain: Optional. In the SDS editor, per product (or resale raw material) and language, enter hazards not otherwise classified, e.g. combustible dust. If missing: 'None known.' in the sheet language. A stored blank row is ignored and the default prints. TextOverrideService::effective :141 (blank rows dropped) > section2.other_hazards (en.php:15); SDSGenerator.php:1296, :1311; getOverrides :4628-4661
Flash point warnings (preview Warnings box only; not printed):	HAZARD ENGINE Shown when any of these applies: flash point below 23 °C with no boiling point; an override Flammable Liquids category that disagrees with the formula flash point; a Section 9 flash or boiling point edit dropped for contradicting the class; a raw material with no flash point carrying a disclosed ingredient (1% or more of the product) classified as a flammable liquid. <i>On a sheet: Flash point warning: raw material(s) R1234 contain an ingredient classified as a flammable...</i> You maintain: Flash points and boiling points on raw materials (Raw materials). Section 9 Flash Point edit (SDS editor) when an override reflects a tested flash point. If missing: Nothing to warn about: no message. Never printed, not shown in publish messages, and never blocks publishing. SDSGenerator::flammabilityWarnings :4152-4212, vetFlashPointEdit :4076, vetBoilingPointEdit :4118; sds['warnings'] shown in preview.php:663, preview-pdf.php:32

SECTION 3: COMPOSITION / INFORMATION ON INGREDIENTS

Section 3 is built from the formula. Each raw material's ingredients (CAS number, name and percentage) are combined across the formula and any nested finished goods. The hazard engine, the hazard-data records and the competent-person determinations then decide which ingredients are listed and which H-codes they carry, and the rows print in a fixed table whose headings and notes come from the translation files. Every concentration prints as a prescribed band. A trade-secret identity is replaced by its description. A Substance sheet always names its main constituent. Operator-only warnings appear in the preview Warnings box and are never printed. A published sheet keeps a frozen copy of this section until a new version is published.

Type:	FIXED TEXT (TRANSLATED) A fixed label from the language file. The layout adds the colon. <i>On a sheet: Type:</i> You maintain: Nothing If missing: The order is: snapshot label > sheet-language translation > English > the literal key "type". labels.type (en.php:593) via SDSGenerator::getLabels:4779; PDFService::label:1338 + labelValue:1358-1377; preview.php:245
Type (value: Mixture / Substance):	FORMULA / RAW MATERIALS The Substance/Mixture setting on the finished good. On Auto it is Substance only when the formula is one raw-material line marked Substance, otherwise Mixture. Resale sheets use the raw material's own setting. <i>On a sheet: Type: Mixture</i> You maintain: Substance / Mixture on the finished-good form. Substance / Mixture on the raw-material form for single-substance raws. If missing: "Mixture" prints. finished_goods.substance_mixture > SubstanceMixtureResolver::resolveForFinishedGood:84-100 (1 RM line) > 'mixture'; resale resolveForRawMaterial; labels.substance/mixture
Hazardous-only disclosure note:	FIXED TEXT (TRANSLATED) A fixed sentence from the language file. It prints above the table on Mixture sheets that list at least one row, and never on Substance sheets. <i>On a sheet: Hazardous ingredients and ingredients with an occupational exposure limit are listed. ...</i> You maintain: Nothing If missing: It does not print when no row is listed or when the sheet is a Substance. Older snapshots without the mixture_notes key are treated as Mixture. labels.hazardous_only_note (en.php:597); sections[3].mixture_notes (SDSGenerator:1490); PDFService.php:763-767; preview.php:247-249
CAS Number Chemical Name Concentration H-Codes (table header):	FIXED TEXT (TRANSLATED) Fixed column headings from the language file, in bold on a grey band. <i>On a sheet: CAS Number Chemical Name Concentration H-Codes</i> You maintain: Nothing If missing: The order is: snapshot label > sheet-language translation > English > the literal key. The table prints only when rows exist. labels.cas_number/chemical_name/concentration/h_codes (en.php:594-596,601); PDFService.php:770-777; preview.php:250-256

CAS Number (ingredient row):	FORMULA / RAW MATERIALS The CAS number on each raw-material ingredient, combined across every formula line and every nested finished-good component. <i>On a sheet: 108-88-3</i> You maintain: A CAS number on every ingredient of every raw material (Raw materials > constituents), and a current formula on every finished good used as a component. If missing: An ingredient with a blank CAS gets no row of its own, and the preview shows a Composition warning (a CAS-less trade secret with H-codes feeds the synthetic TRADE SECRET row instead). A finished-good component with no current formula adds nothing, and only the readiness page warns. A CAS ticked trade secret on any raw material prints "TRADE SECRET". <code>raw_material_constituents.cas_number</code> via <code>Formula::getExpandedComposition</code> (resale: <code>FormulaCalcService::buildResaleComposition:199-350</code>); <code>SDSGenerator:1456</code>
Chemical Name (ingredient row):	FORMULA / RAW MATERIALS Prop 65 list name > CAS master preferred name > name typed on the raw-material ingredient. Resale sheets use the name stored on the ingredient, which is replaced by the canonical CAS name when the raw material is saved. <i>On a sheet: Toluene</i> You maintain: The ingredient name on the raw material, and the Prop 65 list and CAS master names (Regulatory lists). If missing: The name as typed on the ingredient prints, or the cell is blank if every source is blank. It is never translated. <code>COALESCE(prop65_list.chemical_name, cas_master.preferred_name, rmc.chemical_name)</code> <code>Formula.php:292</code>; <code>resale rmc.chemical_name</code>; <code>RawMaterial::canonicalCasName:498</code>
Concentration (ingredient row):	FORMULA / RAW MATERIALS Each contribution's lower and upper bound (line % x ingredient %, an exact % counting as both; resale sheets the same) is summed per CAS, and the widest prescribed band that contains [min, max] prints. If no band contains it, the band that contains the max with the lowest lower end prints. <i>On a sheet: 10 - 30%</i> You maintain: Formula line percentages (Finished goods > formula), and an exact % or a min + max % on each raw-material ingredient. If missing: An ingredient with no % counts as 0 and is dropped. Max only counts as [0, max] and min only as [min, min]. A max above 100 % prints "80 - 100%". "<0.1%" prints only on a Substance identity row whose max is below 0.1 % (other rows that small are not listed). <code>SDSGenerator::formatConcentration:4712-4762</code> over <code>PRESCRIBED_RANGES:4698</code>; <code>Formula::constituentBounds + mergeSubEntryExtras</code>; <code>concentration_min/max</code>
H-Codes (ingredient row):	HAZARD ENGINE The H-codes credited to that CAS before the engine reduces to one entry per class: its own classes (federal data > competent-person determination), mixture-rule contributions, carcinogen-list and powder H351 entries, and its own H224-H227. <i>On a sheet: H225, H304, H315, H336</i> You maintain: Federal hazard data for each CAS, or an active competent-person determination (CAS determinations). Also the carcinogen list and the physical state on each raw material. If missing: The cell is blank. A replace-mode finished-good hazard override blanks every cell except a flammable solvent's own H224-H227, and the rows stay listed. Carbon black and TiO2 keep only the codes left after the wet-mixture rule. <code>TransportClassifier::cashCodeMap:452-500</code> over <code>hazardResult.cas_h_codes</code> (<code>HazardEngine::buildCashCodeMap:3388-3448</code>) + <code>hazard_classes</code>; <code>SDSGenerator:1342-1357,1460</code>
Which ingredients are listed (rows):	HAZARD ENGINE An ingredient is listed when it has a CAS, is at least 0.1 % and has an exposure limit or a hazard class that meets its cut-off or feeds a mixture rule (a flammable solvent counts only when the product flash point classifies the product and the solvent is at least 1 %). An inhalation-only CAS with no remaining H-code and no exposure limit is dropped. <i>On a sheet: Toluene listed; water and talc in a liquid not listed</i> You maintain: Hazard data or a CPD for every CAS, exposure limits, the physical state on raw materials, and the inhalation-only CAS list (Settings). If missing: A CAS with no hazard data, no CPD and no exposure limit is left out. Publishing is blocked only when it is at or above the missing-data threshold (default 1.0 %). An ingredient whose only hazard is Acute Tox. Cat 5 is not listed unless it has an exposure limit. <code>SDSGenerator::section3:1393-1423</code> (<code>hazardous_cas</code>, <code>exposure_limits</code>, <code>getInhalationOnlyCas:4909</code>); <code>HazardEngine recordFlammableIngredient:2820</code>, <code>dropUnadoptedAcuteCategories</code>
Substance identity row:	FORMULA / RAW MATERIALS On a Substance sheet, the CAS with the largest percentage is always listed, even when it is unclassified, has no exposure limit or is below 0.1 %. Other constituents follow the normal rules. <i>On a sheet: 64742-47-8 Distillates (petroleum), hydrotreated light 80 - 100%</i> You maintain: Substance / Mixture setting, and the CAS constituent on the raw material. If missing: With no CAS constituent, the table and notes do not print, and the preview shows the Section 3 Substance warning. <code>SDSGenerator::section3:1379-1391</code> (<code>\$identityCas</code>), <code>:1396,1406-1418</code>; <code>formatConcentration</code>

TRADE SECRET (trade-secret row, CAS cell):	FIXED TEXT (TRANSLATED) A fixed word from the language file. It prints in place of the CAS number on every trade-secret row. <i>On a sheet: TRADE SECRET</i> You maintain: Nothing. The row appears when a listed ingredient (hazardous or with an exposure limit, at least 0.1 %) is ticked trade secret, or when a raw material marked hazardous with no CAS declares hazards. If missing: The order is: sheet-language translation > English > the literal key. labels.trade_secret_cas (en.php:740); SDSGenerator:1430; PDFService.php:785-806 (CAS cell height measured); preview.php:261
Chemical Name (trade-secret row):	FORMULA / RAW MATERIALS The trade-secret description on the raw-material ingredient > the translated words "Trade Secret" (the synthetic row takes the last description entered on a CAS-less trade-secret ingredient), and the real chemical name is never used. There is one row per distinct description, so a blank description and the synthetic row's "Trade Secret" can print as two rows both named "Trade Secret". <i>On a sheet: Proprietary Resin Blend</i> You maintain: The trade-secret description on each trade-secret ingredient (Raw materials > constituents; pick-list in Settings). If missing: "Trade Secret" (translated). SDSGenerator::tradeSecretName:2607-2614 (rmc.trade_secret_description; 'Trade Secret' sentinel -> labels.trade_secret en.php:739); bucket key :1427; Formula.php:485-487
Concentration (trade-secret row):	FORMULA / RAW MATERIALS The summed lower and upper bounds of the trade-secret rows that share one description, banded like normal rows; each trade-secret CAS must reach 0.1 % on its own. CAS-less trade-secret ingredients and hazardous-with-no-CAS raw materials (whole line %, exact) are first summed into one synthetic row, which must reach 0.1 % as a whole. <i>On a sheet: 1 - 5%</i> You maintain: Formula line percentages and ingredient percentages on the raw material. If missing: Same as normal rows. Several trade-secret CAS that are each below 0.1 % are dropped, even when together they reach 0.1 %. SDSGenerator::section3:1438-1444,1470 then formatConcentration; TRADE_SECRET bucket bounds Formula.php:425-437, :477-484
H-Codes (trade-secret row):	HAZARD ENGINE A masked row whose CAS is real shows that CAS's own H-codes, combined across every CAS sharing the description. The synthetic trade-secret row (raw material hazardous with no CAS, or CAS-less trade-secret ingredients) shows the codes of the declared trade-secret classes that survive the one-entry-per-class reduction. <i>On a sheet: H317, H319</i> You maintain: Manual hazard classification on raw materials marked hazardous with no CAS, and trade-secret H-codes on CAS-less trade-secret ingredients (Raw materials). If missing: The cell is blank, including when another ingredient is equally or more severe in every declared class or a replace-mode override applies. Declared H224-H227 never print on the synthetic row; a masked real CAS shows its own H224-H227. SDSGenerator::section3:1360-1369,1448-1451; final hazard_classes cas='TRADE_SECRET' (HazardEngine::consolidateHazardClasses); raw_materials.manual_hazard_json; rmc.trade_secret_h_codes
Trade-secret identity masking:	FORMULA / RAW MATERIALS A CAS ticked trade secret on any raw material in the formula, including one inside a nested finished good, is withheld on the whole sheet. Sections 8, 11, 12 and 15 print the same description and TRADE SECRET label, and Section 13 names only the description. <i>On a sheet: 25068-38-6 on one RM prints TRADE SECRET / Proprietary Resin Blend</i> You maintain: The trade-secret checkbox on each raw-material constituent, kept consistent for the same CAS on every raw material. If missing: Unticked on every raw material: the real CAS and name print. Formula.php:378-383 + sub-FG merge :589-594; SDSGenerator::maskTradeSecretRow:2627-2646; section3:1426; section13 :3881-3891
Row order:	FORMULA / RAW MATERIALS Rows are sorted by the calculated percentage, highest first. A ranged ingredient sorts by its range maximum, and trade-secret rows sort by their summed percentage. You maintain: Formula and ingredient percentages. If missing: Not applicable. usort on components[].concentration_pct desc (SDSGenerator:1476); Formula::resolveConstituentPct (exact > max > min > max-only), resale same (#38)

No-hazardous-ingredients note:	FIXED TEXT (TRANSLATED) A fixed sentence from the language file. It prints instead of the table on an unclassified Mixture sheet that lists no ingredient. <i>On a sheet: No hazardous ingredients above disclosure thresholds.</i> You maintain: Nothing If missing: It shows only when no row is listed and Section 2 is unclassified. It never prints on Substance sheets. sections[3].empty_note = labels.no_hazardous_note (en.php:598) (SDSGenerator:1494-1498); PDFService.php:809-814; preview.php:268-269
Classified-but-no-ingredients note:	FIXED TEXT (TRANSLATED) A fixed sentence from the language file. It prints instead of the table on a Mixture sheet when Section 2 is classified but no single ingredient qualifies for listing. <i>On a sheet: No ingredient is listed: no single ingredient meets the disclosure thresholds. ...</i> You maintain: Nothing If missing: Older snapshots without the key print the no-hazardous-ingredients note. It never prints on Substance sheets. section3.classified_no_ingredients_note (en.php:22) when isClassified(hazardResult) (SDSGenerator:1257,1494-1498); PDFService.php:809-814
Trade-secret / withheld-percentage statement:	FIXED TEXT (TRANSLATED) A fixed sentence from the language file, citing 29 CFR 1910.1200(i). It prints below the table whenever any row is listed, because every row shows a band. <i>On a sheet: Specific chemical identity and/or exact percentage ... withheld as a trade secret ...</i> You maintain: Nothing If missing: It does not print when no row is listed. sections[3].trade_secret_note = section3.trade_secret_note (en.php:20) when components non-empty (SDSGenerator:1506-1508); PDFService:816-821; preview.php:271-273
Section 3 warning: Substance with no CAS constituent (preview only):	FORMULA / RAW MATERIALS An operator warning in the preview Warnings box and the publish flash, shown when a Substance sheet has no CAS constituent to name. It is never printed. <i>On a sheet: Section 3 warning: this sheet is typed Substance but no constituent with a CAS ...</i> You maintain: The CAS constituent on the raw material, or set the product to Mixture. If missing: No warning when the substance has a CAS. Publishing is not blocked. sections[3].identity_missing (SDSGenerator:1501) -> warnings[] :303-306 (base/resale path :705-708); key unset before snapshot
Trade-secret warning: CAS secret on one raw material, disclosed on another (preview only):	FORMULA / RAW MATERIALS An operator warning naming the raw materials that disagree about one CAS. The sheet withholds the identity anyway. <i>On a sheet: Trade-secret warning: CAS 25068-38-6 is marked as a trade secret on raw material(s) ...</i> You maintain: The trade-secret checkbox, made consistent on every raw material that supplies the CAS. If missing: No warning when the raw materials agree. Publishing is not blocked. SDSGenerator::tradeSecretDisclosureWarnings:5556-5584 over contributing_materials[].is_trade_secret; warnings[] :320-322, :722-724
Composition warning: incomplete constituent data (preview only):	FORMULA / RAW MATERIALS An operator warning listing formula raw materials (including those inside nested finished goods) that have no constituents at all, a constituent with no CAS, a CAS-less trade secret with no H-codes, or a constituent with no percentage. Raw materials marked hazardous with no CAS are not checked. <i>On a sheet: Composition warning: 1 raw material has incomplete constituent data — RM-123 (...)</i> You maintain: Every constituent with its CAS number and percentage (Raw Materials > constituents). If missing: No warning when every constituent is complete. Publishing is not blocked. SDSReadinessService::incompleteCompositionWarning:595-612 over TSCAService::incompleteRawMaterials:275-305 (incompleteReasons); SDSGenerator:287-290, :689-692
Trade-secret Prop 65 block message (preview and publish):	REGULATORY LIST Shown when a trade-secret constituent's CAS is on the Prop 65 list. The message appears in the preview Warnings box and every publish path refuses. <i>On a sheet: Publishing blocked: a trade-secret constituent is on the California Prop 65 list: ...</i> You maintain: The trade-secret checkbox and CAS on the raw material, or have the vendor disclose the chemical. If missing: No message when no trade-secret CAS is on the Prop 65 list. Prop65Service trade_secret_conflicts -> SDSReadinessService::tradeSecretProp65Error:384-413; SDSGenerator:323-326, :725-728

SECTION 4: FIRST-AID MEASURES

Section 4 is built almost entirely from the product's final GHS health hazard codes. Each route paragraph, the symptoms line and the notes to physician are put together from translated sentences chosen by those codes. Three other things can change it. An optional text override, stored per product and per language, replaces a whole field; resale raw materials can now have their own overrides. On UV-family products an uncured-product sentence is added to Skin Contact. The symptoms line is also reused in Section 11.

Inhalation:

TEXT CHOSEN BY HAZARD Per-product override > sentences picked by hazard codes. H330 or H331 replaces the base paragraph, otherwise H332 adds a sentence; H334, H335 and H336 each add a sentence in every case.
On a sheet: Move to fresh air. If breathing difficulty persists, seek medical... Harmful if inhaled.
You maintain: Formula lines and raw-material constituents (CAS, %), hazard data per CAS (classification sources, competent-person determinations), trade-secret H-codes and manual hazard JSON on raws, raw physical states with the inhalation-only CAS setting, and any FG hazard override. Optional override on the SDS edit screen (finished good, or resale raw material).
If missing: With no matching codes: "Move to fresh air. If breathing difficulty persists, seek medical attention." A non-blank override prints as typed and replaces every hazard sentence. Blank override rows are ignored.
`TextOverrideService::effective(4.inhalation) > section4.inhalation[_fatal|_toxic] + _harmful(H332) + _resp_sensitizer/_irritant/_narcotic; SDSGenerator.php:1532-1554`

Skin Contact:

TEXT CHOSEN BY HAZARD Per-product override > hazard sentences: H314 corrosive paragraph (plus poison-center sentence for H310/H311, else H312 sentence) > H310/H311 toxic paragraph > base paragraph + H312/H315 sentences. H317 always adds the sensitizer sentence; UV-family products then add the uncured-product sentence (next row).
On a sheet: Remove contaminated clothing. Wash skin thoroughly... May cause an allergic skin reaction...
You maintain: Same hazard inputs as Inhalation. Optional override on the SDS edit screen.
If missing: "Remove contaminated clothing. Wash skin thoroughly with soap and water. If irritation persists, seek medical attention." A non-blank override replaces the whole field, including the UV sentence.
`effective(4.skin) > section4.skin[_corrosive|_toxic] + poison_center_immediate/skin_harmful/_irritant + skin_sensitizer(H317) + UV fragment; SDSGenerator.php:1557-1583`

Skin Contact: UV/EB-curable sentence (end of the field):

PRODUCT FAMILY Added when the UV rule pack is on and the product's resolved family is flagged UV/LED, with or without H317. Acrylates at $\geq 0.1\%$ (known CAS list, or a name with 'acrylate'/'acrylic acid' that is not a polymer or resin) are named in brackets; if none are found, the generic sentence without names prints.
On a sheet: UV/EB-curable product containing acrylate monomers/oligomers (TPGDA): remove uncured...
You maintain: UV/LED flag, active status, rules and manual picks on the Product families screen. The UV rule-pack switch in Settings. Constituent CAS, chemical names and trade-secret descriptions on Raw materials.
If missing: Not added when the pack is off or the family is not UV (a manual pick of an inactive family is ignored). Names: prop65 > CAS master > constituent name (resale: constituent name). A trade secret prints its description or the translated "Trade Secret", never its name or CAS.
`UVAcrylateRulePack::isApplicable :104 + detectAcrylates :128 + section4SkinFragment :171; section4.uv_skin / uv_skin_names; SDSGenerator.php:1580-1582`

Eye Contact:

TEXT CHOSEN BY HAZARD Per-product override > hazard sentences: H314 (30-minute rinse) or H318 (20-minute rinse) replaces the base paragraph. H314, H318 or H319 adds the contact-lens sentence.
On a sheet: Flush eyes with large amounts of water for at least 15 minutes... Remove contact lenses...
You maintain: Same hazard inputs as Inhalation. Optional override on the SDS edit screen.
If missing: The base 15-minute flush sentence prints, with no contact-lens sentence. A non-blank override replaces the field.
`effective(4.eyes) > section4.eyes[_corrosive|_serious_damage] + eyes_contact_lenses(H314/H318/H319); SDSGenerator.php:1586-1598`

Ingestion:

TEXT CHOSEN BY HAZARD Per-product override > hazard sentences: H304 aspiration paragraph (plus poison-center sentence for H300/H301, else the H302 sentence) > H300/H301 toxic paragraph > base paragraph + H302 sentence. H305 no longer selects the aspiration paragraph.
On a sheet: Do not induce vomiting. Rinse mouth with water... Harmful if swallowed. Call a poison...
You maintain: Same hazard inputs as Inhalation. Optional override on the SDS edit screen.
If missing: "Do not induce vomiting. Rinse mouth with water. Seek medical attention if symptoms develop." A non-blank override replaces the field.
`effective(4.ingestion) > section4.ingestion_aspiration(H304) | _toxic(H300/H301) | ingestion; + poison_center_immediate / ingestion_harmful(H302); SDSGenerator.php:1601-1620`

Most Important Symptoms/Effects, Acute and Delayed:

HAZARD ENGINE Per-product override > the product's health hazard statements (H3xx), restated in the sheet language. Sensitisation, CMR, lactation and repeated-exposure codes (H317, H334, H340-H362, H372, H373) are grouped as Delayed and all others as Acute; the same text also prints in Section 11.

On a sheet: Acute: Causes serious eye irritation. Delayed: May cause an allergic skin reaction.

You maintain: Same hazard inputs as Inhalation, plus the carcinogen lists, which add H350/H351 (skipped in 'replace' hazard-override mode). H-statement translations live in the ghs_es/fr/de files.

If missing: With no health codes: "No significant symptoms or effects are known or expected under normal conditions of use." A missing translation prints the English text. A sub-coded statement (e.g. H360D) prints the base code's text. Acute Tox. Cat. 5 (H303/H313/H333) is dropped by the engine and never prints.

effective(4.symptoms) > deriveSymptoms(hazard.h_statements) :1674-1714; DELAYED_EFFECT_H_PREFIXES :98; GHSStatements::hText; Section 11 reuse :3475

Acute: / Delayed: (prefixes inside the symptoms line):

FIXED TEXT (TRANSLATED) Fixed words from the translation file. Each prints only when that group has at least one statement.

On a sheet: Acute: ... Delayed: ...

You maintain: Nothing

If missing: An empty group is left out. If both groups are empty, the 'No significant symptoms...' sentence prints instead.

section4.symptoms_acute_prefix / symptoms_delayed_prefix; SDSGenerator.php:1706-1712; en.php:54-55

Notes to Physician:

TEXT CHOSEN BY HAZARD Per-product override > hazard sentences in this order: aspiration (H304), corrosive burns (H314) or, without H314, serious eye damage (H318), then delayed lung effects (H330/H331). The default closing sentence always follows.

On a sheet: Do not induce vomiting; risk of aspiration into the lungs. Treat symptomatically. ...

You maintain: Same hazard inputs as Inhalation. Optional override on the SDS edit screen.

If missing: "Treat symptomatically. Show this SDS to medical personnel." A non-blank override replaces the field.

effective(4.notes) > notes_aspiration + notes_corrosive | notes_eye_damage + notes_inhalation_delayed + section4.notes; SDSGenerator.php:1637-1652; en.php:30,57-61

SECTION 5: FIRE-FIGHTING MEASURES

Section 5 is standard translated wording. Each paragraph is chosen from the product's final hazard codes, the same codes Section 2 prints. Flammability now has one source: the Flammable Liquids category (Cat 1-4) comes from the product flash point, which is the wt%-weighted average over the raw materials that have one, together with the initial boiling point and the physical state. A finished-good hazard override can replace that category. Sections 2, 5, 7, 9, 10, 13 and 14 therefore agree. A product whose raw materials carry no flash point is treated as not flammable and gets the default text, unless an FG override sets a category. The flash point sentence prints the same string as Section 9. UV/LED-family products get one extra UV sentence inside Specific Hazards. A per-product override can replace any of the four paragraphs, but the flash point sentence inside a Specific Hazards override is kept current.

Suitable Extinguishing Media:

TEXT CHOSEN BY HAZARD One of four fixed sentences, chosen by the final hazard codes. Order: product override > water-reactive (H260/H261) > oxidizer (H271/H272) > flammable liquid (Flammable Liquids Cat 1-4 from the product flash point, or from the finished-good hazard override) > default.

On a sheet: Alcohol-resistant foam, dry chemical, carbon dioxide (CO2), or water spray (fog).

You maintain: Flash point, '>' tick and boiling point on raw materials. Physical state on the finished good. Constituent CAS and % on raw materials, with H-codes on CAS determinations. Optional FG hazard override.

If missing: With no water-reactive, oxidizer or flammable code (including a product with no flash point data): "Water spray, dry chemical, carbon dioxide (CO2), foam."

text_overrides(5,'suitable_media') > section5.suitable_water_reactive > suitable_oxidizer > suitable_flammable > suitable_media; SDSGenerator.php:1878-1909; extractHCodes :1057

Unsuitable Extinguishing Media:

TEXT CHOSEN BY HAZARD Fixed sentence chosen by the final hazard codes. Order: product override > water-reactive (H260/H261) > flammable liquid (flash-point-derived Cat 1-4 or FG override) > default. Oxidizer codes have no tier here, so an oxidizer gets the flammable or default sentence.

On a sheet: Do not use a solid water stream; it may scatter the burning liquid and spread the fire.

You maintain: Same data as Suitable Extinguishing Media.

If missing: "Do not use direct water stream as it may spread fire."

text_overrides(5,'unsuitable_media') > section5.unsuitable_water_reactive > unsuitable_flammable > unsuitable_media; SDSGenerator.php:1911-1921

Specific Hazards - flammable liquid sentence:

HAZARD ENGINE Printed for Flammable Liquids Cat 1-4. Category: FG hazard override with a Flammable Liquids class or H224-H227 > product flash point. FP <23 °C: Cat 1 if IBP <=35 °C, else Cat 2; 23-60 °C Cat 3; 60-93 °C Cat 4.
On a sheet: Flammable liquid and vapor (Flammable Liquids, Category 3). Vapors can form ignitable...
You maintain: Flash point (°C) and '>' tick on each raw material (a high value for water and similar raws), boiling point on solvent raws (Raw materials). Physical state (Finished goods). Optional FG hazard override.
If missing: Omitted, with the flash point line, when no raw material has a flash point, the weighted value is above 93 °C, the state is Solid/Powder/Paste, or a replace-mode FG override has no Flammable Liquids class. Ingredient H224-H227 alone never print it. A Specific Hazards override replaces it.
section5.specific_hazards_flammable_cat{1-4}; SDSGenerator.php:1878-1888, :1936-1937;
HazardEngine::flammableLiquidCategory :2762, applyFlammableLiquidsFromFlashPoint :2849, override :1951

Specific Hazards - Flash point line:

FORMULA / RAW MATERIALS Same string as Section 9: Section 9 Flash Point edit (kept only if it reads as a temperature and gives the same category and D001 result) > wt%-weighted average of raws with a flash point, through sub-formulas, '>' if any is ticked.
On a sheet: Flash point: 38.5 °C (101.3 °F).
You maintain: Flash point (°C) and the '>' tick on raw materials (Raw materials screen). Optionally a Section 9 Flash Point edit with a unit (e.g. '24 °C') in the SDS editor, per language.
If missing: Line omitted when the product is not flammable, or when it is flammable only through the FG override and no raw material or kept edit has a flash point. A contradicting edit is not printed: the formula value prints and the preview warns.
section5.flash_point_line ':fp' :1928-1930; resolveFlashPointDisplay :1742 > flashPointDisplay :1723; vetFlashPointEdit :4076; FormulaCalcService::weightedFlashPoint :690

Specific Hazards - water-reactive sentence:

TEXT CHOSEN BY HAZARD Fixed sentence added when the final hazard codes include H261. The H260 version adds 'which may ignite spontaneously'.
On a sheet: Reacts with water, moisture, and water-based extinguishing agents to release flammable gases.
You maintain: H260/H261 in the constituent CAS hazard data (CAS determinations), a trade-secret manual hazard entry on a no-CAS raw material, or the FG hazard override.
If missing: Omitted when neither H260 nor H261 is present.
section5.specific_hazards_water_reactive / _water_reactive_h260; SDSGenerator.php:1941-1946

Specific Hazards - closing sentence:

TEXT CHOSEN BY HAZARD Always printed after the flammable and water-reactive sentences. Order: oxidizer (H271/H272) 'may cause or intensify fire' > organic peroxide (H240-H242) 'may catch fire or explode' > default combustion sentence. The combustion products named are generic.
On a sheet: Combustion may produce carbon monoxide, carbon dioxide, and other toxic fumes.
You maintain: H271/H272 or H240-H242 in the hazard data where they apply. Otherwise nothing.
If missing: The default combustion sentence prints on every sheet that has no Specific Hazards override.
section5.specific_hazards_oxidizer > specific_hazards_organic_peroxide > specific_hazards; SDSGenerator.php:1947-1953, joined with ' ' at :1957

Specific Hazards - UV/EB sentence:

PRODUCT FAMILY A fixed sentence added at the end of the automatic Specific Hazards text when the UV rule pack is on and the product's resolved family is active and flagged UV/LED. No acrylate detection is needed.
On a sheet: When heated, uncured UV/EB-curable product may polymerize uncontrollably, releasing heat...
You maintain: UV rule pack setting (Settings). UV/LED flag and active status on the family (Product families). Family on the finished good (automatic from the formula, UV families pooled by share, or a manual pick), or on the raw material for a resale sheet.
If missing: Not printed when the pack is off, the family is unresolved, inactive or not UV, or a Specific Hazards override exists.
section5.uv_specific_hazards; SDSGenerator.php:1954-1956; \\${this->uvPack = UVAcrylateRulePack::isApplicable (settings.uv_acrylate_rule_pack + family_is_uv) :208-214, familyFields :1180

Advice for Firefighters:

TEXT CHOSEN BY HAZARD Fixed sentence chosen by the final hazard codes. Order: product override > explosive (H200-H205, plus a no-water sentence if also H260/H261) > water-reactive (H260/H261) > flammable liquid (flash-point-derived Cat 1-4 or FG override) > default.
On a sheet: Wear self-contained breathing apparatus (SCBA)... Eliminate all ignition sources...
You maintain: The hazard codes and flash point data above. Optional per-product override.
If missing: "Wear self-contained breathing apparatus (SCBA) and full protective gear. Cool containers exposed to fire with water spray."
text_overrides(5,'firefighter_advice') > section5.firefighter_advice_explosive (+ firefighter_no_water) > _water_reactive > _flammable > firefighter_advice; SDSGenerator.php:1960-1976

Preview Warnings box - flash point / flammability warnings (not printed):

FORMULA / RAW MATERIALS Operator-only messages, never printed. They cover a dropped or ignored Section 9 flash/boiling point edit, FP below 23 °C with no raw boiling point (Cat 2 assumed), an FG override category the formula FP does not support, and raws with a disclosed flammable-liquid CAS ($\geq 1\%$) but no FP.

On a sheet: Flash point warning: raw material(s) R123 contain an ingredient classified as a flammable...

You maintain: Clear them by entering flash points and boiling points on raw materials, or by correcting or resetting the Section 9 edit in the SDS editor.

If missing: No box when there is nothing to warn about. Publishing is never blocked by these warnings.

vetFlashPointEdit :4076, vetBoilingPointEdit :4118, temperatureOverrideWarnings :1817, flammabilityWarnings :4152 into \sds['warnings'] (:330-337); preview.php:663-667

SECTION 6: ACCIDENTAL RELEASE MEASURES

Section 6 is built from fixed, translated sentences. The hazard codes, the product flash point, the physical state and the UV family decide which sentences print. Operators can replace any of the three fields per product (or per resale raw material) and per language.

Personal Precautions:

FIXED TEXT (TRANSLATED) Field label from the language file. It is stored with the sheet when the sheet is generated.

On a sheet: Personal Precautions:

You maintain: Nothing

If missing: Stored label > sheet-language translation > English > 'Personal Precautions'. No label prints when the value is empty.

labels.personal_precautions -> meta.labels (getLabels :4767); PDFService FIELD_LABEL_MAP :43 + label() :1338; preview \$genericLabelMap :614

Personal Precautions (standard paragraph):

TEXT CHOSEN BY HAZARD Built-in sentence used when the product has no acute toxicity category 1-3 code (H300/H301/H310/H311/H330/H331) and no H314. Operator text > this sentence.

On a sheet: Use appropriate PPE (see Section 8). Avoid contact with skin and eyes. Ensure adequate...

You maintain: Hazard data on the CAS determinations and raw material screens (classifications, ATE values, competent-person determinations, trade-secret H-codes, constituent %), plus any finished-good hazard override.

If missing: This is the default. If hazard data is missing, it also prints for a toxic or corrosive product. Publishing is then blocked when a no-data CAS is at or above the Missing Data Threshold.

text_overrides(6, 'personal_precautions') > section6.personal_precautions; SDSGenerator::section6 :2013-2027; codes via extractHCodes(HazardEngine::classify) :1057

Personal Precautions (acute toxicity paragraph):

TEXT CHOSEN BY HAZARD Replaces the standard paragraph when any of H300, H301, H310, H311, H330 or H331 applies. The acute category is the more severe of the ingredient cut-off result and the ATEmix result.

On a sheet: Use appropriate PPE including self-contained breathing apparatus (SCBA) (see Section 8)...

You maintain: Acute toxicity classifications, ATE values and percentages of constituents (CAS determinations, raw materials); finished-good hazard override.

If missing: If no acute toxicity code results, the corrosive or standard paragraph prints.

section6.precautions_acute_toxic; SDSGenerator :2017-2018 (if branch ahead of the H314 elseif); acute categories from HazardEngine::classify

Personal Precautions (corrosive paragraph):

TEXT CHOSEN BY HAZARD Replaces the standard paragraph when H314 applies and no acute toxicity 1-3 code applies. H318 alone does not trigger it.

On a sheet: Use appropriate PPE including chemical-resistant suit, gloves, and face shield...

You maintain: Skin corrosion classifications of constituents; finished-good hazard override.

If missing: Standard paragraph prints.

section6.precautions_corrosive; SDSGenerator :2023-2024

Personal Precautions (corrosive add-on):

TEXT CHOSEN BY HAZARD Added after the acute toxicity paragraph when H314 also applies, because that paragraph has no skin or eye wording.

On a sheet: ... Wear a chemical-resistant suit, gloves and face shield. Avoid all contact with skin...

You maintain: Acute toxicity and skin corrosion classifications of constituents; finished-good hazard override.

If missing: Not added.

section6.precautions_corrosive_addon; SDSGenerator :2019-2022 (#32)

**Personal Precautions
(flammable add-on):**

FORMULA / RAW MATERIALS Added after the chosen paragraph for H220-H226 or H228. H224-H226 come from the product flash point (wt%-weighted over raw materials that have one: below 23 °C = Cat 1-2, 23-60 °C = Cat 3), not from ingredient codes.
On a sheet: ... Remove all sources of ignition. Use only non-sparking tools and explosion-proof equipment...
You maintain: Flash Point (and the '>' flag) on every raw material on the Raw materials screen; give water and similar raws a high flash point. Also the physical state, and the finished-good hazard override, where a Flammable Liquids entry replaces the derived category.
If missing: Not added when no raw material has a flash point (treated as not flammable), when the average is above 60 °C, or when the product is Solid, Powder or Paste.
section6.precautions_flammable; SDSGenerator :2028-2030; HazardEngine::flammableLiquidCategory :2762 / :2849; FormulaCalcService::weightedFlashPoint :690

**Personal Precautions (ignition
add-on):**

FORMULA / RAW MATERIALS Added in place of the flammable add-on for H227 or H250/H260/H261, when no H220-H226/H228 applies. H227 means the product flash point is above 60 °C and no higher than 93 °C (Flammable Liquids Cat 4); '> 60' also counts.
On a sheet: ... Eliminate all ignition sources (no smoking, sparks, open flames or hot surfaces)...
You maintain: Raw material flash points (Raw materials screen); pyrophoric and water-reactive classifications (CAS determinations); finished-good hazard override.
If missing: Not added.
section6.precautions_ignition; SDSGenerator :2031-2034 (#64); H227 from HazardEngine::flammableLiquidCategory Cat 4

Environmental Precautions:

FIXED TEXT (TRANSLATED) Field label from the language file. It is stored with the sheet when the sheet is generated.
On a sheet: Environmental Precautions:
You maintain: Nothing
If missing: Stored label > sheet-language translation > English > 'Environmental'.
labels.environmental_precautions (data key 'environmental'); PDFService :44 + label() :1338; preview.php :615

**Environmental Precautions
(drains sentence):**

FIXED TEXT Always the first sentence on every sheet. Operator text for this product > this sentence.
On a sheet: Prevent entry into drains, sewers, and waterways.
You maintain: Nothing
If missing: Always prints unless the field has operator text.
text_overrides(6,'environmental') > section6.environmental; SDSGenerator :2037-2040

**Environmental Precautions
(aquatic acute sentence):**

TEXT CHOSEN BY HAZARD One sentence for the acute route, chosen by H400 'Very toxic' > H401 'Toxic' > H402 'Harmful to aquatic life.'. It follows the drains sentence.
On a sheet: Very toxic to aquatic life.
You maintain: Aquatic acute classifications and M-factors of constituents (CAS determinations); finished-good hazard override.
If missing: No acute sentence.
section6.environmental_aquatic_acute / _acute_toxic / _harmful; SDSGenerator :2041-2049; engine keeps one acute code (dropLessSevereAquaticCodes :2117)

**Environmental Precautions
(aquatic chronic sentence):**

TEXT CHOSEN BY HAZARD Chosen separately from the acute sentence: H410 'Very toxic...' > H411 'Toxic...' > H412 'Harmful... with long lasting effects' > H413 'May cause long lasting harmful effects...'. It prints after the acute sentence when both apply.
On a sheet: Harmful to aquatic life with long lasting effects.
You maintain: Aquatic chronic classifications and M-factors of constituents; finished-good hazard override.
If missing: No chronic sentence.
section6.environmental_aquatic_chronic_very / _chronic / _chronic_harmful / _chronic_may_harm; SDSGenerator :2050-2060

**Environmental Precautions
(notify sentence):**

TEXT CHOSEN BY HAZARD Added last whenever any aquatic sentence printed.
On a sheet: Notify authorities if the product enters drains or waterways.
You maintain: Aquatic classifications.
If missing: Not printed for products with no aquatic code.
section6.environmental_notify; SDSGenerator :2061-2064

Containment and Cleanup:

FIXED TEXT (TRANSLATED) Field label from the language file. It is stored with the sheet when the sheet is generated.
On a sheet: Containment and Cleanup:
You maintain: Nothing
If missing: Stored label > sheet-language translation > English > 'Containment'.
labels.containment_cleanup (data key 'containment'); PDFService :45 + label() :1338; preview.php :616

Containment and Cleanup (solid / powder):	FORMULA / RAW MATERIALS Used when the physical state is Solid or Powder. Operator text > finished good's Physical State > state of the largest raw material in the formula that has one > 'Liquid'. <i>On a sheet: Avoid generating dust. Sweep or vacuum up material. Collect in suitable containers...</i> You maintain: Physical State on the finished good, or on its raw materials (the largest one that has a state is used). A resale sheet uses the raw material's own state. If missing: With no state anywhere, 'Liquid' is assumed, the liquid text prints and the preview shows a physical-state warning. section6.containment_solid; SDSGenerator :2068-2072; resolvePhysicalState :2521 (fg > FormulaCalcService formula_props.physical_state :530 > 'Liquid')
Containment and Cleanup (gel / paste):	FORMULA / RAW MATERIALS Used when the resolved physical state is Gel or Paste (same order as above). <i>On a sheet: Stop leak if safe to do so. Scrape up or shovel material and collect in suitable...</i> You maintain: Physical State = Gel or Paste on the finished good, or Paste on the raw material that sets the state (raw materials have no Gel option). If missing: Liquid text prints. section6.containment_paste; SDSGenerator :2073-2074
Containment and Cleanup (gas):	FORMULA / RAW MATERIALS Used when the resolved physical state is Gas. <i>On a sheet: Stop the leak if this can be done without risk. Ventilate the area and allow the gas...</i> You maintain: Physical State = Gas on the finished good, or on the raw material that sets the state. If missing: Liquid text prints. section6.containment_gas; SDSGenerator :2075-2076 (#64)
Containment and Cleanup (liquid / everything else):	FORMULA / RAW MATERIALS Used for Liquid, for custom states such as Aerosol, and when no state is recorded (assumed 'Liquid'). <i>On a sheet: Stop leak if safe to do so. Dam or contain spill. Absorb with inert absorbent material...</i> You maintain: Physical state data as above. When no state is recorded, this text prints and the preview shows a warning. If missing: This is the fallback text. section6.containment_liquid; SDSGenerator :2077-2080
Containment and Cleanup (UV add-on):	PRODUCT FAMILY Appended to whichever containment text printed when the UV acrylate setting is on and the product's resolved family is active and flagged UV/LED. No acrylate needs to be found in the formula. <i>On a sheet: ... Keep collected uncured product in closed containers away from UV light and direct...</i> You maintain: UV/LED flag and Active status on the Product families screen. Family assignment: automatic (UV families pooled by total share) or a manual pick on the finished good or resale raw material. UV acrylate toggle in Settings. If missing: Not added when the family is missing, inactive or not UV, or when the setting is off. Operator text for Containment also removes it. section6.uv_containment; SDSGenerator :2081-2083; uvPack = UVAcrylateRulePack::isApplicable :104 (isEnabled && family_is_uv from familyFields :1180)
Physical-state default warning (preview only):	FORMULA / RAW MATERIALS Shown in the preview Warnings box and never printed. It appears when neither the finished good nor any raw material in its formula has a physical state, so Sections 6, 8, 9 and 14 assume 'Liquid'. <i>On a sheet: Physical state: no physical state is recorded on this product or on any raw material...</i> You maintain: Physical State on the finished good or on its raw materials (on the raw material for a resale sheet). If missing: No warning when any state is recorded. SDSGenerator::PHYSICAL_STATE_DEFAULT_WARNING :2532, physicalStateIsDefault :2535; added to calcResult warnings in generate / computeBase / resale base

SECTION 7: HANDLING AND STORAGE

Section 7 is built automatically from the product's final GHS hazard codes and the P-statements Section 2 prints. Fixed base sentences come first, then a sentence for each matching hazard class, then UV sentences for products in an active UV/LED family. Storage ends by repeating the Section 10 incompatible-materials list. Flammable-liquid wording follows the product flash point (the wt%-weighted average over raws that have one), so a product with no flash point data gets no fire wording. A per-language override in the SDS editor (finished goods and resale raw materials) replaces either paragraph entirely, and a Storage override puts a warning on the preview.

Handling:	FIXED TEXT (TRANSLATED) Fixed label from the language files. <i>On a sheet: Handling:</i> You maintain: Nothing If missing: English 'Handling', then the literal 'labels.handling'. Old snapshots without meta.labels fall back to the sheet-language label, then 'Handling'. meta.labels.handling = labels.handling (en.php:623; getLabels SDSGenerator.php:4767); PDFService.php:47,1266-1281,1338; preview.php:9-13,617,640-643
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Handling (paragraph as a whole):	TEXT CHOSEN BY HAZARD Published sheet snapshot > the Section 7 Handling override for that language (finished good: SDS > Edit; resale raw material: Readiness Check > Edit SDS text), used word for word > base sentence + every matching hazard sentence + UV sentence. Alias and private-label sheets copy the base product's text, and a blank override counts as automatic. <i>On a sheet: Use only in well-ventilated areas. Avoid contact with skin, eyes, and clothing. ...</i> You maintain: GHS classifications on each CAS, flash points and physical state on raw materials, trade-secret hazard codes, the finished good's hazard override, the family UV flag, an optional per-language override, and a republish after any change If missing: Never blank. With no matching hazard the base sentence prints alone. An old stock paragraph saved by the pre-update editor keeps printing until the second cleanup pass (cleanup-legacy-overrides.php --apply) deletes it. snapshot_json > text_overrides(fg.id raw_material_id,7,'handling',lang,sds_version_id NULL) > handling_base + fragments; SDSGenerator.php:2234-2326, 4628-4661
Handling: base sentence (always first):	FIXED TEXT Fixed text that every sheet prints unless Handling is overridden. It ends 'Wear the personal protective equipment specified in Section 8.' <i>On a sheet: Use only in well-ventilated areas. Avoid contact with skin, eyes, and clothing.</i> You maintain: Nothing If missing: English, then the literal key (all four languages define it) section7.handling_base; SDSGenerator.php:2273; en.php:126; es/fr/de.php:116
Handling: heat and ignition sources:	TEXT CHOSEN BY HAZARD Added for flammable (H220-H226, H228), combustible (H227), pyrophoric (H250), self-reactive (H240-H242), oxidizer (H270-H272) or explosive (H200-H205) codes. H224-H227 come only from the product flash point (wt%-weighted average over raws that have one) plus boiling point, or from the finished good's hazard override. <i>On a sheet: Keep away from heat, hot surfaces, sparks, open flames, and other ignition sources. ...</i> You maintain: Flash point (with '>' flag) on every raw material, including a high value for water; physical state; other physical-hazard classifications on CAS records; the FG hazard override If missing: Not printed. No flash point on any raw, or a Solid/Powder/Paste product, gives no H224-H227 and so no fire wording unless another listed class applies. section7.handling_ignition when \$ignition (SDSGenerator.php:2247-2266, 2274-2276); en.php:127; HazardEngine::flammableLiquidCategory :2762
Handling: grounding and static:	TEXT CHOSEN BY HAZARD Added for flammable codes (H220-H226, H228), not for H227 combustible. A flammable-liquid code (H224-H226) means a product flash point at or below 60 °C (a '> n' value counts as just above n), or the FG override. <i>On a sheet: Keep container tightly closed. Ground and bond container and receiving equipment. ...</i> You maintain: Flash points on raw materials (product flash point at or below 60 °C); flammable gas/aerosol/solid classifications on CAS records; the FG hazard override If missing: Not printed section7.handling_flammable_static; SDSGenerator.php:2247, 2277-2279; en.php:128; HazardEngine::flammableLiquidCategory :2762-2777
Handling: aerosol:	TEXT CHOSEN BY HAZARD Added for aerosols (H222, H223, H229). <i>On a sheet: Pressurized container: do not pierce or burn, even after use.</i> You maintain: Aerosol classification, usually through the finished good's hazard override If missing: Not printed section7.handling_aerosol; SDSGenerator.php:2249, 2280-2282; en.php:129
Handling: explosive:	TEXT CHOSEN BY HAZARD Added for explosives (H200-H205). <i>On a sheet: Do not subject to grinding, shock or friction. Keep only in the original packaging.</i> You maintain: Explosive classification on the CAS records or the FG override If missing: Not printed section7.handling_explosive; SDSGenerator.php:2258, 2283-2285; en.php:153
Handling: gases under pressure:	TEXT CHOSEN BY HAZARD Added for gases under pressure (H280, H281). <i>On a sheet: Handle pressurized containers with care; do not drop or damage them. Close the valve ...</i> You maintain: Gas-under-pressure classification on the CAS records or the FG override If missing: Not printed section7.handling_gas_pressure; SDSGenerator.php:2259, 2286-2288; en.php:154
Handling: oxidizer:	TEXT CHOSEN BY HAZARD Added for oxidizers (H270, H271, H272). <i>On a sheet: Keep away from clothing and other combustible materials. Do not mix with flammable ...</i> You maintain: Oxidizer classification on the CAS records or the FG override If missing: Not printed section7.handling_oxidizer_combustibles; SDSGenerator.php:2250, 2289-2291; en.php:130

Handling: self-reactive:	TEXT CHOSEN BY HAZARD Added for self-reactive substances and organic peroxides (H240-H242). <i>On a sheet: Keep cool. Handle and open container with care. Keep only in original container ...</i> You maintain: Self-reactive / organic peroxide classification If missing: Not printed section7.handling_self_reactive; SDSGenerator.php:2251, 2292-2294; en.php:124
Handling: self-heating:	TEXT CHOSEN BY HAZARD Added for self-heating substances (H251, H252). <i>On a sheet: Keep cool. Protect from sunlight. Maintain air gap between stacks or pallets. ...</i> You maintain: Self-heating classification If missing: Not printed section7.handling_self_heating; SDSGenerator.php:2252, 2295-2297; en.php:131
Handling: pyrophoric and water-reactive:	TEXT CHOSEN BY HAZARD Printed instead of the separate air and water sentences when the product is both pyrophoric (H250) and water-reactive (H260/H261), so 'inert gas' appears once. <i>On a sheet: Do not allow contact with air or water. Handle under inert gas.</i> You maintain: Pyrophoric and water-reactive classifications If missing: Not printed section7.handling_pyrophoric_water_reactive; SDSGenerator.php:2298-2299; en.php:152
Handling: pyrophoric (air):	TEXT CHOSEN BY HAZARD Added for pyrophoric substances (H250) that are not also water-reactive. <i>On a sheet: Do not allow contact with air. Handle under inert gas.</i> You maintain: Pyrophoric classification If missing: Not printed section7.handling_pyrophoric_air; SDSGenerator.php:2253, 2300-2301; en.php:132
Handling: water-reactive:	TEXT CHOSEN BY HAZARD Added for water-reactive substances (H260, H261) that are not also pyrophoric. <i>On a sheet: Do not allow contact with water. Handle under inert gas.</i> You maintain: Water-reactive classification If missing: Not printed section7.handling_water_reactive_water; SDSGenerator.php:2254, 2302-2303; en.php:133
Handling: moisture:	TEXT CHOSEN BY HAZARD Printed once when the product is pyrophoric (H250) or water-reactive (H260/H261). <i>On a sheet: Protect from moisture.</i> You maintain: Pyrophoric / water-reactive classification If missing: Not printed section7.handling_moisture; SDSGenerator.php:2305-2307; en.php:134
Handling: corrosive:	TEXT CHOSEN BY HAZARD Added for skin-corrosive products (H314). <i>On a sheet: Do not breathe mist, vapors, or spray. Do not get in eyes, on skin, or on clothing.</i> You maintain: Skin Corr. 1 classification (including the engine's summation) If missing: Not printed section7.handling_corrosive; SDSGenerator.php:2255, 2308-2310; en.php:135
Handling: corrosive to metals:	TEXT CHOSEN BY HAZARD Added for products corrosive to metals (H290). <i>On a sheet: Keep only in original container.</i> You maintain: Met. Corr. 1 classification If missing: Not printed section7.handling_corrosive_metals; SDSGenerator.php:2256, 2311-2313; en.php:136
Handling: repeated-exposure organ toxicity:	TEXT CHOSEN BY HAZARD Added for STOT RE (H372, H373) unless the product is also H314, whose sentence already says 'Do not breathe'. <i>On a sheet: Do not breathe dust, mist, vapors, or spray.</i> You maintain: STOT RE classification on the CAS records If missing: Not printed section7.handling_stot_re; SDSGenerator.php:2260, 2314-2316; en.php:155
Handling: sensitizer:	TEXT CHOSEN BY HAZARD Added for skin or respiratory sensitizers (H317, H334). When H314 or H372/H373 already printed a 'Do not breathe' sentence, only the work-clothing half prints. <i>On a sheet: Avoid breathing vapors, mist, or spray. Contaminated work clothing must not be ...</i> You maintain: Skin / respiratory sensitizer classification If missing: Not printed section7.handling_sensitizer handling_sensitizer_clothing; SDSGenerator.php:2257, 2268 (\$breatheSaid), 2317-2320; en.php:137,156

Handling: UV uncured product:	PRODUCT FAMILY Added when the UV acrylate rule pack is on in admin settings and the product's resolved family is active and flagged UV/LED. No acrylate needs to be detected. Alias, private-label and resale sheets follow the same rule. <i>On a sheet: Avoid prolonged or repeated skin contact with uncured product.</i> You maintain: Product family UV/LED flag, active flag and membership rules or manual picks (Product Families); the admin 'UV acrylate rule pack' toggle If missing: Not printed. It is also dropped when Handling is overridden. section7.uv_handling when \$this->uvPack (UVAcrylateRulePack::isApplicable :104; familyFields :1180); SDSGenerator.php:2321-2323; en.php:160
Storage:	FIXED TEXT (TRANSLATED) Fixed label from the language files. <i>On a sheet: Storage:</i> You maintain: Nothing If missing: English 'Storage', then the literal 'labels.storage'. Old snapshots fall back to the sheet-language label, then 'Storage'. meta.labels.storage = labels.storage (en.php:624; getLabels SDSGenerator.php:4767); PDFService.php:48,1266-1281; preview.php:618,640-643
Storage (paragraph as a whole):	TEXT CHOSEN BY HAZARD Published sheet snapshot > the Section 7 Storage override for that language (finished good or resale raw material), used word for word, which also replaces the incompatibles list > base sentence + hazard sentences + UV sentence + 'Store locked up.' + Section 10 incompatibles. Alias and private-label sheets copy the base product, and a blank override counts as automatic. <i>On a sheet: Store in a cool, dry, well-ventilated area. ... (see Section 10): strong oxidizing ...</i> You maintain: CAS GHS classifications, raw-material flash points, the FG hazard override, the family UV flag, optional Section 7 Storage and Section 10 Incompatible Materials overrides, and a republish after changes If missing: Never blank. The minimum is the base sentence plus 'Store away from incompatible materials (see Section 10): strong oxidizing agents, strong acids, strong bases.' An old stock override prints until the second cleanup pass deletes it. snapshot_json > text_overrides(fg.id raw_material_id,7,'storage',lang,NULL) > storage_base + fragments + storage_locked + storage_incompatible; SDSGenerator.php:2328-2378
Storage: base sentence (always first):	FIXED TEXT Fixed text that every sheet prints unless Storage is overridden. <i>On a sheet: Store in a cool, dry, well-ventilated area. Keep container tightly closed when not ...</i> You maintain: Nothing If missing: English, then the literal key section7.storage_base; SDSGenerator.php:2330; en.php:138; es/fr/de.php:128
Storage: heat and ignition sources:	TEXT CHOSEN BY HAZARD Same trigger as the Handling ignition sentence: flammable, combustible (H227), pyrophoric, self-reactive, oxidizer or explosive. H224-H227 come only from the product flash point or the FG override. <i>On a sheet: Keep away from heat and ignition sources.</i> You maintain: Flash points on raw materials; physical-hazard classifications on the CAS records; the FG hazard override If missing: Not printed section7.storage_ignition when \$ignition; SDSGenerator.php:2266, 2331-2333; en.php:139
Storage: flammables storage area:	TEXT CHOSEN BY HAZARD Added for flammable codes (H220-H226, H228), not for H227. H224-H226 mean a product flash point at or below 60 °C (a '> n' value counts as just above n), or the FG override. <i>On a sheet: Store in an approved flammable-materials storage area. Ground and bond containers ...</i> You maintain: Flash points on raw materials; flammable gas/aerosol/solid classifications; the FG hazard override If missing: Not printed section7.storage_flammable_area; SDSGenerator.php:2247, 2334-2336; en.php:140; HazardEngine::flammableLiquidCategory :2762-2777
Storage: aerosol:	TEXT CHOSEN BY HAZARD Added for aerosols (H222, H223, H229). <i>On a sheet: Protect from sunlight. Do not expose to temperatures exceeding 50 °C / 122 °F.</i> You maintain: Aerosol classification If missing: Not printed section7.storage_aerosol; SDSGenerator.php:2337-2339; en.php:141
Storage: explosive:	TEXT CHOSEN BY HAZARD Added for explosives (H200-H205). <i>On a sheet: Store in accordance with the applicable explosives storage regulations, away from ...</i> You maintain: Explosive classification If missing: Not printed section7.storage_explosive; SDSGenerator.php:2340-2342; en.php:157

Storage: gases under pressure:	TEXT CHOSEN BY HAZARD Added for gases under pressure (H280, H281). <i>On a sheet: Protect from sunlight. Store in a well-ventilated place. Secure cylinders against falling.</i> You maintain: Gas-under-pressure classification If missing: Not printed section7.storage_gas_pressure; SDSGenerator.php:2343-2345; en.php:158
Storage: oxidizer:	TEXT CHOSEN BY HAZARD Added for oxidizers (H270-H272). <i>On a sheet: Store separately from combustible materials, reducing agents, and flammable substances.</i> You maintain: Oxidizer classification If missing: Not printed section7.storage_oxidizer_separate; SDSGenerator.php:2346-2348; en.php:142
Storage: self-reactive:	TEXT CHOSEN BY HAZARD Added for self-reactive substances and organic peroxides (H240-H242). <i>On a sheet: Store at the recommended temperature. Keep away from heat and sunlight.</i> You maintain: Self-reactive classification If missing: Not printed section7.storage_self_reactive; SDSGenerator.php:2349-2351; en.php:143
Storage: self-heating:	TEXT CHOSEN BY HAZARD Added for self-heating substances (H251, H252). <i>On a sheet: Keep cool and protect from sunlight. Maintain air gap between stacks or pallets.</i> You maintain: Self-heating classification If missing: Not printed section7.storage_self_heating_cool; SDSGenerator.php:2352-2354; en.php:144
Storage: pyrophoric (air):	TEXT CHOSEN BY HAZARD Added for pyrophoric substances (H250). <i>On a sheet: Protect from air.</i> You maintain: Pyrophoric classification If missing: Not printed section7.storage_pyrophoric_air; SDSGenerator.php:2355-2357; en.php:145
Storage: water-reactive:	TEXT CHOSEN BY HAZARD Added for water-reactive substances (H260, H261). <i>On a sheet: Protect from water.</i> You maintain: Water-reactive classification If missing: Not printed section7.storage_water_reactive_water; SDSGenerator.php:2358-2360; en.php:146
Storage: inert gas and moisture:	TEXT CHOSEN BY HAZARD Printed once for pyrophoric (H250) or water-reactive (H260/H261) products. <i>On a sheet: Store under inert gas. Protect from moisture.</i> You maintain: Pyrophoric / water-reactive classification If missing: Not printed section7.storage_inert_moisture; SDSGenerator.php:2361-2363; en.php:147
Storage: corrosive to metals:	TEXT CHOSEN BY HAZARD Added for products corrosive to metals (H290). <i>On a sheet: Store in a corrosion-resistant container with a resistant inner liner.</i> You maintain: Met. Corr. 1 classification If missing: Not printed section7.storage_corrosive_metals; SDSGenerator.php:2364-2366; en.php:148
Storage: UV light:	PRODUCT FAMILY Added when the product's resolved family is active and flagged UV/LED. The admin rule-pack toggle is ignored, as for the Section 10 UV condition. <i>On a sheet: Store away from UV light sources and direct sunlight to prevent premature polymerization.</i> You maintain: Product family UV/LED flag, active flag and membership rules or manual picks (Product Families) If missing: Not printed. It is also dropped when Storage is overridden. section7.uv_storage when \$this->uvFamily (UVAcrylateRulePack::familyIsUv :94); SDSGenerator.php:2367-2370; en.php:161
Storage: Store locked up:	TEXT CHOSEN BY HAZARD Added when the P-statements Section 2 prints include P405, alone or in a combined code. P405 comes from the class defaults for Acute Tox. 1-3, Skin Corr. 1, CMR, STOT SE 1-3 and Asp. Tox. 1-2, or from source data or the FG override. <i>On a sheet: Store locked up.</i> You maintain: Health-hazard classifications on the CAS records; P-codes in the FG hazard override If missing: Not printed section7.storage_locked when \$lockUp = P405 in resolvedPCodes() (SDSGenerator.php:1082-1095, 2264, 2371-2373); en.php:149; GHSazardData p_codes

Storage: incompatible materials (see Section 10):	TEXT CHOSEN BY HAZARD The Section 10 Incompatible Materials override (trailing full stop removed; first letter lower-cased except on German sheets or acronyms like PVC) > a list built from the hazard classes. The list always has acids and bases, has oxidizing agents unless the product is an oxidizer, and adds items per class (e.g. halogens for H220-H228). Not printed when Storage is overridden. <i>On a sheet: Store away from incompatible materials (see Section 10): strong oxidizing agents, ...</i> You maintain: Physical-hazard classifications and raw-material flash points; optional Section 10 Incompatible Materials override in the SDS editor If missing: Always printed: with no physical hazard the list is 'strong oxidizing agents, strong acids, strong bases' code: section7.storage_incompatible :list = incompatibleMaterialsInline() :2224-2232 + inlineCase() :1235; incompatibleMaterialItems() :2160-2205; en.php:150
Section 7 warning (preview only):	PRODUCT OVERRIDE Shown in the preview Warnings box, never printed on the SDS, whenever a non-blank Section 7 Storage override is stored for the sheet language. It says to list the incompatibles in the override or reset Storage to automatic. <i>On a sheet: Section 7 warning: the Storage text is a per-product override, so it no longer names ...</i> You maintain: The Section 7 Storage override (SDS editor) If missing: No warning when Storage is automatic code: storageOverrideWarning() SDSGenerator.php:1220-1227, appended at :275-278 (generate) and :677-680 (generateFromBase); preview.php:663-672

SECTION 8: EXPOSURE CONTROLS / PERSONAL PROTECTION

Section 8 lists the federal and CAS-determination exposure limits for each ingredient, with the Section 3 concentration band and trade-secret identities masked, or prints a fixed 'no exposure limits' sentence. The engineering-control and PPE sentences are chosen automatically from the product's final hazard codes (flammability comes from the product flash point) and its physical state. UV-family products get extra PPE sentences on hazard-driven lines, and each line can be replaced by a per-product override.

Exposure limits table (which rows appear):	HAZARD ENGINE One row per exposure limit on file for each ingredient at 0.01% or more: federal limits (NIOSH, ACGIH, OSHA) always, limits from the active CAS determination only when that CAS has no current federal hazard classification, and limits in a vendor trade-secret hazard declaration that carries hazards. Inhalation-only CAS (carbon black, TiO2) and ingredients supplied only by solid/powder raw materials get no rows when any raw material in the product is not solid or powder, and a limit repeated with the same CAS, type, value and units prints once. <i>On a sheet: 108-88-3 Toluene PEL-TWA 754 mg/m3 10 - 30% Skin; 29 CFR 1910.1000 Table Z-1</i> You maintain: Keep the federal data imports current. Add exposure limits on Admin > CAS determinations for chemicals with no federal data, and on the vendor hazard declaration of trade-secret raw materials. Set the physical state on every raw material, keep constituent CAS numbers and percentages complete, and keep the inhalation-only CAS list in Admin Settings up to date. If missing: When no rows remain, the table is replaced by the italic sentence "No occupational exposure limits apply to the components of this product in the form supplied." code: HazardEngine::classify :714-735 (hsr.is_current=1), trade-secret JSON :765-806, CPD :2998; excludeInhalationOnlyCas :586; dedupeExposureLimits :1224; SDSGenerator :5223
Exposure limits table headers: CAS Chemical Type Value Units Conc% Notes:	FIXED TEXT (TRANSLATED) Fixed column headings from the language file. <i>On a sheet: CAS Chemical Type Value Units Conc% Notes</i> You maintain: Nothing If missing: English heading, then the key name code: labels.el_cas/el_chemical/el_type/el_value/el_units/el_conc_pct/el_notes; PDFService.php:824-837 widths [22,42,22,20,17,15,37] mm; preview.php:276-287
CAS (exposure limits table):	FORMULA / RAW MATERIALS The ingredient's CAS number from the raw-material constituent. If any raw material marks that CAS as a trade secret, or the limit comes from a vendor trade-secret hazard declaration, it prints "TRADE SECRET" instead. <i>On a sheet: 108-88-3</i> You maintain: CAS number and the trade-secret box on each raw-material constituent (Raw materials screen) If missing: Constituents with no CAS never get a row; vendor trade-secret declarations print "TRADE SECRET" code: rmc.cas_number via Formula::getExpandedComposition; SDSGenerator::maskTradeSecretRow :2627-2646 -> labels.trade_secret_cas (is_trade_secret or CAS 'TRADE_SECRET')

Chemical (exposure limits table):	FORMULA / RAW MATERIALS Ingredient name: Prop 65 list name > CAS master preferred name > constituent name typed on the raw material, never translated. For a trade-secret ingredient: its trade-secret description > "Trade Secret". <i>On a sheet: Toluene</i> You maintain: Preferred name in the CAS master, or the constituent name on the raw material; the trade-secret description on trade-secret constituents If missing: Blank cell. A trade-secret row with no description prints "Trade Secret" (translated). COALESCE(prop65_list.chemical_name, cas_master.preferred_name, rmc.chemical_name) Formula.php:293; HazardEngine.php:830; tradeSecretName() SDSGenerator.php:2607
Type (exposure limits table):	REGULATORY LIST Limit type as stored in the federal data or chosen on the determination. It is printed exactly as stored. <i>On a sheet: TLV-TWA</i> You maintain: Federal data imports; the limit type on each determination row If missing: Blank cell (determination rows only) exposure_limits.limit_type (NIOSH REL-/PEL-, ACGIH TLV-, OSHA PEL-...); CPD exp_limit_type[] PEL-TWA/TLV-TWA/REL-TWA/STEL/Ceiling/IDLH/Other (determination-form.php:311)
Value (exposure limits table):	REGULATORY LIST Numeric limit from the federal data or the determination, printed as stored. <i>On a sheet: 20</i> You maintain: Federal data imports; the value on each determination limit If missing: A determination row with no value is dropped exposure_limits.value (NOT NULL); CPD determination_json.exposure_limits[].value, rows with empty() value (incl. '0') skipped HazardEngine.php:3305-3310
Units (exposure limits table):	REGULATORY LIST Unit stored with the limit. Importers keep mg/m3 when the source gives it, otherwise ppm, so only one unit is printed per limit. <i>On a sheet: mg/m3</i> You maintain: Federal data imports; the units on each determination limit If missing: "mg/m3" exposure_limits.units (DB default 'mg/m3'; NIOSHConnector and load-seed-data prefer *_mgm3); CPD units mg/m3/ppm/f/cc/mppcf, default 'mg/m3' HazardEngine.php:3316
Conc% (exposure limits table):	FORMULA / RAW MATERIALS The ingredient's prescribed concentration band, the same band Section 3 prints, never the exact %. It is the widest band containing the ingredient's summed lower and upper bounds (an exact % adds the same value to both), else the band containing the upper bound with the lowest lower end. <i>On a sheet: 10 - 30%</i> You maintain: Exact % or min/max % on raw-material constituents; formula line percentages If missing: "<0.1%" when the upper bound is below 0.1%; "80 - 100%" when it exceeds 100%. Blank on snapshots published before #8. section8() SDSGenerator.php:2397-2410 -> formatConcentration :4712 (PRESCRIBED_RANGES :4698); bounds Formula::constituentBounds; concentration_pct unset
Notes (exposure limits table):	REGULATORY LIST ACGIH rows: the notation; OSHA rows: the notation plus the CFR reference; determination rows: free text; NIOSH rows: blank. When the same limit is on file twice, the first row's notes print, or the first non-blank notes of a duplicate. <i>On a sheet: Skin; A4</i> You maintain: Federal seed files; notes on determination limits If missing: Blank cell exposure_limits.notes from load-seed-data.php:159-170 (notation [+ ' ; ' + cfr_ref]); CPD exposure_limits[].notes; merged in dedupeExposureLimits HazardEngine.php:1224
No occupational exposure limits apply to the components of this product in the form supplied.:	FIXED TEXT (TRANSLATED) Fixed sentence from the language file, printed in italics in place of the table when no exposure-limit row remains. <i>On a sheet: No occupational exposure limits apply to the components of this product in the form...</i> You maintain: Nothing (the exposure-limit rows decide whether it prints) If missing: Not printed when any limit row exists. Snapshots published before the change print nothing in its place. section8.no_exposure_limits -> exposure_limits_none SDSGenerator.php:2427; PDFService.php:871-877; preview.php:302-303

Engineering Controls:	TEXT CHOSEN BY HAZARD A standard ventilation sentence, plus a dust sentence when the product's physical state is Powder, an explosion-proof and grounding sentence when the product carries H224-H226 (Flammable Liquid Cat 1-3 from its flash point, or a finished-good hazard override), and an eyewash sentence for H314/H318. Precedence: per-product override > automatic sentences. <i>On a sheet: Use local exhaust ventilation or other engineering controls to maintain airborne...</i> You maintain: Physical state on the finished good (else the largest raw material that has one). Flash point on raw materials, the finished-good hazard override, and the hazard data that produces H314/H318. Optional override in the SDS editor. If missing: Only the base sentence. With no state on the product or any raw material the state counts as Liquid (no dust sentence) and the preview warns; no flash point data means no flammable sentence. <pre>non-blank text_overrides[8]['engineering'] (TextOverrideService::effective) > engineeringControls() SDSGenerator.php:2449-2465; state = resolvePhysicalState :2521</pre>
Respiratory Protection:	TEXT CHOSEN BY HAZARD One of five sentences chosen from the final H-codes (SCBA, sensitizer, cartridge respirator, general, none), plus the UV sentence on UV products when the tier is hazard-driven or the product has H317. Precedence: per-product override > automatic. <i>On a sheet: If exposure limits are exceeded or irritation is experienced, use a NIOSH-approved...</i> You maintain: Hazard data that drives the H-codes (federal data, determinations, vendor trade-secret declarations, finished-good hazard override). Optional override per language in the SDS editor (on the raw material for resale sheets). If missing: "No special respiratory protection required under normal conditions of use with adequate ventilation." <pre>non-blank text_overrides[8]['respiratory'] > section8.ppe.respiratory.<tier> (HazardEngine::derivePPE :1279) in SDSGenerator::resolvePPE :2483 (+section8.uv_respiratory)</pre>
Hand Protection:	TEXT CHOSEN BY HAZARD Tier sentence chosen from the final H-codes (impervious, sensitizer, resistant, general, none), plus the UV sentence on UV products when the tier is hazard-driven or the product has H317. Precedence: per-product override > automatic. <i>On a sheet: Chemical-resistant gloves (nitrile recommended) meeting 29 CFR 1910.138...</i> You maintain: Hazard data that drives the H-codes; optional override in the SDS editor If missing: "No special hand protection required under normal conditions of use. Gloves may be worn as good industrial hygiene practice." <pre>non-blank text_overrides[8]['hand_protection'] > section8.ppe.hand_protection.<tier> (H310/H311/H314 impervious; H317 sensitizer; H312/H313/H315/H316/systemic resistant)</pre>
Eye Protection:	TEXT CHOSEN BY HAZARD Tier sentence chosen from the final H-codes (goggles and face shield, goggles, glasses, general, none), plus the UV sentence on UV products when the tier is hazard-driven or the product has H317. Precedence: per-product override > automatic. <i>On a sheet: Chemical splash goggles or safety glasses with side shields meeting ANSI/ISEA Z87.1...</i> You maintain: Hazard data that drives the H-codes; optional override in the SDS editor If missing: "No special eye protection required under normal conditions of use. Safety glasses may be worn as good industrial hygiene practice." <pre>non-blank text_overrides[8]['eye_protection'] > section8.ppe.eye_protection.<tier> (H314/H318 goggles_faceshield; H319 goggles; H320 glasses)</pre>
Skin Protection:	TEXT CHOSEN BY HAZARD Tier sentence chosen from the final H-codes (suit, clothing, general, none), plus the UV sentence on UV products when the tier is hazard-driven or the product has H317. Precedence: per-product override > automatic. <i>On a sheet: Wear protective clothing (long sleeves and an impervious apron) to prevent skin contact...</i> You maintain: Hazard data that drives the H-codes; optional override in the SDS editor If missing: "No special protective clothing required under normal conditions of use." <pre>non-blank text_overrides[8]['skin_protection'] > section8.ppe.skin_protection.<tier> (H310/H311/H314 suit; H312/H313/H315/H316/H317/systemic clothing)</pre>
UV acrylate sentences (added to the end of each PPE line):	PRODUCT FAMILY Added to a PPE line when the UV rule pack is on and the product's resolved, active family is marked UV/LED, but only where that line's tier is hazard-driven or the product has H317. No acrylate ingredient is required, the same text prints in Section 2 PPE, and a line with an override gets none. <i>On a sheet: ... For uncured acrylate product, wear chemical-resistant gloves (nitrile, minimum 0.4 mm...</i> You maintain: The UV/LED flag and Active status on the product family (Product families screen) and the product's family (rules, content share with UV families pooled, or manual pick). The UV rule pack toggle in Admin Settings. If missing: Nothing is added. An unclassified UV product without H317 keeps its "No special ... required" sentences. <pre>UVAcrylateRulePack::isApplicable :104 (settings.uv_acrylate_rule_pack && family_is_uv, SDSGenerator::familyFields :1180) -> resolvePPE :2502; section8.uv_<field></pre>

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Section 9 prints every HazCom Appendix D property, followed by VOC and solids. Eleven lines come from the finished good and its raw materials: the product's own fields, the largest raw material, the lowest value, a weighted average or weighted sums. Twelve Appendix D properties have no data field, so they print "Not determined" unless a value is typed for the product. Seventeen lines can be replaced per product and per language. Labels, the title and the fallback wording come from the translation files.

Physical State:	FORMULA / RAW MATERIALS The product's own physical state > the state of the largest raw material (by summed wt%) that has one > 'Liquid'. The six standard states are translated. <i>On a sheet: Physical State: Liquid</i> You maintain: Physical state on the finished good (Finished Goods > Edit), or on the raw materials (Raw Materials > Edit). If missing: 'Liquid' (Líquido / Liquide / Flüssig), and the preview shows a warning. A raw with no state is skipped and the next-largest raw with a state is used. <pre>resolvePhysicalState (SDSGenerator.php:2521-2529): finished_goods.physical_state > formula_props.physical_state (FormulaCalcService.php:604-623) > 'Liquid'; localizeSection9Enum :2579</pre>
Color:	FORMULA / RAW MATERIALS The product's colour dropdown. The seven default colours are translated; a custom colour prints as entered. <i>On a sheet: Color: Black</i> You maintain: Colour on the finished good (Finished Goods > Edit). If missing: The Color line is left off. It is always left off on resale raw-material sheets, because raw materials have no colour field. <pre>finished_goods.color (vocab setting sds.color_options) -> localizeSection9Enum('color') (SDSGenerator.php:2659-2662, 2579-2597); PDF labelValue skips ''</pre>
Appearance:	FORMULA / RAW MATERIALS Per-product edit > product colour + resolved state, in each language's word order > the state alone. Resale sheets use the raw material's appearance text before falling back to the state. <i>On a sheet: Appearance: Black liquid (ES: Líquido, negro)</i> You maintain: Colour and physical state on the finished good. For resale items, appearance on the raw material. Optional edit in SDS > Edit Text, per language. If missing: The state alone, e.g. 'liquid' (EN, lower case) or 'Líquido' (ES). Never blank. <pre>text_overrides(9,'appearance',lang) > section9.appearance_color_state if finished_goods.color > (is_resale) formula_props.appearance > section9.appearance_state (SDSGenerator.php:2697-2715)</pre>
Odor:	FORMULA / RAW MATERIALS Per-product edit > odour of the raw material with the highest summed wt% > 'Not determined'. Raw-material text is not translated. <i>On a sheet: Odor: Mild solvent</i> You maintain: Odor on the raw material that dominates each formula (Raw Materials > Edit). Optional edit in SDS > Edit Text. If missing: 'Not determined' (No determinado / Non déterminé / Nicht bestimmt). The second-largest raw is not tried. <pre>text_overrides(9,'odor',lang) > formula_props.odor (dominant raw_materials.odor; FormulaCalcService.php:567-624) > labels.not_determined (SDSGenerator.php:2721-2727)</pre>
Odor Threshold:	PRODUCT OVERRIDE Per-product value typed in the editor > 'Not determined'. No raw-material or formula data feeds it. <i>On a sheet: Odor Threshold: Not determined</i> You maintain: Only when a tested value exists: SDS > Edit Text, Section 9, per language (resale items: Edit Resale SDS, stored on the raw material). If missing: 'Not determined' <pre>SECTION9_APPENDIX_D_FIELDS (SDSGenerator.php:2548-2552); text_overrides(9,'odor_threshold',lang) > labels.not_determined (:2740-2750)</pre>
pH:	PRODUCT OVERRIDE Per-product value typed in the editor > 'Not determined'. No raw-material or formula data feeds it. <i>On a sheet: pH: Not determined</i> You maintain: Only when a tested value exists: SDS > Edit Text, Section 9, per language. If missing: 'Not determined' <pre>SECTION9_APPENDIX_D_FIELDS (SDSGenerator.php:2548-2552); text_overrides(9,'ph',lang) > labels.not_determined (:2740-2750)</pre>
Melting Point/Freezing Point:	PRODUCT OVERRIDE Per-product value typed in the editor > 'Not determined'. No raw-material or formula data feeds it. <i>On a sheet: Melting Point/Freezing Point: Not determined</i> You maintain: Only when a tested value exists: SDS > Edit Text, Section 9, per language. If missing: 'Not determined' <pre>SECTION9_APPENDIX_D_FIELDS (SDSGenerator.php:2548-2552); text_overrides(9,'melting_point',lang) > labels.not_determined (:2740-2750)</pre>

Initial Boiling Point and Boiling Range:	FORMULA / RAW MATERIALS Per-product edit, if it reads as a temperature and does not move a Category 1/2 product across 35 °C > lowest boiling point of any raw in the full formula (wt% ignored) > 'Not determined'. <i>On a sheet: Initial Boiling Point and Boiling Range: 78.4 °C (173.1 °F)</i> You maintain: Boiling point (°C) on solvent and other volatile raw materials (Raw Materials > Edit). Pigments and resins can stay blank. Optional edit in SDS > Edit Text, with a °C/°F unit. If missing: 'Not determined'. If the flash point is below 23 °C, the product is assumed Flammable Liquids Category 2 / PG II and the preview warns. resolveBoilingPointDisplay (SDSGenerator.php:1776-1784) after vetBoilingPointEdit (:4118-4136); MIN(raw_materials.boiling_point_c) (FormulaCalcService.php:534-553); TemperatureParser
Flash Point:	FORMULA / RAW MATERIALS Per-product edit, kept only if it is a temperature giving the same flammability result > wt%-weighted average over raws with a flash point, with '>' if any is flagged > 'Not determined'. Section 5 prints the same text, but only on a flammable product. <i>On a sheet: Flash Point: > 71.3 °C (160.3 °F)</i> You maintain: Flash point and '>' flag on every raw material, including a high value (e.g. 100) for water and other non-flammable liquid raws (Raw Materials > Edit). If missing: 'Not determined'. The product is then treated as not flammable (no Flammable Liquids class, Section 14 not Class 3) unless a finished-good hazard override sets one. Publishing is not blocked. resolveFlashPointDisplay (SDSGenerator.php:1742-1751) after vetFlashPointEdit (:4076-4107); FormulaCalcService::weightedFlashPoint $\Sigma(\text{pct} \cdot \text{FP}) / \Sigma \text{pct}$, round 0.1 (:690)
Evaporation Rate:	PRODUCT OVERRIDE Per-product value typed in the editor > 'Not determined'. No raw-material or formula data feeds it. <i>On a sheet: Evaporation Rate: Not determined</i> You maintain: Only when a tested value exists: SDS > Edit Text, Section 9, per language. If missing: 'Not determined' SECTION9_APPENDIX_D_FIELDS (SDSGenerator.php:2548-2552); text_overrides(9,'evaporation_rate',lang) > labels.not_determined (:2740-2750)
Flammability (Solid, Gas):	PRODUCT OVERRIDE Per-product value typed in the editor > 'Not applicable' when the resolved physical state is Liquid > 'Not determined' (Solid, Powder, Paste, Gel, Gas). <i>On a sheet: Flammability (Solid, Gas): Not applicable</i> You maintain: Physical state on the finished good or its raws. Optional value in SDS > Edit Text. If missing: 'Not applicable' for liquids; otherwise 'Not determined'. SDSGenerator.php:2740-2750: text_overrides(9,'flammability_solid_gas') > strcasecmp(physicalState,'Liquid')==0 -> labels.not_applicable > labels.not_determined
Upper/Lower Flammability or Explosive Limits:	PRODUCT OVERRIDE Per-product value typed in the editor > 'Not determined'. No raw-material or formula data feeds it. <i>On a sheet: Upper/Lower Flammability or Explosive Limits: Not determined</i> You maintain: Only when a tested value exists: SDS > Edit Text, Section 9, per language. If missing: 'Not determined'. The label is too wide for the column, so the value prints on the next line. SECTION9_APPENDIX_D_FIELDS (SDSGenerator.php:2548-2552); text_overrides(9,'flammability_limits',lang) > labels.not_determined (:2740-2750)
Vapor Pressure:	PRODUCT OVERRIDE Per-product value typed in the editor > 'Not determined'. No raw-material or formula data feeds it. <i>On a sheet: Vapor Pressure: Not determined</i> You maintain: Only when a tested value exists: SDS > Edit Text, Section 9, per language. If missing: 'Not determined' SECTION9_APPENDIX_D_FIELDS (SDSGenerator.php:2548-2552); text_overrides(9,'vapor_pressure',lang) > labels.not_determined (:2740-2750)
Vapor Density:	PRODUCT OVERRIDE Per-product value typed in the editor > 'Not determined'. No raw-material or formula data feeds it. <i>On a sheet: Vapor Density: Not determined</i> You maintain: Only when a tested value exists: SDS > Edit Text, Section 9, per language. If missing: 'Not determined' SECTION9_APPENDIX_D_FIELDS (SDSGenerator.php:2548-2552); text_overrides(9,'vapor_density',lang) > labels.not_determined (:2740-2750)
Specific Gravity:	FORMULA / RAW MATERIALS Weight-averaged mixture specific gravity, $\Sigma w / \Sigma (w / \text{SG})$, from every raw material's specific gravity. A raw with no value counts as 1.0. <i>On a sheet: Specific Gravity: 1.052</i> You maintain: Specific gravity on every raw material (Raw Materials > Edit). If missing: Each missing raw counts as 1.0, and the preview names it in a warning. 'Not determined' never prints in practice. VOC Calculator::getMixtureSG (VOC Calculator.php:248-279), applyDefaults SG NULL -> 1.0 (:613-622); round(,3) (SDSGenerator.php:2768); dataGapWarnings (FormulaCalcService.php:658)

Solubility:	FORMULA / RAW MATERIALS Per-product edit > band from the soluble fraction (Soluble 1, Partially 0.5, Negligible 0.03, Insoluble 0; raws without a value skipped): ≥90 / ≥5 / ≥1 / below > 'Not determined'. <i>On a sheet: Solubility: Negligible solubility in water</i> You maintain: Solubility dropdown on raw materials (Raw Materials > Edit). Optional edit in SDS > Edit Text. If missing: 'Not determined' when no raw material in the formula has a solubility value. <code>text_overrides(9, 'solubility') (SDSGenerator.php:2683-2689) > section9.solubility_<key>; SOLUBILITY_WEIGHTS (FormulaCalcService.php:30-35), F :557-588, bands :636-648</code>
Partition Coefficient: n-Octanol/Water:	PRODUCT OVERRIDE Per-product value typed in the editor > 'Not determined'. No raw-material or formula data feeds it. <i>On a sheet: Partition Coefficient: n-Octanol/Water: Not determined</i> You maintain: Only when a tested value exists: SDS > Edit Text, Section 9, per language. If missing: 'Not determined'. The label is too wide for the column, so the value prints on the next line. <code>SECTION9_APPENDIX_D_FIELDS (SDSGenerator.php:2548-2552); text_overrides(9, 'partition_coefficient', lang) > labels.not_determined (:2740-2750)</code>
Auto-Ignition Temperature:	PRODUCT OVERRIDE Per-product value typed in the editor > 'Not determined'. No raw-material or formula data feeds it. <i>On a sheet: Auto-Ignition Temperature: Not determined</i> You maintain: Only when a tested value exists: SDS > Edit Text, Section 9, per language. If missing: 'Not determined' <code>SECTION9_APPENDIX_D_FIELDS (SDSGenerator.php:2548-2552); text_overrides(9, 'auto_ignition_temp', lang) > labels.not_determined (:2740-2750)</code>
Decomposition Temperature:	PRODUCT OVERRIDE Per-product value typed in the editor > 'Not determined'. No raw-material or formula data feeds it. <i>On a sheet: Decomposition Temperature: Not determined</i> You maintain: Only when a tested value exists: SDS > Edit Text, Section 9, per language. If missing: 'Not determined' <code>SECTION9_APPENDIX_D_FIELDS (SDSGenerator.php:2548-2552); text_overrides(9, 'decomposition_temp', lang) > labels.not_determined (:2740-2750)</code>
Viscosity:	PRODUCT OVERRIDE Per-product value typed in the editor > 'Not determined'. No raw-material or formula data feeds it. <i>On a sheet: Viscosity: Not determined</i> You maintain: Only when a tested value exists: SDS > Edit Text, Section 9, per language. If missing: 'Not determined' <code>SECTION9_APPENDIX_D_FIELDS (SDSGenerator.php:2548-2552); text_overrides(9, 'viscosity', lang) > labels.not_determined (:2740-2750)</code>
VOC (lb/gal) (EPA Method 24):	FORMULA / RAW MATERIALS VOC weight fraction (normalised to the formula total) × mixture specific gravity × 8.345. A raw's VOC % is 0.99 if flagged '<1%', otherwise its VOC %, and 0 when blank. <i>On a sheet: VOC (lb/gal) (EPA Method 24): 2.15</i> You maintain: VOC % (or the <1% flag) and specific gravity on every raw material (Raw Materials > Edit). If missing: Missing VOC counts as 0 and missing specific gravity as 1.0, and the preview names each such raw. A number always prints (e.g. '0'). <code>getVOClbPerGal (VOCCalculator.php:289-306) = getTotalVOCWeightPercent Σ(pct·effVOC)/Σpct (:181-200) /100 · getMixtureSG · 8.345; effVOC :626-671; round(,2) SDSGenerator.php:2774</code>
VOC (wt%):	FORMULA / RAW MATERIALS '<1' when every raw material in the formula is flagged '<1% VOC'. Otherwise the weighted average of raw VOC %, normalised to the formula total (0.99 for flagged raws, 0 when blank). <i>On a sheet: VOC (wt%): 12.35 or VOC (wt%): <1</i> You maintain: VOC % or the <1% flag on every raw material (Raw Materials > Edit). If missing: Missing VOC counts as 0, and the preview names each such raw. A formula with no valid lines prints '<1'. <code>formula_props.all_voc_less_than_one ? '<1' : round((total_voc_wt_pct,2) (SDSGenerator.php:2675-2678); getTotalVOCWeightPercent Σ(pct·effVOC)/Σpct (VOCCalculator.php:181-200)</code>
Solids (wt%):	FORMULA / RAW MATERIALS Weighted average of each raw's solids %, normalised to the formula total. When a raw has no solids %, it is calculated as 100 – VOC – exempt solvent – water. <i>On a sheet: Solids (wt%): 64.2</i> You maintain: Solids % on every raw material, or accurate VOC/exempt/water % and constituents (Raw Materials > Edit). Keep the exempt VOC CAS list current. If missing: Back-calculated with no preview warning; only the formula Calculate page lists it as an assumption. A number always prints. <code>getSolidsWeightPercent Σ(pct·effSolids)/Σpct (VOCCalculator.php:390-404); effSolids :709-724, water 7732-18-5 :684-707; exempt list (FormulaCalcService.php:452-469)</code>

'Not determined' value:	FIXED TEXT (TRANSLATED) Fixed no-data wording from the language file. It prints for Odor, Boiling Point, Flash Point, Solubility and the Appendix D properties that have no value. <i>On a sheet: Not determined</i> You maintain: Nothing If missing: English wording, then the key text. labels.not_determined via TranslationService (SDSGenerator.php:2653); en.php:734, es/fr.php:704, de.php:701
'Not applicable' value:	FIXED TEXT (TRANSLATED) Fixed wording from the language file, printed for Flammability (Solid, Gas) on a liquid product. <i>On a sheet: Not applicable (ES: No aplica; FR: Non applicable; DE: Nicht zutreffend)</i> You maintain: Nothing If missing: English wording, then the key text. labels.not_applicable (SDSGenerator.php:2746); en.php:736, es/fr.php:706, de.php:703
Section 9 preview warnings (not printed):	PUBLISH RECORD The preview Warnings box lists Section 9 data gaps and dropped edits: raws missing SG or VOC, the 'Liquid' default, edits that are not temperatures or contradict the classification, and flash points without a boiling point or that are missing from the average. <i>On a sheet: Raw material RM123: no specific gravity entered; ... count it as 1.0.</i> You maintain: Act on each warning on the screen it names (Raw Materials > Edit, Finished Goods > Edit, or SDS > Edit Text). If missing: Nothing prints on the sheet. None of these warnings blocks publishing. FormulaCalcService::dataGapWarnings (:658); PHYSICAL_STATE_DEFAULT_WARNING (SDSGenerator.php:2532); temperatureOverrideWarnings :1817; vetFlashPointEdit :4076; flammabilityWarnings :4152

SECTION 10: STABILITY AND REACTIVITY

Section 10 is built mostly from the product's final hazard classification (H-codes) and the nitrogen, sulfur and halogen flags on its constituent CAS numbers. Flammability now has one source, the Flammable Liquids category worked out from the product flash point (wt%-weighted over the raws that have one) and the initial boiling point, so Section 10 agrees with Sections 2, 5, 7, 9, 13 and 14. The product family adds the UV item, the translation files supply the wording, and a per-product, per-language override can replace any of the five lines.

Reactivity / Chemical Stability / Conditions to Avoid / Incompatible Materials / Hazardous Decomposition Products (labels):	FIXED TEXT (TRANSLATED) Fixed label text in the sheet's language. The decomposition label is wider than the 50 mm PDF label column, so its value prints on the next line. <i>On a sheet: Hazardous Decomposition Products:</i> You maintain: Nothing If missing: A label key missing from every translation file prints the raw key, e.g. "labels.decomposition_products."; an old snapshot without the label and no translation prints the field key in title case, e.g. "Decomposition". All five values are always generated and blank override rows are ignored, so no Section 10 line is left out. meta.labels[key] = t->get('labels.<key>') (getLabels :4767) > SDSDocumentStrings::translate (old snapshots) > ucwords(field key); PDFService FIELD_LABEL_MAP :51-56; preview:620-624
Reactivity:	TEXT CHOSEN BY HAZARD Comes from the product's hazard classification. Each reactive class present adds its own sentence: explosive, self-reactive substance or organic peroxide, pyrophoric, self-heating, water-reactive, chemically unstable gas, oxidizer. Precedence: per-product override > class sentences > standard sentence. <i>On a sheet: Oxidizer: may cause or intensify fire in contact with combustible materials.</i> You maintain: Hazard data that produces H200-H205, H230-H232, H240-H242, H250-H252, H260/H261 or H270-H272. CAS hazard data and competent-person determinations (/determinations) count only at 1% or more of the formula. Manual hazards on no-CAS raws and trade-secret H-codes (Raw materials screen) count at any level, as does the finished-good hazard override (Finished goods screen). If missing: "No dangerous reaction known under conditions of normal use." override(10,'reactivity') > section10.reactivity_<class> joined ' ' (H200-205/230-232/240-242/250-252/260-261/270-272) > section10.reactivity; SDSGenerator.php:2799-2851
Chemical Stability:	TEXT CHOSEN BY HAZARD The same reactive classes as Reactivity except oxidizers, listed as the conditions under which the product is unstable. Precedence: per-product override > 'Unstable under ...' sentence > standard sentence. <i>On a sheet: Unstable under the following conditions: contact with water or moisture (releases ...).</i> You maintain: Same hazard inputs as Reactivity. Oxidizer codes H270-H272 do not change this line. If missing: "Stable under recommended storage conditions." override(10,'stability') > section10.stability_unstable(:conditions = stability_cond_<class> joined ' '; ', no oxidizer) > section10.stability; SDSGenerator.php:2841-2851

Conditions to Avoid:	TEXT CHOSEN BY HAZARD A list of items picked by the hazard classification. Heat is always included. Storage temperature, ignition sources, air, water and shock are added for the classes that need them, oxidizers get combustibles instead of oxidizing agents, and UV families get UV light. Precedence: per-product override > generated list. <i>On a sheet: Excessive heat, sparks, open flames and other ignition sources, water and moisture, ...</i> You maintain: Same hazard inputs as Reactivity, plus the flash-point inputs behind the ignition-sources item, the flammable gas, aerosol and solid codes (H220-H223, H228) in hazard data, and the family UV/LED flag (Product families screen) If missing: "Excessive heat, contact with strong oxidizing agents." (never empty) <code>override(10,'conditions_avoid') > section10.cond_<key> in COND_ORDER (SDSGenerator.php:2099-2101), items chosen at :2854-2893 from extractHCodes (:1057)</code>
Conditions to Avoid: sparks, open flames and other ignition sources:	HAZARD ENGINE Added for a flammable liquid (Cat 1-4 from the product flash point, which is the wt%-weighted average over raws that have one, plus the initial boiling point), a flammable gas, aerosol or solid, or a pyrophoric, self-reactive, oxidizing, explosive or water-reactive product. Precedence for the liquid category: finished-good hazard override > flash-point category; ingredient H224-H227 are ignored. <i>On a sheet: Excessive heat, sparks, open flames and other ignition sources, contact with strong ...</i> You maintain: Flash point (with its '>' flag) and physical state on every raw material (Raw materials screen). Physical state on the finished good (Finished goods screen); when it is blank, the state of the raw with the largest share is used. A Flammable Liquids class or H224-H227 in the finished-good hazard override replaces the derived category. If missing: Item omitted when no raw has a flash point (not flammable), the weighted flash point is above 93 °C, or the product is Solid, Powder or Paste, unless another ignition class applies. A Section 9 flash point override does not change it. <code>cond_ignition on H220-H228/H200-205/H240-242/H250/H260-261/H270-272 (:2862); H224-H227 only via HazardEngine::applyFlammableLiquidsFromFlashPoint or FG override</code>
Conditions to Avoid: UV light item:	PRODUCT FAMILY Added at the end of Conditions to Avoid when the product's assigned family is active and flagged UV/LED. It does not depend on the UV acrylate setting or on acrylate detection. <i>On a sheet: ..., exposure to UV light and direct sunlight (risk of premature polymerization; ...).</i> You maintain: The UV/LED flag and Active status of each family (Product families screen), plus family rules or manual picks so that finished goods and resale raws land in the right family. Recompute families after imports or rule changes, because the stored family_id is used; content resolution pools the share of all UV families. Use 'Reset legacy manual picks to Auto' (Product families screen) for picks carried over from the old family name. If missing: Item omitted when there is no family, the family is inactive, or it is not flagged UV. A conditions_avoid override also removes it. <code>fg.family_is_uv from attachFamily/familyFields (SDSGenerator.php:1165-1197; inactive family -> null); UVAcrylateRulePack::familyIsUv :94-97; section10.cond_uv</code>
Incompatible Materials:	TEXT CHOSEN BY HAZARD A list picked by the hazard classification: acids and bases always, oxidizing agents unless the product is an oxidizer, halogens for flammables (including flash-point Flammable Liquids Cat 1-4), and extra items for the self-reactive, pyrophoric, water-reactive, oxidizer and H290 classes (none for explosive, self-heating or unstable gas). Precedence: per-product override > generated list, and the same value is repeated in Section 7 Storage. <i>On a sheet: Strong oxidizing agents, strong acids, strong bases, halogens.</i> You maintain: Same hazard inputs as Reactivity, plus the flash-point inputs (a flammable product adds halogens) and corrosive-to-metals (H290) in hazard data. An override here also changes Section 7 Storage. If missing: "Strong oxidizing agents, strong acids, strong bases." (never empty) <code>resolveIncompatibleMaterials: override(10,'incompatible') > section10.incompat_<key> in INCOMPAT_ORDER; SDSGenerator.php:2144-2217; Section 7 inline copy :2224-2232</code>
Hazardous Decomposition Products:	FORMULA / RAW MATERIALS Comes from the nitrogen, sulfur and halogen flags on each ingredient's CAS number, at any concentration. One element names that gas and two or more list them all. Precedence: per-product override > element sentence > standard sentence. <i>On a sheet: Carbon monoxide, carbon dioxide, nitrogen oxides, and other toxic gases may be ...</i> You maintain: Element flags for every constituent CAS (/determinations, CAS Descriptions tab, 'Flags'), and the molecular formula in the CAS registry, which the seeder uses first. A CAS first saved on a raw material gets flags guessed from its name. Re-run the element-flag seeder (/determinations/element-flags or scripts/seed-cas-element-flags.php) after CAS rows arrive through CAS Descriptions or TSCA saves, seed data or direct inserts, which leave the flags at 0. If missing: "Carbon monoxide, carbon dioxide, and other toxic gases may be released upon thermal decomposition." This prints when no constituent is flagged, the CAS is not in the registry or its flags are 0, the constituent is a no-CAS trade secret or sits in a hazardous no-CAS raw, or the CAS is blank (that constituent is dropped). A trade secret with a CAS keeps its flags; only the name is masked. <code>override(10,'decomposition') > decomposition_<el> / decomposition_multi / decomposition; cas_master has_* via Formula::getExpandedComposition; SDSGenerator.php:2899-2915</code>

Decomposition base wording (carbon monoxide, carbon dioxide, other toxic gases):

FIXED TEXT Fixed wording inside every decomposition sentence. There is no carbon flag, so it prints even for wholly inorganic products unless overridden.
On a sheet: Carbon monoxide, carbon dioxide, ...
You maintain: Nothing
If missing: Always printed unless overridden
Fixed text inside section10.decomposition, decomposition_el, decomposition_multi (en.php:241-244, :272; ES/FR/DE word order differs)

SECTION 11: TOXICOLOGICAL INFORMATION

Section 11 opens with the likely routes of exposure and the same symptoms line as Section 4. Three acute-toxicity route lines, the chronic-effect fragments and the carcinogen and component block follow. All of it comes from the hazard classification (federal data, CPDs, trade-secret declarations, ATE summation, Finished good hazard override), the carcinogen registry and the exposure-limit datasets. Trade-secret identities are masked. Per product, the Acute Toxicity, Chronic Effects, Carcinogenicity and UV Acrylate Information texts can each be overridden, and the symptoms line follows the Section 4 override.

Likely Routes of Exposure:

FIXED TEXT (TRANSLATED) Fixed label from the translation files.
On a sheet: Likely Routes of Exposure:
You maintain: Nothing
If missing: Sheet language > English > the literal key 'labels.routes_of_exposure'. On snapshots made before this line existed, the whole line is skipped.
labels.routes_of_exposure (en.php:680; getLabels :4767); PDFService::renderSection11 :924 labelValue (skips a blank value)

Likely routes of exposure (text):

TEXT CHOSEN BY HAZARD Skin Contact and Eye Contact always print. Inhalation is added for H330-H332 or H334-H336, or when any ingredient keeps an exposure limit after the physical-state filters; Ingestion for H300-H302, H304 or H305 (not H303). Route names reuse the Section 4 labels; there is no override.
On a sheet: Inhalation, Skin Contact, Eye Contact, Ingestion.
You maintain: Nothing directly. It follows the hazard data on ingredient CAS and the exposure-limit data.
If missing: Never blank. At minimum it prints 'Skin Contact, Eye Contact.'
SDSGenerator::deriveRoutesOfExposure :3221-3238 (extractHCodes, hazard.exposure_limits); labels.inhalation/skin_contact/eye_contact/ingestion (en.php:604-607)

Most Important Symptoms/Effects, Acute and Delayed:

FIXED TEXT (TRANSLATED) Fixed label shared with Section 4. It is wider than the label column, so it prints on its own line above the text.
On a sheet: Most Important Symptoms/Effects, Acute and Delayed:
You maintain: Nothing
If missing: Sheet language > English > the literal key 'labels.symptoms_effects'. On older snapshots the line is skipped.
labels.symptoms_effects (en.php:608); PDFService::renderSection11 :925; labelValue :1358-1377 (long label goes on its own line)

Symptoms/effects (text):

TEXT CHOSEN BY HAZARD The product's health H-statements (H3xx) are split into 'Acute:' and 'Delayed:' groups (sensitisation, CMR, lactation and STOT RE count as delayed). The text is identical to Section 4. Section 4 Symptoms override > derived text > the no-symptoms sentence.
On a sheet: Acute: Causes skin irritation. Causes serious eye irritation. Delayed: May cause an...
You maintain: Health-hazard data on ingredient CAS (federal, CPD or trade-secret declaration). Optionally, the Section 4 Symptoms override in the SDS editor.
If missing: 'No significant symptoms or effects are known or expected under normal conditions of use.'
overrides[4]['symptoms'] > deriveSymptoms :1674-1714 (DELAYED_EFFECT_H_PREFIXES :98); section4.symptoms_none/_acute_prefix/_delayed_prefix (en.php:53-55)

Acute Toxicity:

FIXED TEXT (TRANSLATED) Fixed label from the translation files.
On a sheet: Acute Toxicity:
You maintain: Nothing
If missing: Sheet language > English > the literal key 'labels.acute_toxicity'. The value is never blank because three route lines always print.
labels.acute_toxicity (en.php:676; getLabels :4767); PDFService::renderSection11 :926 labelValue

Acute toxicity (oral):

HAZARD ENGINE Most severe oral Category 1-4 in the classification: ingredient cut-off or ATEmix, whichever is more severe, or an H300-H302 that prints in Section 2. Per-product Acute Toxicity override > classification > 'Not classified' sentence.
On a sheet: Acute toxicity (oral): Category 4 — Harmful if swallowed (H302).
You maintain: Oral acute-toxicity data on each ingredient CAS (federal data, or a CPD on CAS determinations where there is none); constituent % on Raw materials; optional hazard override on the Finished good.
If missing: 'Acute toxicity (oral): Not classified based on available data.' Category 5 (H303) prints the same sentence.
text_overrides(11,acute_toxicity) > buildAcuteToxicity :3037-3213 (canonical acute_toxicity_oral; H-code fallback #40; GHSStatements::hText); H300 alone prints no category

Acute toxicity (dermal):	HAZARD ENGINE Same as the oral line, for the skin route (H310-H312). Per-product override > classification > 'Not classified' sentence. <i>On a sheet: Acute toxicity (dermal): Category 3 — Toxic in contact with skin (H311).</i> You maintain: Dermal acute-toxicity data on ingredient CAS (federal or CPD); constituent %. If missing: 'Acute toxicity (dermal): Not classified based on available data.' buildAcuteToxicity canonical acute_toxicity_dermal; defaults H310/H310/H311/H312; section11.acute_route_dermal / acute_route_line(_no_category) en.php:297-300
Acute toxicity (inhalation):	HAZARD ENGINE Same as the oral line, for inhalation (H330-H332). Per-product override > classification > 'Not classified' sentence. <i>On a sheet: Acute toxicity (inhalation): Category 4 — Harmful if inhaled (H332).</i> You maintain: Inhalation acute-toxicity data on ingredient CAS (federal or CPD); constituent %. If missing: 'Acute toxicity (inhalation): Not classified based on available data.' buildAcuteToxicity canonical acute_toxicity_inhalation; defaults H330/H330/H331/H332; section11.acute_route_inhalation en.php:298
ATEmix value (after a classified route):	HAZARD ENGINE The calculated mixture acute toxicity estimate, appended when the ATEmix falls in the printed category, including a tie with the ingredient cut-off; not for a bare H-code route. Explicit LD50/LC50 on the ingredient > GHS category default estimate. <i>On a sheet: ATEmix = 1250 mg/kg bw. / ATEmix = 0.8 mg/L (4 h, dusts/mists).</i> You maintain: LD50/LC50 (ATE) values on ingredient CAS data or in the CPD; constituent %; trade-secret raw material line % (counted at its real share). Product physical state Solid/Powder, or an ingredient with only a dust/mist LC50, switches inhalation to the dust unit. If missing: No number. The line ends with 'Classified based on ingredient concentration.' instead (next row). hazard_classes ate_mix > ate_results at same cat (outcome classified/already_classified); HazardEngine::applyATECalculation :2303; CATEGORY_DEFAULT_ATES :207; inhalationAteRoute :1527; formatAte :3243
Classified based on ingredient concentration.:	TEXT CHOSEN BY HAZARD Added to a classified route when no ATEmix at its category exists: ingredient cut-off with a lower, dominated or no ATEmix, or a trade-secret declaration. Not added when the category came only from the Finished good hazard override or a bare H-code. <i>On a sheet: Classified based on ingredient concentration.</i> You maintain: Nothing If missing: Not printed on unclassified routes or when an ATEmix prints. buildAcuteToxicity :3200-3206 (override_only / source_h_code skip); section11.acute_basis_concentration (en.php:308)
Chronic Effects:	FIXED TEXT (TRANSLATED) Fixed label from the translation files. <i>On a sheet: Chronic Effects:</i> You maintain: Nothing If missing: Sheet language > English > the key name. labels.chronic_effects (en.php:677); PDFService::renderSection11 :927
Chronic effects: sensitisation:	TEXT CHOSEN BY HAZARD Standard sentence printed when the product carries H334 or H317. A per-product Chronic Effects override replaces the whole paragraph. <i>On a sheet: Skin sensitizer: may cause an allergic skin reaction (H317). Repeated or prolonged...</i> You maintain: Sensitisation data on ingredient CAS (federal, CPD or trade-secret declaration), or a Finished good hazard override. If missing: Fragment omitted. text_overrides(11,chronic_effects) > buildChronicEffects :2940-3017 (H334 chronic_resp_sens, H317 chronic_skin_sens; en.php:335-336)
Chronic effects: germ cell mutagenicity:	TEXT CHOSEN BY HAZARD Standard sentence when the product carries H340, otherwise H341. <i>On a sheet: Germ cell mutagenicity: suspected of causing genetic defects (H341).</i> You maintain: Mutagenicity data on ingredient CAS. If missing: Fragment omitted. buildChronicEffects :2967-2972; section11.chronic_muta_1 / chronic_muta_2 (en.php:337-338)
Chronic effects: carcinogenicity:	TEXT CHOSEN BY HAZARD H350 sentence > H351 sentence > 'listed' sentence when a reportable registry listing survives. 'See Carcinogenicity below' is added only when that paragraph lists components (no override), and H350/H351 come from hazard data, the registry (CAS with no other hazard class) or the all-powder inhalation-only rule. <i>On a sheet: Carcinogenicity: suspected of causing cancer (H351). See Carcinogenicity below for...</i> You maintain: Carcinogen registry (seed CSV only, no admin screen); carcinogenicity data on CAS or CPD; physical state on Raw materials; Settings: Airborne/Unbound Particles Override (sds.inhalation_only_cas). If missing: Fragment omitted. With a 'replace' Finished good hazard override, the registry and the powder rule add no H350/H351. buildChronicEffects :2973-2985 (chronic_carc_1/2/listed + _no_ref, #66); applyPostClassifyCarcinogenSteps :5014; applyCarbonBlackLogic :5037; applyCarcinogenFindings :5400

Chronic effects: reproductive toxicity / lactation:	TEXT CHOSEN BY HAZARD Standard sentence for any H360 code, otherwise any H361 code, plus a lactation sentence for H362. <i>On a sheet: Reproductive toxicity: suspected of damaging fertility or the unborn child (H361).</i> You maintain: Reproductive-toxicity data on ingredient CAS. If missing: Fragment omitted. <code>buildChronicEffects :2987-2995 (prefix match H360x/H361x); chronic_repr_1/2, chronic_lactation (en.php:346-348)</code>
Chronic effects: STOT repeated exposure:	TEXT CHOSEN BY HAZARD Standard sentence for H372, otherwise H373. Target organs are not named here; the text points to Section 2. <i>On a sheet: Specific target organ toxicity, repeated exposure (Category 2): may cause damage... (H373).</i> You maintain: STOT RE data on ingredient CAS. If missing: Fragment omitted. <code>buildChronicEffects :2997-3002; chronic_stot_re_1/2 (en.php:349-350)</code>
Chronic effects: repeated skin / eye contact:	TEXT CHOSEN BY HAZARD Skin sentence for H314/H315; eye sentence for H318/H319. <i>On a sheet: Repeated or prolonged skin contact may cause irritation, dryness, or dermatitis.</i> You maintain: Skin/eye irritation data on ingredient CAS. If missing: Fragment omitted. <code>buildChronicEffects :3004-3010; chronic_repeated_skin / chronic_repeated_eye (en.php:351-352)</code>
Chronic effects: none known:	FIXED TEXT (TRANSLATED) Fixed sentence printed only when none of the chronic sentences above applies and there is no override. <i>On a sheet: None known. Based on available data for the mixture and its components, the...</i> You maintain: Nothing If missing: This is itself the fallback. <code>buildChronicEffects :3012-3014; section11.chronic_none (en.php:353)</code>
Carcinogenicity:	FIXED TEXT (TRANSLATED) Fixed label, always printed, with the text on the lines below. <i>On a sheet: Carcinogenicity:</i> You maintain: Nothing If missing: Sheet language > English > the key name. <code>labels.carcinogenicity (en.php:678); PDFService::renderSection11 :929-936</code>
Carcinogenicity: no listed components:	FIXED TEXT (TRANSLATED) Fixed sentence printed when no reportable registry listing (IARC 1/2A/2B, NTP Known/RAHC, OSHA) at 0.1% or more survives filtering. Per-product Carcinogenicity override > this sentence. <i>On a sheet: No components present at or above 0.1% are listed as carcinogens by IARC, NTP, or OSHA.</i> You maintain: Carcinogen registry kept current (storage/data/seed/carcinogens.csv + load-seed-data.php). If missing: Default text. <code>text_overrides(11,carcinogenicity) > section11.carcinogenicity (:threshold = CarcinogenService::LISTING_THRESHOLD_PCT 0.1); SDSGenerator.php:3441-3468</code>
Carcinogenicity: listed-components intro:	REGULATORY LIST Printed when any ingredient at 0.1% or more has a reportable registry listing after the physical-state filters. Override > registry result. <i>On a sheet: The following component(s), present at or above 0.1%, are listed as carcinogens or...</i> You maintain: Carcinogen registry; ingredient CAS and %; physical state on Raw materials; Settings: sds.inhalation_only_cas. If missing: The negative sentence prints instead. <code>CarcinogenService::buildSummaryText :285-303; analyse :146 (isReportableListing); applySolidPowderLiquidFiltering :5181; section11.carcinogenicity_listed_intro</code>
Carcinogenicity: listed component line:	REGULATORY LIST One line per listed ingredient: name, CAS, Section 3 band and each reportable agency classification in the sheet language. Name: Prop 65 name > CAS master name > constituent name > registry name; trade secrets are masked (see the trade-secret row). <i>On a sheet: Ethylbenzene (CAS 100-41-4, 1 - 5%) — IARC: Group 2B</i> You maintain: Carcinogen registry (one correct row per CAS+agency in the seed CSV); constituent name, CAS and exact or min/max % on Raw materials. If missing: IARC Group 3 and unrecognised classifications are never reported, so a CAS that has only those gets no line. <code>CarcinogenService::buildComponentTexts :233-280; section11.carcinogenicity_listed_line/_listing/carcinogen_class_*; formatConcentration(compByCas) :4712</code>

Carcinogen inhalable-dust note:	FORMULA / RAW MATERIALS Added after the listing of an inhalation-only substance (Settings list; default carbon black, TiO₂), in the Carcinogenicity line and the component block, when any raw material is not Solid/Powder. The listing adds no H350/H351, exposure limit or Prop 65 entry. <i>On a sheet:</i> (The listing applies to inhalable dust only; in this product the substance is bound... You maintain: Settings: Airborne/Unbound Particles Override (sds.inhalation_only_cas); physical state on Raw materials. If missing: Not printed for all-powder/solid products, which get Carc. 2 (H351) unless H350/H351 is already present or a 'replace' override applies. A blank raw-material state counts as not powder. removeInhalationOnlyFromResults :5239-5290 (inhalable_dust_only); buildComponentTexts; section11.carcinogenicity_inhalable_dust_note (en.php:315); PDFService.php:968-971
Component Toxicological Data:	FIXED TEXT (TRANSLATED) Fixed sub-heading, printed only when at least one component row qualifies. <i>On a sheet:</i> Component Toxicological Data: You maintain: Nothing If missing: Not printed when no component at 0.1% or more has an exposure limit or a reportable carcinogen listing. labels.component_tox_data (en.php:679); PDFService::renderSection11 :939-944
Component row (name, CAS, band):	FORMULA / RAW MATERIALS Each ingredient at 0.1% or more that has an exposure limit or a reportable carcinogen listing, banded as in Section 3. Name: Prop 65 name > CAS master name > constituent name. <i>On a sheet:</i> Toluene (CAS 108-88-3) — 10 - 30% You maintain: Constituent CAS, name and pct_exact / pct_min / pct_max on Raw materials; formula lines on Finished goods. If missing: Row omitted. Solid/powder-only ingredients in a liquid product lose their limits and listings, so they have no row; Settings inhalation-only substances keep their listing and row. section11() :3381-3436 over calcResult.composition; formatConcentration :4712 (summed min/max bounds); literals 'CAS ', ' - ' PDFService.php:950-956
Trade-secret component (masked name and CAS):	FORMULA / RAW MATERIALS A constituent flagged trade secret, or a no-CAS trade-secret raw material, prints its trade-secret description and 'TRADE SECRET' in place of its name and CAS. This applies in the component block and the Carcinogenicity lines. Trade-secret description > translated 'Trade Secret'; the band still prints and English registry descriptions are dropped. <i>On a sheet:</i> Acrylate monomer (CAS TRADE SECRET) — 5 - 10% You maintain: Trade-secret flag and trade-secret description on each constituent on Raw materials, or the no-CAS trade-secret raw material and its hazard declaration. If missing: A blank description prints 'Trade Secret (CAS TRADE SECRET)'. A CAS flagged on any one raw material is masked on the whole sheet. maskTradeSecretRow :2627-2646; tradeSecretName :2607; labels.trade_secret / trade_secret_cas (en.php:739-740); section11() :3418-3428, :3454
Component carcinogen listing line:	REGULATORY LIST Each reportable registry listing for the CAS: agency and translated classification. English sheets add the registry description; other languages add a translated gloss. <i>On a sheet:</i> IARC: Group 1 — Carcinogenic to humans. You maintain: Carcinogen registry rows (agency, classification, description). If missing: The ' — description' part is omitted when the description is empty or the component is a masked trade secret. Group 3 and unknown classes are not listed. CarcinogenService::localiseListings :113-135; carcinogen_list.agency/classification/description; PDFService.php:958-967
Component exposure limit line:	HAZARD ENGINE All exposure limits found for the CAS, combined and printed as stored (untranslated): federal datasets plus CPD limits (only for a CAS with no federal hazard data) plus trade-secret declarations. <i>On a sheet:</i> PEL-TWA: 435 mg/m³ (29 CFR 1910.1000 Table Z-1) You maintain: Federal OEL dataset refresh (OSHA PEL / ACGIH TLV seed, NIOSH connector); CPD exposure limits for CAS without federal data; physical state on Raw materials. If missing: No line. '(notes)' is omitted when notes are blank. An inhalation-only CAS in a non-powder product and a solid-only CAS in a liquid product print no limits. hazardResult.exposure_limits by CAS: exposure_limits+hazard_source_records.is_current CPD exposure_limits manual JSON; dedupeExposureLimits; PDFService.php:974-985
UV Acrylate Information:	FIXED TEXT (TRANSLATED) Fixed label, printed only together with the UV note. <i>On a sheet:</i> UV Acrylate Information: You maintain: Nothing If missing: Not printed without a note. labels.uv_acrylate_note (en.php:741); PDFService::renderSection11 :988-993

UV acrylate note text:

PRODUCT FAMILY Printed on every product whose resolved family is an active UV/LED family while the UV acrylate rule pack is on. Uses the sensitizer wording when the product has H317, otherwise the unclassified wording. Per-product UV Acrylate Information override > that text.
On a sheet: Contains acrylate monomers that are known skin sensitizers. Repeated exposure may...
You maintain: Family on the Finished good (manual pick, rule match or content share, with UV families pooled) and the family's UV flag and active status on Product families; Settings: UV acrylate rule pack.
If missing: Not printed, override included, when the pack is off or no active UV family resolves. A missing settings row counts as enabled.
UVAcrylateRulePack::isApplicable :104 (isEnabled + family_is_uv), section11Note :191;
SDSGenerator.php:216, :262-270; familyFields :1180

Preview warning: trade secret on one raw material, disclosed on another:

FORMULA / RAW MATERIALS Shown only in the preview Warnings box, never on the SDS. It appears when one raw material flags a CAS as trade secret and another discloses it; Section 11 then masks that CAS.
On a sheet: Trade-secret warning: CAS 123-45-6 is marked as a trade secret on raw material(s) ...
You maintain: Trade-secret flags on Raw materials kept consistent across raw materials.
If missing: No warning.
SDSGenerator::tradeSecretDisclosureWarnings :5556-5582 (contributing_materials.is_trade_secret);
added to sds.warnings :320-322

SECTION 12: ECOLOGICAL INFORMATION

Section 12 is built from each constituent's aquatic classifications and M-factors, as summed by the hazard engine or set by a finished-good hazard override, and from the EPA PBT designation on the SARA 313 list. The component table lists only constituents that Section 3 lists, and vendor trade-secret identities are masked. Fixed translation sentences fill anything with no data, and a per-product text override can replace any of the four paragraphs.

Ecotoxicity:

TEXT CHOSEN BY HAZARD Per-product text override > the product's aquatic hazard statements (lead-in + H statements + release warning) > a not-classified sentence when aquatic data exists or a replace-mode override is set > the standard no-data sentence. The aquatic H codes come from the M-factor summation of the constituents' aquatic categories, or from a finished-good hazard override.
On a sheet: Classified as hazardous to the aquatic environment by the GHS summation method ... H410
You maintain: Aquatic categories and M-factors on each constituent CAS (federal hazard data, or a CAS determination when no federal data exists). Constituent percentages on raw materials and formula lines. The finished-good hazard override if used. Section 12 text overrides in the SDS editor.
If missing: "No data available on the mixture. Avoid release to the environment."
text_overrides[12]['ecotoxicity'] (blank ignored) > H4(0[0-2]|1[0-3]) in h_statements
(lead-in+codes+environmental_warning) > *not_classified* > section12.ecotoxicity;
SDSGenerator:3563-3609

Ecotoxicity: lead-in when classified:

TEXT CHOSEN BY HAZARD Fixed sentence chosen by where the printed codes came from. The manufacturer's-assessment version prints when a replace-mode override is set, or when a printed code came only from the override. The summation version ("listed below") prints when every code came from the summation and the table prints. Otherwise a neutral version that names no method prints.
On a sheet: Classified as hazardous to the aquatic environment (GHS Rev. 7, Chapter 4.1):
You maintain: Nothing beyond the aquatic data and the finished-good hazard override (mode + JSON).
If missing: Not printed when the product has no aquatic H code; a not-classified or no-data sentence is used instead.
hazard aquatic_basis (override_mode, override_codes, summation_codes) →
ecotoxicity_classified_manufacturer > ecotoxicity_classified > _no_table; SDSGenerator:3572-3591

Ecotoxicity: aquatic H statements:

HAZARD ENGINE The aquatic hazard statements the product resolved to, in code order and in standard GHS wording for the sheet's language. Only the most severe code per route prints (at most one acute and one chronic). Summation alone can produce only H400, H410, H411 and H412. H401, H402 and H413 come from a finished-good override, or rarely from a CAS determination or no-CAS hazard JSON that names an aquatic class with no category (trade-secret H-code lists cannot add them).
On a sheet: H410: Very toxic to aquatic life with long lasting effects.
You maintain: Constituent aquatic categories and M-factors. Finished-good hazard override (mode + JSON) where used.
If missing: Not printed when the product has no aquatic H code. All seven aquatic codes have English, Spanish, French and German wording, so a printed code always carries its full statement.
h_statements ^H4(0[0-2]|1[0-3])\$ ksort, one per route (HazardEngine::dropLessSevereAquaticCodes :2117); EN GHSStatements, non-EN ghs_{lang}.php > EN; SDSGenerator:3502-3511

Ecotoxicity: release warning:	TEXT CHOSEN BY HAZARD Fixed warning sentence added after the H statements, only when the product has an aquatic H code. <i>On a sheet: Avoid release to the environment. Prevent entry into waterways, sewers, and soil.</i> You maintain: Nothing If missing: Not printed on unclassified products. The not-classified and no-data sentences carry their own "Avoid release" clause. section12.environmental_warning; SDSGenerator.php:3592-3594; en.php:371
Ecotoxicity: not-classified sentence:	TEXT CHOSEN BY HAZARD Fixed sentence printed when the product has no aquatic H code. There are three versions. A replace-mode override gives the manufacturer's-assessment version. When the component table prints, the "Based on the component data below" version is used. When aquatic data exists only on constituents that Section 3 does not list, a version with no table reference is used. Typical causes are a total below the thresholds, or constituents with only Acute 2/3 or Chronic 4, which are never summed. <i>On a sheet: Based on the component data below, the mixture does not meet the GHS criteria ...</i> You maintain: Constituent aquatic classifications; finished-good hazard override mode. If missing: Falls back to the no-data sentence when no constituent has aquatic data. no aquatic code: replace override → ecotoxicity_not_classified_manufacturer; table → ecotoxicity_not_classified; else aquatic_components → _no_table; SDSGenerator:3595-3606
Ecotoxicity: no-data sentence:	FIXED TEXT (TRANSLATED) Fixed sentence from the translation file, used when no constituent has aquatic data, no replace-mode override is set and there is no text override. <i>On a sheet: No data available on the mixture. Avoid release to the environment.</i> You maintain: Nothing If missing: This is the fallback itself. section12.ecotoxicity; SDSGenerator.php:3607; en.php:361; es/fr/de.php:342
Component Aquatic Hazard Data:	FIXED TEXT (TRANSLATED) Fixed sub-heading, printed only when at least one constituent that Section 3 lists has an aquatic classification. It never prints under a replace-mode override. <i>On a sheet: Component Aquatic Hazard Data:</i> You maintain: Nothing If missing: The heading and table are omitted when there are no rows. labels.component_ecotox_data; printed when component_aquatic non-empty; PDFService.php:1035 (adds ':'), preview.php:431 (no colon)
Component table headers: Chemical Name CAS Number Conc% Aquatic Acute Aquatic Chronic:	FIXED TEXT (TRANSLATED) Fixed column labels from the translation file. Conc% is the same label used in Section 8. <i>On a sheet: Chemical Name CAS Number Conc% Aquatic Acute Aquatic Chronic</i> You maintain: Nothing If missing: Not printed when the table is empty. Long French and German headers wrap inside their cells. labels.chemical_name, cas_number, el_conc_pct, aquatic_acute, aquatic_chronic; PDFService.php:1015-1046 (68/25/17/32/33 mm; headers wrap in one shared-height row)
Component table: Chemical Name:	FORMULA / RAW MATERIALS Prop 65 list name > CAS master preferred name > the constituent name entered on the raw material. A trade-secret constituent shows its trade-secret description instead (see the trade-secret row). <i>On a sheet: Zinc oxide</i> You maintain: CAS master preferred names and raw-material constituent names (Raw materials screen). If missing: Blank cell when all name sources are empty. aquatic_components[].name = composition chemical_name: COALESCE(NULLIF(p65.chemical_name,''),NULLIF(cm.preferred_name,''),rmc.chemical_name) Formula.php:293
Component table: CAS Number:	FORMULA / RAW MATERIALS The constituent's CAS number from the raw material. Trade-secret constituents are masked. <i>On a sheet: 1314-13-2</i> You maintain: CAS numbers and trade-secret flags on raw-material constituents. If missing: A constituent with no CAS prints no row of its own. No-CAS trade-secret constituents with H codes and no-CAS hazardous raw materials share one "TRADE SECRET" row; a flagged constituent with a real CAS gets its own row showing "TRADE SECRET". component_aquatic[].cas_number = aquatic_components[].cas (rmc.cas_number or TRADE_SECRET sentinel) → labels.trade_secret_cas when masked; SDSGenerator:3520-3537
Component table: Conc%:	FORMULA / RAW MATERIALS The same prescribed concentration band that Section 3 prints for that constituent. The exact percentage is never printed. <i>On a sheet: 10 - 30%</i> You maintain: Constituent percentages (exact or min/max) on raw materials and formula line percentages. If missing: Always a band, because rows under 0.1% are not listed. Only a Substance sheet's identity constituent could print "<0.1%". formatConcentration(compByCas[cas]) on concentration_min/max → widest PRESCRIBED_RANGES band containing it; SDSGenerator:3530-3536, :4712-4762

Component table: Aquatic Acute / Aquatic Chronic:	HAZARD ENGINE Each constituent's aquatic category per route. The source order is current federal hazard data (an unreadable row's category can be inferred from its H codes) > the CAS determination, used only when no federal data exists > the trade-secret hazard list. Category 1 also shows its M-factor. <i>On a sheet: Category 1 (M = 1, default) Category 1 (M = 10)</i> You maintain: Aquatic acute/chronic categories and M-factors on each constituent CAS: the federal hazard data import, or the CAS determinations screen for any CAS with no federal data. If missing: An em dash (—) when there is no class on that route. Category 1 with no M-factor on file prints "(M = 1, default)" (translated). <code>aquatic_components[.]{acute,chronic}_category + _m_factor/_source (HazardEngine::resolveMFactor :1410) → formatAquaticCategory :3674; default → section12.m_factor_default</code>
Component table: trade-secret constituent:	FORMULA / RAW MATERIALS A vendor-claimed trade secret prints its trade-secret description in place of the name, "TRADE SECRET" in place of the CAS, and the Section 3 band of its total. The real identity never prints. Its aquatic categories count toward the summation at the raw material's real share of the product. <i>On a sheet: Biocide blend TRADE SECRET 1 - 5% — Category 1 (M = 1, default)</i> You maintain: Trade-secret flag, trade-secret description and trade-secret H codes on the constituent, or the manual hazard JSON on a no-CAS hazardous raw material (Raw materials screen). If missing: A blank description (or one that just says "Trade Secret") prints the translated "Trade Secret". A no-CAS trade secret whose only declared hazards are aquatic gets no row, because Section 3 does not list it, but it still counts in the summation. <code>cas TRADE_SECRET or composition is_trade_secret → tradeSecretName(trade_secret_description) :2607, labels.trade_secret_cas, Section 3 band; SDSGenerator:3525-3532</code>
Component table: which rows print:	HAZARD ENGINE One row for each constituent that has an aquatic class and that Section 3 lists. Section 3 lists constituents at 0.1% or more that are classified, including as contributors to a summation that fired, or that have an exposure limit, plus a Substance sheet's identity. Rows are sorted by concentration from highest. Under a replace-mode override the whole table is dropped. You maintain: Constituent aquatic data and percentages; current flags on hazard source records; the finished-good hazard override mode. If missing: Table omitted when there are no rows. Inhalation-only CAS (e.g. carbon black) bound in a non-powder product never appear. A no-CAS trade secret with only aquatic hazards never gets a row. <code>HazardEngine::buildAquaticComponentSummary :1468 (conc desc, then CAS) n section3ListedCas :3705; emptied when aquatic_basis.override_mode = 'replace' (:3560)</code>
Persistence and Degradability:	REGULATORY LIST Per-product override > a sentence naming every PBT-flagged constituent at any concentration above 0 > "No data available.". The flag comes from the SARA 313 list or the TRI lead/mercury compound categories. TRI PFAS, which are of special concern but not PBT, are not named here. <i>On a sheet: Contains component(s) identified as persistent, bioaccumulative and toxic (PBT) ...</i> You maintain: SARA 313 list kept current with its PBT flag (Regulatory lists import). CAS master molecular formula or names for category matching. Constituent CAS numbers. If missing: "No data available." <code>text_overrides[12]['persistence'] > section12.pbt_components (SARA313Service::evaluate is_pbt: sara313_list or regulatory_categories, conc>0) > section12.persistence; :3626-3646</code>
PBT component item:	REGULATORY LIST For each PBT constituent: the SARA list name, or the TRI category name such as "Lead compounds", with footnote markers stripped, falling back to the composition name, followed by the CAS number. A trade-secret constituent shows its description and "TRADE SECRET". <i>On a sheet: Lead compounds (CAS 1314-41-6)</i> You maintain: SARA 313 list chemical names and PBT flags; seeded TRI categories (migration 058). If missing: Not printed when there is no PBT constituent. <code>section12.pbt_component_item ':name (CAS :cas)'; name = sara_name (list row > TRI category name, **stripped) > composition name; TS masked; SDSGenerator:3628-3641</code>
Bioaccumulative Potential:	REGULATORY LIST Per-product override > "See Persistence and Degradability above." when the PBT sentence prints (the full PBT sentence if Persistence is overridden) > "No data available." <i>On a sheet: See Persistence and Degradability above.</i> You maintain: Same SARA 313 PBT flags as Persistence. If missing: "No data available." <code>No PBT: text_overrides[12]['bioaccumulation'] > section12.bioaccumulation. PBT: override > (persistence overridden ? pbtLine : section12.pbt_see_persistence); :3647-3652</code>

Mobility in Soil:	FIXED TEXT (TRANSLATED) Per-product override > fixed "No data available.". No data source exists for this line. <i>On a sheet: No data available.</i> You maintain: Nothing (an override only). If missing: "No data available." On old snapshots the PDF leaves the line out, while the HTML preview prints an empty label. text_overrides[12]['mobility'] > section12.mobility; SDSGenerator.php:3660; PDFService.php:1078; preview.php:456
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SECTION 13: DISPOSAL CONSIDERATIONS

Section 13 prints one fixed disposal sentence, which can be replaced for each product (finished goods and resale raw materials). It also prints an RCRA hazardous-waste line that is built when the sheet is generated. That line uses the sheet's final hazard codes, including the Flammable Liquids category derived from the product flash point. It also uses the product's physical state, and a match of each component's CAS number against the admin-maintained EPA RCRA waste-code list, extended to metal compounds through each CAS's RCRA metals flags.

Disposal Methods:	PRODUCT OVERRIDE A fixed sentence from the language file, unless someone has typed replacement text for this product and language. Per-product text > standard sentence. Resale raw-material sheets now keep their own text, stored against the raw material. <i>On a sheet: Dispose of in accordance with all applicable federal, state, and local regulations. Do not...</i> You maintain: Nothing for the standard text. Optional wording per language: SDS > Edit, Section 13 (finished goods), or SDS Creation Readiness Check > 'Edit SDS text' (resale) If missing: The standard sentence: "Dispose of in accordance with all applicable federal, state, and local regulations. Do not dump into sewers, drains, or waterways." Blank text, or text equal to it, stores no override TextOverrideService::effective(text_overrides 13/'methods', FG or resale RM, lang, sds_version_id NULL) > section13.methods (lang > EN > key); SDSGenerator.php:3929
RCRA Hazardous Waste Classification (lead-in sentence):	TEXT CHOSEN BY HAZARD Fixed wording that prints only when at least one as-sold item applies: D001, D002, D003, a D-code component, or a trade-secret toxicity-characteristic constituent. The items follow, joined by '; ' and ending with ' '; in the PDF the long label sits on its own line. <i>On a sheet: If discarded as sold, this product may be regulated as hazardous waste under US EPA RCRA (40 CFR 261):</i> You maintain: Nothing directly. It depends on the hazard, flash point, physical state and RCRA-list inputs below If missing: Replaced by one of the two 'No RCRA hazardous waste characteristic...' sentences section13.rcra_intro + ' '; '-joined items + '.' (SDSGenerator.php:3914-3916); label labels.rcra_classification (en.php:694); PDFService::labelValue :1358
Ignitable (D001) item:	HAZARD ENGINE Prints when any D001 reason applies: a Flammable Liquids Category 1-3 code (H224-H226) on the sheet, or an oxidizer, pyrophoric/self-heating, flammable gas/aerosol or flammable solid code. The reasons are joined with commas. <i>On a sheet: ignitable (D001) — flash point below 60 °C (140 °F)</i> You maintain: Flash points and physical state on raw materials, finished-good physical state, CAS determination hazard data, finished-good hazard override If missing: No D001 item. Without another D001 code, none prints for a Category 4 (H227) product, a Category 3 product at exactly 60 °C, or a product with no flash point data unless the hazard override sets H224-H226 section13.rcra_d001 :reasons; d001FlashReasonKey (SDSGenerator.php:4052) + H270-H272, H250-H252, H220-H223, H228 checks (:3806-3830)
D001 reason: flash point below 60 °C / minimum value below 60 °C:	FORMULA / RAW MATERIALS Prints only when the sheet carries H224-H226 and the liquid product's flash point is below 60 °C. That flash point is the wt%-weighted average over the raw materials that have one, through sub-formulas; a '>' on any of them prints the 'reported only as a minimum value' wording. <i>On a sheet: flash point below 60 °C (140 °F)</i> You maintain: Flash point (°C) and '>' flag on raw materials (Raw materials screen), with a high value for water and similar raws. Physical State on the finished good or its main raw material If missing: No reason when no raw material has a flash point (treated as not flammable), when the state is Solid/Powder/Paste, or at exactly 60 °C. A Section 9 flash-point edit is never used hazardResult['flammability'] (flammabilityFromProps < FormulaCalcService::weightedFlashPoint); fp<60 && !gt > rcra_reason_flash_point; gt > rcra_reason_flash_point_gt

D001 reason: classified as a flammable liquid (Category 1, 2 or 3):	PRODUCT OVERRIDE Prints when H224-H226 are on the sheet but no formula flash point below 60 °C on a liquid supports them, which in practice means the finished-good hazard override set the class. <i>On a sheet: classified as a flammable liquid (Category 1, 2 or 3)</i> You maintain: Finished-good hazard override (Finished goods screen). Enter the tested flash point as a Section 9 edit so Sections 5 and 9 agree If missing: Not printed d001FlashReasonKey: H224-H226 present, no liquid fp<60, engine flammability category not 1-3 > section13.rcra_reason_flammable_liquid (en.php:401)
D001 reasons: oxidizer / pyrophoric or self-heating material:	HAZARD ENGINE Prints when the final hazard codes contain H270-H272 (oxidizer) or H250-H252 (pyrophoric or self-heating). These codes also make Section 14 'Not determined', which blocks publishing. <i>On a sheet: oxidizer</i> You maintain: Hazard classification inputs for oxidizing or self-heating constituents (CAS determinations, vendor data) If missing: Not printed section13.rcra_reason_oxidizer / rcra_reason_ignitable_solid; SDSGenerator.php:3813-3818; TransportClassifier::UNSUPPORTED_CODES
D001 reasons: ignitable compressed gas / flammable solid:	HAZARD ENGINE Prints for H220-H223 (flammable gas or aerosol) or H228 (flammable solid). It is a safety net only: the company sells none, and these codes also make Section 14 'Not determined', which blocks publishing. <i>On a sheet: ignitable compressed gas (flammable gas or aerosol)</i> You maintain: Hazard classification inputs (CAS determinations, vendor data) If missing: Not printed section13.rcra_reason_flammable_gas / rcra_reason_flammable_solid (en.php:404-405); SDSGenerator.php:3822-3827
Corrosive (D002) item and reasons:	HAZARD ENGINE Prints when the product is H314 and/or H290, with each named as a reason, unless its physical state is Solid, Powder or Paste. The state is taken in this order: finished good's own > highest-wt% raw material with one > 'Liquid'. <i>On a sheet: corrosive (D002) — classified as corrosive to skin (H314)</i> You maintain: Skin- and metal-corrosion classification inputs (CAS determinations, vendor data, finished-good hazard override). Physical State on the finished good or raw materials If missing: No D002 item. pH is never considered rcraIsNonLiquidState (:3970, resolvePhysicalState :2521); H314 > rcra_reason_skin_corrosion, H290 > rcra_reason_metal_corrosion; rcra_d002 (:3832-3846)
Reactive (D003) item and reasons:	HAZARD ENGINE Prints when the product has a water-reactive (H260/H261), explosive (H200-H205) or self-reactive/organic peroxide (H240-H242) code, with the reasons joined by commas. All of these codes also make Section 14 'Not determined', which blocks publishing. <i>On a sheet: reactive (D003) — reacts with water, explosive</i> You maintain: Hazard classification inputs (CAS determinations, vendor data) If missing: No D003 item section13.rcra_d003 + rcra_reason_water_reactive rcra_reason_explosive rcra_reason_unstable; SDSGenerator.php:3848-3860
Toxicity-characteristic component (D code) item:	REGULATORY LIST Each disclosed component's CAS is matched against the RCRA waste-code list, and a compound also inherits the D codes of every metal in its RCRA metals flags. D codes print at any concentration above zero with the list's TCLP limit; name: Prop 65 name > CAS preferred name > constituent name > CAS (resale sheets: constituent name > CAS). <i>On a sheet: contains Lead chromate (D007 if the toxicity characteristic regulatory level of 5 mg/L...</i> You maintain: RCRA Waste Codes list (Regulatory Data > RCRA Waste Codes). RCRA metals box per CAS (CAS Determinations > CAS Descriptions, or 'Seed RCRA metal flags'). Constituent CAS numbers and percentages If missing: No item. A constituent with no percentage, or a compound with neither its own row nor metal flags, is not reported. A blank TCLP limit prints '...if the toxicity characteristic regulatory level is exceeded' RCRAService::analyse/match: rcra_waste_codes kind D + cas_master.tc_metals > TC_METAL_ELEMENT_CAS rows; formatRcraCode rcra_code_d/_nolimit; name Formula.php:293

Trade-secret component fragments:

REGULATORY LIST A matched constituent flagged as a trade secret is never identified: no waste code and no TCLP level, only its trade-secret description. A D match prints in the as-sold list, and an F/K/P/U match prints '...a listed hazardous waste constituent (code withheld as a trade secret)' in the reference sentence.

On a sheet: contains a toxicity characteristic constituent (Trade Secret) and may exhibit the...

You maintain: Trade-secret flag and description on the raw-material constituent (Raw materials screen)

If missing: A blank description, or one reading 'Trade Secret', prints the translated 'Trade Secret'. Vendor trade-secret materials with no CAS are never matched

is_trade_secret on composition row > section13.rcra_component_ts_tc / rcra_component_ts_listed;
tradeSecretName() :2607; duplicates printed once (:3881-3899)

'No characteristic identified' sentence:

TEXT CHOSEN BY HAZARD Prints when no as-sold item applies. If only listed (F/K/P/U) components matched, the shorter version prints without the clause 'no component has been identified as...'.
On a sheet: No RCRA hazardous waste characteristic (40 CFR 261 Subpart C) has been identified...

You maintain: A complete RCRA waste-code list, RCRA metals flags, and complete constituent CAS and percentage data, because this sentence is a claim based on finding no match

If missing: This sentence is itself the default when nothing matches
section13.rcra_none (nothing matched) / rcra_none_characteristic (only F/K/P/U matched);
SDSGenerator.php:3917-3920

Listed-waste codes of components, for reference:

REGULATORY LIST Prints when a component matches an F, K, P or U code on the RCRA list and is at 0.1 % or more in total across the formula. Each constituent counts at its exact %, else its range maximum. Metal elements' F/K/P/U rows are never inherited by compounds.
On a sheet: Listed-waste codes of components, for reference (they do not apply to this product as sold):...

You maintain: F/K/P/U rows in the RCRA Waste Codes list (Regulatory Data > RCRA Waste Codes). Constituent CAS numbers and percentages

If missing: Sentence omitted
section13.rcra_listed_intro + rcra_component + rcra_code_f/k/p/u; RCRAService::MIN_CONCENTRATION_PCT 0.1 on summed concentration_pct; SDSGenerator.php:3921-3923

Generator-responsibility sentence:

FIXED TEXT (TRANSLATED) Fixed wording from the language file. It is always the last sentence of the RCRA line and cannot be edited per product.

On a sheet: The waste generator is responsible for the final hazardous waste determination at the time...

You maintain: Nothing

If missing: The English text, or the raw key

section13.rcra_generator (en.php:395) appended unconditionally; SDSGenerator.php:3924

SECTION 14: TRANSPORT INFORMATION

Section 14 is worked out for each product by the transport classifier. Class 3 follows the product's Flammable Liquids category, which comes from the wt%-weighted flash point and the lowest initial boiling point of the raw materials, so Section 14 always agrees with Sections 2, 5, 7, 9, 10 and 13. Classes 8, 6.1 and 9 come from the hazard engine's corrosion, toxicity and aquatic statements, and the UN entry depends on the product's transport type or description and its physical state. Per-product overrides, entered separately for each language (also for resale sheets), win line by line. Every language is checked before publishing. The fixed wording (entry names, notes, bulk and special-precautions lines, labels) comes from the translation files.

UN Number:

TRANSPORT RULES Worked out from the product's Flammable Liquids category (wt%-weighted flash point and lowest boiling point of the raws), its corrosion, toxicity and aquatic hazard statements, its physical state and its transport product type (or description keywords). Per-product override > classifier result > "Not regulated" / "Not determined".

On a sheet: UN1210

You maintain: Flash point on every raw that has one, including a high value for water-like raws (Raw materials). Boiling point on volatile raws. Hazard data on each CAS (CAS determinations). Transport product type or description, and physical state (Finished goods).

If missing: No flash point on any raw = not flammable, so "Not regulated" unless another class applies. "Not determined" (publish blocked) only for gas/aerosol/oxidizer-type codes, Gas state, or 3+8+6.1 precedence.

text_overrides(14,un_number,lang) > TransportClassifier::classify() un_number (TC.php:182-372);
inputs SDSGenerator::section14() :4271-4300 via flammabilityOf()

Proper Shipping Name:

TRANSPORT RULES The official DOT name for the chosen UN entry, from the translation file. ES/FR/DE keep the English name and add a local translation in brackets. Override > classifier entry name > "Not regulated" / "Not determined".

On a sheet: Printing ink, flammable

You maintain: Same inputs as UN Number. The names themselves are kept in the translation files.

If missing: "Not regulated" / "Not determined". A name missing in one language falls back to the English text.
text_overrides(14,proper_shipping_name) > section14.psn_{psn_key} (en.php:429-449; es/fr/de:401-420) + technical names; SDSGenerator.php:4310-4316

Proper Shipping Name: technical names in brackets (n.o.s. entries only):	FORMULA / RAW MATERIALS For n.o.s. entries, up to two disclosed ingredients are added in brackets, highest wt% first: those whose CAS carries the primary hazard's codes > (Class 3 only) ingredients of raws with a flash point of 60 °C or below, never water > ingredients carrying any transport hazard. The name is the Prop 65 list name > CAS preferred name > raw-material constituent name; trade secrets are never named and names are not translated. <i>On a sheet: Flammable liquids, n.o.s. (Toluene)</i> You maintain: Ingredient names (CAS determinations > preferred name; Raw materials > constituent name; a Prop 65 list name wins on finished goods). Flash point on solvent raws. Corrosion, toxicity, flammability and aquatic determinations on each CAS. Trade-secret flags. If missing: No brackets are printed, and the preview Warnings box shows an English transport warning (publishing is not blocked). Never added to UN1210/UN1263 or when the PSN is overridden. TC::technicalNames() > technicalNamesFromCas(flammable_cas) > any-hazard pass (TC.php:377-396,606-641); cashCodeMap(); name COALESCE Formula.php:293
Hazard Class:	TRANSPORT RULES Class 3 when the product is Flammable Liquids Cat 1-3 (flash point 60 °C or below on a liquid, or set by the finished-good hazard override). Class 8 for Skin Corr. 1, or for H290 on a liquid. Division 6.1 for acute toxicity Cat 1-3, and Class 9 for aquatic H400/H410/H411 alone. Primary and secondary classes follow 49 CFR 173.2a. Override > classifier > "Not regulated" / "Not determined". <i>On a sheet: 3 (8)</i> You maintain: Flash point and boiling point on the raws. Skin-corrosion, metal-corrosion and acute-toxicity categories on each CAS. Physical state, which decides whether 8 or 6.1 comes first. If missing: "Not regulated" / "Not determined" text_overrides(14,hazard_class) > TC primary + ' (' + subsidiaries + ')' (TC.php:291-375); UNSUPPORTED_CODES TC.php:138-147; SDSGenerator.php:4316,4359
Packing Group:	TRANSPORT RULES The most stringent of the Class 3 group (I when the lowest raw boiling point is 35 °C or below, II for Cat 1/2, III for Cat 3), the Class 8 group (1A = I, 1B or 1 = II, 1C = III, H290 alone = III) and the 6.1 group (Cat 1/2/3 = I/II/III). Class 9 alone is III. Override > classifier > "Not applicable" / "Not determined". <i>On a sheet: II</i> You maintain: Boiling point on every volatile raw (Raw materials). Flash point. Skin-corrosion sub-category and acute-toxicity category on each CAS. A Section 9 boiling-point edit is not used here. If missing: With no boiling point, a flash point below 23 °C gives PG II (with a preview warning). "Not applicable" when not regulated; "Not determined" when undetermined. text_overrides(14,packing_group) > mostStringent(pg3, pg8, pg61) (TC.php:256-287,502-559); IBP = FormulaCalcService lowest raw boiling_point_c
Environmental Hazards:	TRANSPORT RULES "Marine pollutant: Yes" when the product carries H400, H410 or H411; otherwise "No". Override > engine aquatic codes > "Not determined". <i>On a sheet: Environmental Hazards: Marine pollutant: No</i> You maintain: Aquatic toxicity data and M-factors on each CAS (CAS determinations). If missing: "Marine pollutant: No" when no aquatic code is present. "Not determined" while Section 14 is undetermined, unless any core line (UN, name, class, packing group) is overridden (then Yes/No from the codes). text_overrides(14,environmental_hazards) > section14.marine_pollutant_yes no (en.php:450-451) on H400/H410/H411; SDSGenerator.php:4302,4340-4341,4361
Transport in Bulk:	FIXED TEXT (TRANSLATED) Fixed sentence from the translation file, printed on every sheet. It cannot be overridden. <i>On a sheet: Not intended for transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code.</i> You maintain: Nothing If missing: Always printed (skipped only on snapshots made before the line existed). section14.transport_in_bulk_text (en.php:456; es/fr/de:426) set at SDSGenerator.php:4363; PDFService.php:1090; preview.php:467-469
Special Precautions for User:	FIXED TEXT (TRANSLATED) Fixed sentence from the translation file, printed on every sheet. It cannot be overridden. <i>On a sheet: Read this safety data sheet before handling. Keep containers closed, upright and secured...</i> You maintain: Nothing If missing: Always printed (skipped only on snapshots made before the line existed). section14.special_precautions_text (en.php:457; es/fr/de:427) set at SDSGenerator.php:4364; PDFService.php:1091; preview.php:470-472

Note: combustible-liquid sentence (conditional):

TRANSPORT RULES Printed when the product is not regulated, is not Solid/Powder/Paste, and carries H227 with no flash point of 93 °C or more: a flash point above 60 °C and below 93 °C (a "> n" value counts as just above n, so "> 60" qualifies), or H227 from the hazard override with no flash point. Dropped when any Section 14 core line is overridden.
On a sheet: Combustible liquid; regulated in bulk packaging (> 450 L) only.
You maintain: Flash point on the raws (Raw materials). Physical state (Finished goods).
If missing: Not printed
TC combustible (TC.php:203,248-250) > section14.note_combustible (en.php:452);
SDSGenerator.php:4344-4353; engine Cat 4 HazardEngine.php:2762-2777

Note: viscous Packing Group III sentence (conditional):

TRANSPORT RULES Printed only when the derived result is Class 3, Packing Group II, with no secondary class. Dropped when any Section 14 core line is overridden.
On a sheet: If viscous, may be assigned to Packing Group III under 49 CFR 173.121(b)(1)...
You maintain: Same inputs as Packing Group.
If missing: Not printed
TC notes[]='viscous' (TC.php:397-401) > section14.note_viscous (en.php:454);
SDSGenerator.php:4344-4353

Note: carrier verification sentence:

FIXED TEXT (TRANSLATED) Fixed sentence from the translation file. It is always printed last on the Note line.
On a sheet: Transport classification should be verified with the carrier and against current 49 CFR...
You maintain: Nothing
If missing: Always printed
section14.note (en.php:426) appended unconditionally at SDSGenerator.php:4353; PDFService.php:1093; preview.php:473-475

Outcome wording: Not regulated / Not applicable / Not determined:

FIXED TEXT (TRANSLATED) Fixed words from the translation file. No class: "Not regulated" on the UN, name and class lines and "Not applicable" on Packing Group. Classifier cannot decide: "Not determined" on all five lines. A per-line override still wins.
On a sheet: Not regulated
You maintain: Nothing (the outcome depends on flash point and hazard data)
If missing: n/a
labels.not_regulated / labels.not_applicable / labels.not_determined (en.php:734-736);
SDSGenerator.php:4299-4309

Preview Warnings box: Section 14 messages (operator only, not printed):

TRANSPORT RULES English messages in every preview (finished good, resale, private label): the publish-block reason, missing n.o.s. technical names, PG II assumed for a missing boiling point, raws with a flammable ingredient but no flash point, a hazard override that disagrees with the flash point, a Section 9 flash-point or boiling-point edit dropped for contradicting the class, and default physical state.
On a sheet: Publishing blocked: the Section 14 transport classification for 12345 (EN sheet) is "Not determined"...
You maintain: Nothing directly; each message names the data to enter.
If missing: No message when nothing applies.
SDSGenerator generate():129,292-314,335 / generateFromBase():694-716,736;
vetFlashPointEdit/vetBoilingPointEdit :4076-4136; flammabilityWarnings() :4152-4210;
SDSReadinessService :630-674

SECTION 15: REGULATORY INFORMATION

Section 15 is built in SDSGenerator::section15(). It checks the product's composition against the regulatory lists you maintain: the TSCA inventory, SARA 313 (with its metal-compound and glycol-ether categories), HAP, SNUR and Prop 65. It also uses the Section 2 hazard result, one per-product editor note (State Regulations) and fixed translated wording. OSHA and TSCA Status are automatic and cannot be edited per product. Trade-secret ingredient identities are masked on every line.

OSHA Status:

HAZARD ENGINE Follows the same classification Section 2 prints: 'classified' when there is a signal word, pictogram, hazard class or H-statement, otherwise 'not classified'. It cannot be edited per product, and stored notes are ignored.
On a sheet: This product is not classified as hazardous under OSHA HazCom 2024 (29 CFR 1910.1200).
You maintain: The hazard data behind Section 2: constituent classifications and CPDs on CAS Determinations; flash point and boiling point on raw materials and the product's physical state (Flammable Liquids category); the finished-good hazard override (Finished goods). The sentence itself has no per-product edit.
If missing: Nothing classified: 'This product is not classified as hazardous under OSHA HazCom 2024 (29 CFR 1910.1200).' If no raw material has a flash point, the product counts as not flammable.
resolveOshaStatus: isClassified(hazard) ? section15.osha_status :
section15.osha_status_not_classified (SDSGenerator.php:4500-4505, :1257-1263); 15.osha_status retired

TSCA Status:	REGULATORY LIST Every ingredient CAS is checked against the EPA TSCA inventory and the per-CAS TSCA Review, and every raw material in the formula must have constituents with a CAS. Sentence: a CAS marked Not listed > all listed/exempt > not verified. It cannot be edited per product. <i>On a sheet: One or more components of this product are not listed on the TSCA inventory.</i> You maintain: TSCA Inventory imported from EPA (Regulatory lists > TSCA Inventory). TSCA Review on CAS Determinations for exempt, confidential, crossover, absent or INACTIVE CAS (for INACTIVE, Listed + NOA note). Constituents with a CAS on every raw material. If missing: 'TSCA inventory status has not been verified for all components.' prints when a CAS is unchecked, a trade secret has no CAS, or a raw material has no constituents or a blank-CAS constituent. A TSCA warning also shows (not printed). TSCAService::analyse(composition, formulaRawMaterialIds): cas_master.tscs_status<>'auto' > tscs_inventory > not_verified; applyIncomplete; statusKey (SDSGenerator.php:4478-4483)
SARA 313 / TRI Reporting:	FIXED TEXT (TRANSLATED) Fixed heading from the translation file. Always printed on new sheets. <i>On a sheet: SARA 313 / TRI Reporting:</i> You maintain: Nothing If missing: Older saved sheets without SARA data leave out the whole block. labels.sara_313_title; printed when sections[15].sara_313.reportable is an array (PDFService.php:1107-1111; preview.php:486-488)
SARA 313 supplier-notification statement / 'none' sentence:	REGULATORY LIST Fixed wording. Which sentence prints depends on whether any ingredient is reportable: at or above its de minimis, or a PBT or TRI PFAS chemical at any amount. <i>On a sheet: This product contains the following toxic chemical(s) subject to ... Section 313 ...</i> You maintain: SARA 313 list (Regulatory lists > SARA 313) and ingredient CAS numbers If missing: Nothing reportable: 'This product does not contain any toxic chemicals subject to ... Section 313 ... at or above the applicable de minimis concentration.' labels.sara_313_statement when reportable non-empty, else labels.sara_313_none (PDFService.php:1112-1135; preview.php:489-504)
SARA 313 chemical lines (name, CAS, band, threshold or no-de-minimis note):	REGULATORY LIST Ingredients on the SARA 313 list by CAS, or in a listed metal-compound or glycol-ether category, at or above the de minimis; PBT and TRI PFAS (special concern) at any amount. Name: list name (footnote marks stripped) > category name > ingredient name. Band: Section 3 range. <i>On a sheet: • Toluene (CAS 108-88-3) — 10 - 30% (de minimis threshold: 1%)</i> You maintain: SARA 313 list rows: CAS, name, de minimis % (0.1 for carcinogens), PBT flag, special-concern flag (CSV column 7 or migration 058). Category definitions come from migration 058. Ingredient CAS and percentages on raw materials. If missing: PBT prints 'PBT chemical; no de minimis exemption'; PFAS prints 'Chemical of special concern (40 CFR 372.28); no de minimis exemption'. Blank list name: category or ingredient name. CSV with no de minimis: 1%. Old sheets: empty band. SARA313Service::evaluate (sara313_list + RegulatoryCategoryService LIST_SARA; is_pbt is_special_concern → threshold 0); bandRegulatoryEntries; PDFService.php:1114-1128
SARA 313 range note:	FIXED TEXT (TRANSLATED) Fixed italic note. Printed only when SARA chemicals are listed. <i>On a sheet: Concentrations are stated as the prescribed concentration ranges of 29 CFR 1910.1200(i)(1)...</i> You maintain: Nothing If missing: Not printed when no SARA chemical is listed. labels.sara_313_range_note (PDFService.php:1129-1131; preview.php:500)
EPA HAPs (heading):	FIXED TEXT (TRANSLATED) Fixed heading. English prints 'EPA HAPs'; Spanish, French and German print a long Clean Air Act 112(b) title. <i>On a sheet: EPA HAPs:</i> You maintain: Nothing If missing: Older saved sheets without HAP data leave out the block. labels.hap_title (en.php:714; es/fr.php:684; de.php:681); PDFService.php:1139-1177; preview.php:508-535
HAP table: Triggering HAP Chemical Wt% in Formula:	REGULATORY LIST Ingredients at 0.01% or more whose CAS is on the HAP list or in a HAP compound category (metal compounds, glycol ethers), plus HAPs typed on raw materials. Name: HAP list name > category name. Amount: Section 3 range band. <i>On a sheet: Toluene 10 - 30%</i> You maintain: HAP List (Regulatory lists > HAPs) with a name for every CAS. HAPs typed on the raw material (name and weight %) for any HAP not matched by CAS. A molecular formula or recognisable name on the CAS so category members are found. If missing: Blank HAP-list name: empty name cell. Typed HAP with a blank name or 0%: left out. No HAP at all: 'This product does not contain any EPA HAPs listed under Clean Air Act Section 112(b).' HAPService::evaluate (hap_list + RegulatoryCategoryService LIST_HAP, ≥0.01%) + getManualHaps (haps_data × summed RM %); bandRegulatoryEntries; sorted by exact % desc

Total HAP Content:	FORMULA / RAW MATERIALS The exact percentages of all HAP rows are added up and printed as the prescribed-range band that contains the total. <i>On a sheet:</i> Total HAP Content: 10 - 30% You maintain: Same as the HAP rows If missing: Printed only when there are HAPs. Older saved sheets that stored an exact total print no total row. total_hap_range = formatConcentration(total_hap_pct) (SDSGenerator.php:4434-4437); labels.hap_total; PDFService.php:1162-1169; preview.php:522-530
EPA Significant New Use Rules (SNUR):	REGULATORY LIST Ingredients whose CAS is on the SNUR list, at any concentration. Name, CAS, rule citation and description come from the SNUR list exactly as typed (English). <i>On a sheet:</i> • Chemical X (CAS 123-45-6) — 40 CFR 721.1234 You maintain: SNUR List (Regulatory lists > SNUR List, /admin/snur-list) with name, CAS, rule citation and description If missing: No match: no SNUR block at all. Blank name: the line starts '(CAS ...)'. Blank citation or description: that part is left out. SDSGenerator::analyseSnur (:5736-5788) on snur_list.cas_number; chemical_name, rule_citation, description; labels.snur_title; PDFService.php:1179-1199
Trade-secret ingredients in SARA 313, HAP and SNUR lines:	FORMULA / RAW MATERIALS A constituent flagged trade secret prints its trade-secret description (else 'Trade Secret') and 'TRADE SECRET' instead of its name and CAS; its band still prints. A SNUR line also drops the rule citation and prints a fixed sentence saying the identity is withheld. <i>On a sheet:</i> • Acrylic monomer (CAS TRADE SECRET) — 1 - 5% (de minimis threshold: 1%) You maintain: Trade-secret tick and trade-secret description on the raw-material constituent (Raw materials). If missing: Blank description: 'Trade Secret' prints. Trade-secret constituents with no CAS are never checked against SARA 313, HAP or SNUR. maskTradeSecretRow (SDSGenerator.php:2627-2646; called :4442-4456): name → trade_secret_description ?? labels.trade_secret; CAS → labels.trade_secret_cas; section15.snur_trade_secret
California Proposition 65 (heading):	FIXED TEXT (TRANSLATED) Fixed heading, always printed. <i>On a sheet:</i> California Proposition 65: You maintain: Nothing If missing: Uses the English wording if the sheet language lacks it. labels.prop65_title (PDFService.php:1203-1206, :1241-1244; preview.php:567, :580)
Prop 65 warning symbol:	ADMIN SETTING Yellow warning triangle, shown only when a warning prints. Admin-uploaded PROP65 image > image shipped with the system. You maintain: Optional PROP65 image upload (Regulatory lists > Pictograms); otherwise the shipped image must exist If missing: No image file: no symbol prints, but the warning text still prints. PictogramHelper getPngPath/getWebPath('PROP65'): uploads/pictograms/PROP65.(png jpg gif) > assets/pictograms/png/PROP65.png; PDFService.php:1209-1220; preview.php:563-566
Prop 65 WARNING text:	REGULATORY LIST Cancer and reproductive chemicals found by matching ingredient CAS (0.01% or more) to the Prop 65 list, plus Prop 65 entries on raw materials, minus inhalation-only chemicals when the product is not all solid. Trade secrets are never named (a match blocks publishing); the text is the cancer, reproductive or combined template in the sheet language. <i>On a sheet:</i> WARNING: This product can expose you to chemicals including Toluene, which is/are known... You maintain: Prop 65 list (Regulatory lists > Prop 65: CAS, name, toxicity types). Prop 65 entries on raw materials (a listed CAS, or Override with name and types). The inhalation-only CAS setting. Physical state on every raw material. If missing: No cancer or reproductive chemical left: 'This product is not known to contain any chemicals listed under California Proposition 65.' Types match in any case. Old name-only entries are used as typed. Prop65Service::analyse (prop65_list ≥0.01%, normaliseTypes; prop65_data; trade_secret_conflicts); removeInhalationOnlyFromResults; rebuildProp65Warning (SDSGenerator.php:5327)
Prop 65 '(trace)' marker:	ADMIN SETTING Added to a chemical name when every occurrence is below the admin trace threshold (CAS path) or ticked 'trace' on the raw material. <i>On a sheet:</i> Benzene (trace) You maintain: Prop 65 trace threshold in Admin Settings; trace tick-boxes on raw-material Prop 65 entries If missing: Setting blank or 0: 0.1% is used. setting prop65.auto_trace_threshold_pct (default 0.1; Prop65Service.php:123-132); prop65_data[.is_trace; section15.prop65_trace_name (Prop65Service.php:420-445)

Listed chemicals:	REGULATORY LIST One line per Prop 65 chemical with its listing types, printed under the warning. The types come from the Prop 65 list or the raw-material override. <i>On a sheet:</i> • Toluene (CAS 108-88-3) — developmental toxicity, female reproductive toxicity You maintain: Prop 65 list toxicity types: cancer, developmental, female reproductive, male reproductive or reproductive. Any case is accepted, as are a 'toxicity' suffix and 'female'/'male'. If missing: No CAS: the line prints without '(CAS ...)'. A type outside that list is skipped; a chemical with no recognised type gets no line. No warning: no list. buildProp65ListedLines (SDSGenerator.php:3312-3367); Prop65Service::normaliseTypes; labels.prop65_listed; section15.prop65_listed_line / _no_cas / prop65_type_*
State Regulations:	PRODUCT OVERRIDE Only from the note the operator types in the SDS editor for this product or resale raw material, separately for each language. Prints whenever present. <i>On a sheet: State Regulations: New Jersey Right-to-Know: Toluene (CAS 108-88-3)</i> You maintain: SDS editor > Section 15 > Additional State Regulations (/sds/{fg}/edit or /sds/resale/{rm}/edit), per language If missing: Line left out. There is no fallback from another language. trim(overrides[15][state_regs]) via getOverrides(fg resale rm, lang) (SDSGenerator.php:4416); labels.state_regulations; PDFService.php:1252-1255; preview.php:584-590
Sections 12–15 footnote:	ADMIN SETTING Fixed translated note, switched on or off for all sheets by one admin checkbox (on by default). <i>On a sheet: Sections 12–15 are included as required by OSHA HazCom (29 CFR 1910.1200(g)(2))...</i> You maintain: Admin Settings checkbox 'Print the Sections 12–15 footnote' If missing: Checkbox off: nothing prints. settings sds.show_ghs_section_note (missing = on, '0' = off) → document.ghs_section_note (SDSGenerator.php:4609-4619; PDFService.php:1288-1301; preview.php:650-651)
Preview Warnings box: TSCA, incomplete composition, trade-secret Prop 65 (not printed):	REGULATORY LIST Operator messages shown above the preview and never printed on the sheet. They cover a TSCA gap or INACTIVE inventory entry, raw materials with incomplete constituent data, and a trade-secret constituent on the Prop 65 list (which also blocks publishing). <i>On a sheet: TSCA warning: Section 15 prints "One or more components ... not listed ..." — ...</i> You maintain: The same data as TSCA Status and Prop 65: TSCA inventory and reviews, constituents with CAS and percentage on raw materials, trade-secret ticks. If missing: No message when everything is covered. TSCAService::warningText; SDSReadinessService::incompleteCompositionWarning / tradeSecretProp65Error (SDSGenerator.php:282-291, :319-326, :684-693, :721-728)

SECTION 16: OTHER INFORMATION

Section 16 is built entirely from publish metadata (the version number and effective date stamped when a sheet is published), the per-language translation files (labels, draft text, abbreviation master table) and the Admin or manufacturer disclaimer text. Nothing in it can be typed per product.

Version:	PUBLISH RECORD The number the system assigns at publish: the next number for that finished good or resale raw material, alias (when a sender publishes it, pack variants share one counter), or private-label item. Unpublished previews print the translated draft text. <i>On a sheet: Version: 3 (unpublished: Version: Draft (not yet published))</i> You maintain: Nothing to enter. Never hard-delete published SDS rows (soft-deleted rows still count toward the next number). If missing: Draft or preview: 'Draft (not yet published)' (translated). A pre-version snapshot, now re-rendered only in a private-label record preview, prints no Version line. section16() :4519 = section16.draft; stampPublishedVersion() :1024-1043 sets it; MAX(sds_versions.version)+1 per base / alias id / alias group (AliasPublisher :216), or private_label_sds
Effective Date:	PUBLISH RECORD The publish date (today in the admin time zone) written to the version record. It prints as month/day/year in every language. <i>On a sheet: Effective Date: 10/09/2026</i> You maintain: Nothing If missing: Draft or preview: the line is not printed. A pre-version private-label snapshot in its record preview has no line; the old 'Revision Date' line prints instead. stampPublishedVersion :1030 format_date(...,'m/d/Y'); publishers pass PublishClock::todayLocal() (alias publishes in SDSController/SDSUpdateController: localDateOfUtc(\$now))

Abbreviations:	FIXED TEXT (TRANSLATED) A master list of abbreviations for each language. It is filtered to the terms that print elsewhere on that sheet (including the alias or private-label code and the disclaimer) and printed in file order. <i>On a sheet: Abbreviations: CAS = Chemical Abstracts Service registry number; DOT = ...; IATA =</i> You maintain: Developers keep the four translation tables (en/es/fr/de section16.abbreviation_table) complete and alphabetical. Any new abbreviation in generated text, labels, disclaimers or overrides must be added to all four. If missing: No term matched: the line is not printed. A term missing from the table gets no definition (today SDS/HDS/FDS in the EN/ES/FR Section 4 note, plus ISEA, ASTM, PBT when printed). AbbreviationService::build :46 (generate :342, generateFromBase :743, alias :835, PL :977); section16.abbreviation_table[lang] > EN; PATTERNS :27; collectCorpus :107-210
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DOCUMENT HEADER, FOOTER AND DISCLAIMER

Document chrome is the page-1 header, section banners, footer, disclaimer, PDF properties, file names and the preview page around the PDF. It comes mainly from company or manufacturer settings, the per-language translation files and the version and date stamped at publish, plus a few fixed layout constants. Alias sheets print the alias base code, and senders always use an alias's own published PDF.

Header logo (page 1) - standard, alias and resale SDS:	COMPANY / MANUFACTURER The company logo uploaded in Admin Settings. It prints on page 1 only. <i>On a sheet: Company logo top-left of page 1, about 51 mm wide</i> You maintain: Upload the company logo (PNG/JPG/GIF, 2 MB max) in Admin > Settings > Company Logo. The file must stay on the server. If missing: No logo prints, but the title and rule still print. Re-renders of older snapshots use today's logo file, or none if the logo was removed or its file type changed. settings.company.logo_path (AdminController::processLogoUpload) > meta.company_logo_path > PDFService::resolveLogoPath :1384-1391 > SDSTcpdf::Header :97-111
Header logo (page 1) - private label SDS:	COMPANY / MANUFACTURER The private label manufacturer's own logo. The company logo is never used in its place. <i>On a sheet: Manufacturer's logo top-left of page 1</i> You maintain: Upload a logo on each private label manufacturer's form. If missing: No logo prints. The company logo is not used instead. manufacturers.logo_path ('/uploads/manufacturers/mfg-logo-<id>.<ext>') > SDSGenerator::createManufacturerVariant :950 sets meta.company_logo_path
SAFETY DATA SHEET (header title):	FIXED TEXT (TRANSLATED) Fixed title from the translation file for the sheet language, printed bold and right-aligned on page 1. Re-renders of older snapshots that lack it get the sheet-language title. <i>On a sheet: SAFETY DATA SHEET / HOJA DE DATOS DE SEGURIDAD</i> You maintain: Nothing If missing: "SAFETY DATA SHEET" (the English default) only if no translation file defines it. meta.document.title > translation document.title (sheet lang; SDSDocumentStrings::forLanguage) > DEFAULTS; SDSTcpdf::Header :88-95
Header rule; no header on pages 2+:	FIXED TEXT A navy rule sits under the page-1 header. Pages 2 onward have no header and the text starts near the top edge. You maintain: Nothing If missing: n/a SDSTcpdf::Header :70-72 returns when page > 1, rule :113-117; PDFService MARGIN_TOP 30 / MARGIN_TOP_CONTINUATION 15 (:19-21), SetTopMargin :299
SECTION N: TITLE banners (all 16):	FIXED TEXT (TRANSLATED) The word 'SECTION' and each section title come from the translation file and print in capitals, accented letters included. Older snapshots without a stored prefix get the sheet-language word. <i>On a sheet: SECTION 1: IDENTIFICATION / SECCIÓN 1: IDENTIFICACIÓN</i> You maintain: Nothing If missing: The sheet-language "SECTION" plus the translation title meta.document.section_prefix (forLanguage) + sections[N].title (translation sectionN.title); mb_strtoupper in PDFService::renderSection :317-325; preview.php :18/:45
Sections 12-15 footnote (after Section 15):	ADMIN SETTING One shared footnote in the sheet language, printed once after Section 15 when the admin toggle is on (the default). <i>On a sheet: Sections 12–15 are included as required by OSHA HazCom (29 CFR 1910.1200(g)(2))...</i> You maintain: Admin > Settings: tick 'Print the Sections 12–15 footnote' to print it (on by default). If missing: Toggle off: nothing prints. settings.sds.show_ghs_section_note ('0' = off) gates document.ghs_section_note > sections[15].ghs_note (SDSGenerator :4490, :4609-4619); renderGhsSectionNote :1288-1300

DISCLAIMER (bar):	FIXED TEXT (TRANSLATED) A navy bar after Section 16, labelled from the translation file. It prints only when there is disclaimer text. <i>On a sheet: DISCLAIMER / DESCARGO DE RESPONSABILIDAD / HAFTUNGS AUSSCHLUSS</i> You maintain: Nothing If missing: Newly generated sheets always print it. A re-render of a pre-#34 snapshot with no stored disclaimer prints no bar and no text. meta.labels.disclaimer > translation labels.disclaimer (sheet lang, then EN) > key; PDFService::renderLegalDisclaimer :1396-1415; preview.php :656-659
Disclaimer text - standard, alias and resale SDS:	ADMIN SETTING Admin per-language disclaimer > built-in translation text for that language. It is frozen into the snapshot at publish. <i>On a sheet: The information provided in this Safety Data Sheet is correct to the best of our...</i> You maintain: Optional. Admin > Settings > Legal / Disclaimer Statement, one box per language (EN was seeded by migration 052). If missing: The built-in translation paragraph for the sheet language (never blank on new sheets). Re-renders of pre-#34 snapshots keep the old value, which may be blank or English. settings.sds.legal_disclaimer.<lang> > translation section16.disclaimer; SDSGenerator::resolveLegalDisclaimer :4586-4593; migration 052 seeded .en only
Disclaimer text - private label SDS:	COMPANY / MANUFACTURER Manufacturer's disclaimer for that language > admin per-language disclaimer > built-in translation text. <i>On a sheet: Manufacturer's own disclaimer paragraph in the sheet language</i> You maintain: Optional. A per-language disclaimer on the manufacturer form. If missing: The admin text for that language, else the built-in translation text. manufacturers.disclaimer_json[lang] (blank = inherit) > settings.sds.legal_disclaimer.<lang> > section16.disclaimer; createManufacturerVariant :955-959
Bold field labels throughout the sheet:	FIXED TEXT (TRANSLATED) Labels stored with the sheet > translation label in the sheet language > English translation > a capitalized key name. <i>On a sheet: Product Identifier: / Identificador del producto:</i> You maintain: Nothing If missing: The translation label (sheet language, then English) meta.labels (SDSGenerator::getLabels :4767) > labels.<key> via SDSDocumentStrings::translate > key; PDFService::label :1338-1343
Footer left: product code (every page):	FORMULA / RAW MATERIALS The code of the product the sheet is issued for. Alias and resale sheets print the code without its pack extension; private label custom codes print exactly as typed. <i>On a sheet: UVNG009 / BK1080 (alias BK1080-5G) / AC-100 (private label custom code)</i> You maintain: The product code on the finished good, the customer code on each alias, the internal code on resale raw materials, and the custom code on private label items. If missing: The cell is empty. meta.product_code = FG code; alias/resale stripPack(code); PL custom_code > stripped alias > FG (PrivateLabelPublisher::resolveIdentity :51); SDSTcpdf::Footer :144-146
Footer centre: Page X of Y:	FIXED TEXT (TRANSLATED) The words come from the translation file for the sheet language (also on re-renders of older snapshots) and the numbers from the PDF engine. <i>On a sheet: Page 2 of 5 / Página 2 de 5 / Page 2 sur 5 / Seite 2 von 5</i> You maintain: Nothing If missing: "Page X of Y" in English only if no translation file defines the words. meta.document.page / page_of (SDSDocumentStrings::forLanguage) > DEFAULTS; SDSTcpdf::Footer :148-153
Footer right: Rev. version and effective date (published):	PUBLISH RECORD The version number and publish date stamped automatically at publish, the same values as Section 16. The date is today in the admin time zone and always prints as m/d/Y. <i>On a sheet: Rev. 3 — 10/09/2026 / Rév. 3 — 10/09/2026</i> You maintain: Nothing. It is automatic at publish. The date follows the time zone set in Admin > Settings. If missing: Only the version prints: "Rev. 3". PDFService::footerRevision :1309-1323: revision_prefix + meta.sds_version + (s16.effective_date > meta.effective_date > s16.revision_date); PublishClock::todayLocal
Footer right: Draft (unpublished):	FIXED TEXT (TRANSLATED) Prints on previews and ad-hoc renders that have no version stamp. A published private label record re-rendered because its snapshot is missing prints its own version instead. <i>On a sheet: Draft (not yet published) / Borrador (aún no publicado)</i> You maintain: Nothing If missing: n/a sections[16].version = translation section16.draft when meta.sds_version is 0/absent; footerRevision :1329; PrivateLabelController preview :975-979 stamps

Footer right: legacy generation date (old snapshots):	PUBLISH RECORD Re-renders of snapshots made before #32 print the old generation date instead of a version. <i>On a sheet: Rev. 10/01/2026</i> You maintain: Nothing If missing: The Draft text footerRevision :1325-1327 uses sections[16].revision_date when meta.sds_version is absent
Footer rule and typography; page size and font:	FIXED TEXT Every sheet in every language uses a grey rule 15 mm from the bottom, 8 pt Helvetica in three equal cells, and US Letter portrait pages. The core font prints Latin-1 and Windows-1252 characters only. You maintain: Nothing If missing: Any other character (for example '←' or '→') prints as "?". SDSTcpdf::Footer :123-158; new SDSTcpdf('P','mm','LETTER') PDFService :261; core font helvetica; MARGIN_BOTTOM 20; tests/Services/CoreFontCharsetTest.php
PDF property: Title:	FIXED TEXT (TRANSLATED) A translated prefix plus the code printed in the footer. <i>On a sheet: SDS - UVNG009 / HDS - BK1080 / FDS - ... / SDB - ...</i> You maintain: Nothing beyond the product code If missing: English "SDS - <code>" only if no translation file defines the prefix. Re-renders of older non-English snapshots get their own-language prefix. meta.document.pdf_title (forLanguage) with :code = meta.product_code; PDFService :270
PDF property: Author:	COMPANY / MANUFACTURER The supplier name printed in Section 1: the company name from Admin Settings, or the private label manufacturer's name. <i>On a sheet: Acme Inks, Inc.</i> You maintain: Company name in Admin > Settings, and the name on each manufacturer (required; a blank name cannot be saved). If missing: "SDS System", with a preview warning. If the company name was never saved, the config value (placeholder "Your Ink Company, Inc.") is used, also with a warning. trim(sections[1].manufacturer_name): settings.company.name (config company.name only if never stored) / manufacturers.name; PDFService :268-269
PDF property: Subject:	FIXED TEXT (TRANSLATED) Fixed text from the translation file for the sheet language. <i>On a sheet: Safety Data Sheet / Hoja de datos de seguridad</i> You maintain: Nothing If missing: English "Safety Data Sheet" only if no translation file defines it. meta.document.pdf_subject (SDSDocumentStrings::forLanguage) > DEFAULTS; PDFService :271
PDF properties: Creator, Producer, dates, XMP, language:	FIXED TEXT Fixed program names and the render time, with an embedded XMP copy of Title, Author and Subject. No document language is declared. <i>On a sheet: Creator 'SDS System'; Producer 'TCPDF 6.11.3 (http://www.tcpdf.org)'</i> You maintain: Nothing If missing: n/a PDFService :264 SetCreator; TCPDF _putinfo / _putXMP; no /Lang (setLanguageArray never called); no Keywords
Stored PDF file name (published):	PUBLISH RECORD The code as printed in the footer (characters other than letters, digits, '.', '-' and '_' become '_'), then the PL tag, the version and the language (the default language has no suffix). A name clash adds _2, _3. <i>On a sheet: UVNG009_v3.pdf / BK1080_v2_es.pdf / AC-100_PL_Acme_Coatings_v2.pdf</i> You maintain: Product and alias codes, private label custom codes, and the manufacturer name for private label. If missing: An empty code gives "unnamed_file". A blank manufacturer name gives the bare tag "PL", but such a sheet is refused at publish. PDFService::generate :113-124 (sanitize_filename), versionedBaseName :159-168, reserveUniquePath :178-193; meta.filename_tag (name cut to 40 chars) createManufacturerVariant :967-969
Stored PDF file name (unversioned render):	FIXED TEXT Code, an optional PL tag, the language and a timestamp in the admin time zone, used when the sheet has no version. <i>On a sheet: UVNG009_SDS_en_20261009_143012.pdf</i> You maintain: Nothing If missing: n/a PDFService::generate :122 date('Ymd_His')

Download file names:	PUBLISH RECORD Built when a user downloads; standard and Lookup names strip the pack extension, and only Lookup leaves out the language. Preview names use the printed code. <i>On a sheet:</i> SDS_UVNG009_v3_en.pdf / SDS_BK1080_v3.pdf / PL_SDS_AC_100_Acme_v2_en.pdf You maintain: Nothing If missing: No code: "SDS_v3_en.pdf" / Lookup "SDS_SDS_v3.pdf" / private label "PL_SDS_SDS_..." / preview "SDS_draft_en.pdf" SDSController::download :852-866; LookupController :87-95 (no lang); PrivateLabelController :912-915; SDSPreviewResponse::filename :56-68 (v{n} or draft)
Email attachment and report ZIP names:	PUBLISH RECORD Item code + '_SDS' (+ language when not English), with no version. Auto-send cuts an alias item code at its first hyphen (R1055-84 becomes R1055); the send queue (English only) and report ZIP keep the code as shipped. <i>On a sheet:</i> R1055_SDS.pdf / R1055_SDS_ES.pdf / BK1080-5G_SDS.pdf You maintain: Nothing If missing: Auto-send skips a language with no published PDF on disk and attaches nothing for an alias that cannot be published. The report ZIP logs the item in _MISSING_ITEMS.csv. SDSAutoSendService::sendOrderEmail :384-432 (alias: code up to first '-'); SDSSendQueueController::sendEmail :267-273 (item_name or item_code); ReportController ZIP :972-1084
Alias PDFs in sends (send queue, auto-send, report ZIP):	PUBLISH RECORD The alias's own published PDF; pack variants of one alias share one sheet. If a wanted language is missing or its PDF is gone from disk, the alias is published first from current data. <i>On a sheet:</i> Footer 'BK1080' with the alias's own 'Rev. 2 — 10/09/2026' You maintain: Nothing beyond the alias code and description. If missing: If the alias cannot be published: the queue shows the reason, auto-send attaches nothing for it, and the report ZIP lists it in _MISSING_ITEMS.csv. AliasPublisher::ensurePublished :112-146 > publish :164-270 (createAliasVariant(stripPack code) + stampPublishedVersion); never a snapshot re-render
Preview page: browser title, back links, status line, buttons:	FIXED TEXT English application text around the embedded real PDF. The private label back link names the manufacturer. <i>On a sheet:</i> EN — Draft: this is the PDF Publish will produce... You maintain: Nothing If missing: Links and buttons that do not apply are left out. layouts/main.php:7; SDSController::preview :58-111, previewResale :121-200; PrivateLabelController preview; preview-pdf.php; SDSPreviewResponse
Preview page: Warnings box:	FORMULA / RAW MATERIALS Data-quality and publish-block messages from the generator on every preview, plus supplier checks per path. They never print on the PDF; the private label stored preview shows those frozen at publish. <i>On a sheet:</i> Composition warning: 1 raw material has incomplete constituent data — ... You maintain: The underlying data: emergency phones, company name and phone, raw material data, transport data, TSCA list, product status. If missing: The box is not shown. sds.warnings: calc (incl. missing SG/VOC), UV, TSCA, composition, S14, trade secret, flash point; FG adds phone/supplier/inactive (SDSController :76-90); PL mfg checks

CALCULATED VALUES AND THEIR INPUTS

In this scope, every number, range and derived classification on the sheet is calculated at generation time. The inputs are the formula lines and each raw material's own data: constituent percentages, flash and boiling points, specific gravity, VOC, solubility and physical state. These pass through the hazard engine, the transport classifier and the admin regulatory lists (RCRA, SARA 313/TRI, HAP, carcinogens). Batch E changed several of these rules. The product flash point is now the wt%-weighted average of the raw materials that have one, and a product with no flash point is not flammable. The Flammable Liquids category comes only from that flash point and the initial boiling point. Every Section 9 property in Appendix D prints, with 'Not determined' where there is no data. Trade-secret identities are masked in every section. Missing raw-material data raises warnings in the preview Warnings box, which are never printed. Per-product text overrides still win where they exist. Section 9 flash point and boiling point overrides are printed only when they read as a temperature and agree with the classification.

Flammable Liquids class line and H224-H227 (Section 2; drives Sections 5-8, 10, 13, 14):

FORMULA / RAW MATERIALS A Flammable Liquids class or H224-H227 in the finished-good hazard override > the category from the product flash point and initial boiling point (Cat 1 below 23 °C with IBP 35 °C or less, Cat 2 below 23 °C, Cat 3 up to 60 °C, Cat 4 up to 93 °C). Ingredient flammable classifications are ignored, and Solid, Powder or Paste products are never classified.
On a sheet: Flammable Liquids, Category 3 — H226 Flammable liquid and vapor
You maintain: Flash Point (°C) and Boiling Point (°C) on raw materials and Physical State (Raw Materials / Finished Goods forms); a Flammable Liquids class in the finished good's hazard override only for a tested product
If missing: No flash point on any raw material: not classified as flammable. Flash point below 23 °C with no boiling point: Category 2 is assumed and a preview warning is shown
HazardEngine::applyFlammableLiquidsFromFlashPoint :2849, flammableLiquidCategory :2762;
SDSGenerator::flammabilityInputs :4009; FG override :1950-1990

Carbon black / titanium dioxide cancer hazard (Sections 2, 3, 8, 11 and 15 Prop 65):

FORMULA / RAW MATERIALS When every raw material line is Solid or Powder, each inhalation-only CAS at 0.1% or more adds Carcinogenicity Cat 2 (H351) unless H350/H351 is already present or a replace-mode hazard override is set, and Prop 65 keeps it. Otherwise these CAS are kept out of classification and exposure limits, Prop 65 drops them, and Section 11 keeps the listing with an inhalable-dust note.
On a sheet: H351 added only when every raw material in the formula is Solid or Powder
You maintain: Physical State on every raw material (Raw Materials form); the inhalation-only CAS list (Admin > Settings; default carbon black 1333-86-4 and TiO2 13463-67-7)
If missing: A raw material with a blank Physical State counts as bound (not dry), so H351 and the Prop 65 listing are suppressed
SDSGenerator::isBoundMixture :4969, inhalationOnlyCasToExclude :4988, applyCarbonBlackLogic :5037, removeInhalationOnlyFromResults :5239; setting sds.inhalation_only_cas

Section 3 Type (Substance / Mixture):

FORMULA / RAW MATERIALS Substance/Mixture setting on the finished good > on Auto, 'Substance' only for a single-ingredient formula whose raw material is marked Substance, otherwise 'Mixture'. Resale sheets use the raw material's own setting.
On a sheet: Mixture
You maintain: Substance/Mixture on the finished good, or on the raw material for single-ingredient and resale products
If missing: 'Mixture'
SubstanceMixtureResolver; finished_goods.substance_mixture > raw_materials.substance_mixture (single line); SDSGenerator :133-139, :465, section3 :1478-1482

Section 3 component rows (which ingredients are listed):

FORMULA / RAW MATERIALS A constituent is listed when its amount (the exact %, else the top of its range) is 0.1% or more and it is classified, has an exposure limit (ignored for an ingredient supplied only by Solid or Powder raws when the formula has another state), or (at 1% or more) is a flammable-liquid ingredient behind the flash-point class. A Substance sheet always lists its largest constituent; inhalation-only CAS with no remaining hazard are dropped, and trade-secret rows are merged by description.
On a sheet: Toluene | 108-88-3 | 10 - 30% | H225, H304, H315
You maintain: Constituent CAS and % on each raw material (Raw Materials > Constituents); classifications per CAS (CAS determinations)
If missing: Rows below 0.1% are left out, even when several trade-secret rows with one description together reach 0.1%. The non-hazardous checkbox has been removed and is not read
SDSGenerator::section3 :1315-1510; engine hazardous_cas (flash-point contributors :2923) / exposure_limits (solid-in-liquid filter :5181); sds.inhalation_only_cas

Section 3 Concentration column:

FORMULA / RAW MATERIALS Each constituent's lower and upper % (an exact value counts as both) times its line % are summed per CAS and placed in the widest prescribed HazCom range that contains that span. If no range contains it, the range containing the maximum with the lowest lower end is used, so the upper end is never below the maximum present.

On a sheet: 10 - 30%

You maintain: Constituent % (exact, or min and max) on each raw material (Raw Materials > Constituents) and the line % on each formula

If missing: A constituent with no % counts as 0 and is dropped. '<0.1%' prints only when the maximum is below 0.1%; above 100% prints '80 - 100%'

SDSGenerator::formatConcentration :4712-4762, PRESCRIBED_RANGES :4698; Formula::constituentBounds :814, resolveConstituentPct :784; resale FormulaCalcService :353

Trade-secret ingredient identity (Sections 3, 8, 11, 12, 13 and 15):

FORMULA / RAW MATERIALS A constituent marked trade secret on any raw material prints its trade-secret description (else 'Trade Secret') and 'TRADE SECRET' for the CAS, with the band only, in every section. Section 13 uses generic RCRA wording with no code, and the SARA category code and SNUR citation are blanked.

On a sheet: Acrylic resin | TRADE SECRET | 10 - 30%

You maintain: Trade-secret flag and description (plus declared H-codes on blank-CAS rows) on the raw material's constituents (Raw Materials > Constituents)

If missing: A blank description prints 'Trade Secret'. A trade-secret CAS on the Prop 65 list blocks publishing
SDSGenerator::maskTradeSecretRow :2627, tradeSecretName :2607; section3 :1418-1441; section13 :3889-3903; section15 :4436-4452

Section 5 Specific Hazards (flammable sentence and 'Flash point:' line); flammable media and firefighter advice:

FORMULA / RAW MATERIALS Per-product override, with its 'Flash point:' sentence refreshed each time > the sentence for the sheet's Flammable Liquids category (H224-H227) followed by the Section 9 flash point. Extinguishing media and firefighter advice switch to the flammable wording on the same test.

On a sheet: Flammable liquid and vapor (Flammable Liquids, Category 3)... Flash point: 45.3 °C (113.5 °F).

You maintain: Flash and boiling points on raw materials (they set the category); Physical State

If missing: No flammable sentence and no 'Flash point:' line when the product is not a flammable liquid (including when there is no flash point data); a stale 'Flash point:' sentence in an edit is removed

SDSGenerator::section5 :1866-1987, refreshFlashPointSentence :1837;
section5.specific_hazards_flammable_cat1-4, flash_point_line; resolveFlashPointDisplay :1742

Section 6 Methods and Material for Containment and Cleaning Up:

FORMULA / RAW MATERIALS Per-product override > wording for the resolved physical state: Solid or Powder sweep, Gel or Paste scrape, Gas ventilate, anything else the liquid absorbent text.

On a sheet: Stop leak if safe to do so. Dam or contain spill. Absorb with inert absorbent material...

You maintain: Physical State on the finished good, or on its highest-weight raw material that has one

If missing: Liquid wording

SDSGenerator::section6 :2066-2080; text_overrides(6,'containment') >
section6.containment_solid|paste|gas|liquid; state via resolvePhysicalState :2521

Section 8 exposure-limit table, Conc. % column:

FORMULA / RAW MATERIALS The same band as the CAS row in Section 3 > the limit's own concentration when the CAS is not in the composition. Trade-secret rows are masked.

On a sheet: 10 - 30%

You maintain: As Section 3 (constituent %)

If missing: '<0.1%' for a component whose maximum is below 0.1%

SDSGenerator::section8 :2397-2414 formatConcentration(compByCas[cas]) else
exposure_limits[].concentration_pct; maskTradeSecretRow

Section 8 Engineering Controls (dust and explosion-proof sentences):

FORMULA / RAW MATERIALS Per-product override > the general sentence plus: the dust sentence when the resolved physical state is Powder; the explosion-proof sentence when the sheet carries H224-H226; and the eyewash sentence for H314/H318.

On a sheet: Adds the dust sentence for powders and explosion-proof ventilation for Flam. Liq. 1-3

You maintain: Physical State on the finished good or raw material; raw-material flash and boiling points

If missing: No dust sentence (Solid and blank states do not get it); no explosion-proof sentence without a flash point

SDSGenerator::section8 :2428, engineeringControls :2449-2466; section8.engineering_dust, engineering_flammable, engineering_corrosive

Physical State:	FORMULA / RAW MATERIALS The finished good's Physical State (the raw material's own on resale sheets) > the state of the highest-weight raw material that has one > 'Liquid'. The same value drives Sections 6, 8, 13 (D002) and 14, the flammability test and the ATE dust route. <i>On a sheet: Liquid</i> You maintain: Physical State on the finished good (Finished Goods form) or on its raw materials (Raw Materials form) If missing: 'Liquid', plus a preview warning that the default was used. A raw material with a blank state is skipped in favour of the next-largest one SDSGenerator::resolvePhysicalState :2521, physicalStateIsDefault :2535; FormulaCalcService::deriveFormulaProperties :604-623
Color:	FORMULA / RAW MATERIALS Only the finished good's Color dropdown. Resale raw-material sheets never show a color. <i>On a sheet: Black</i> You maintain: Color on the finished good (Finished Goods form) If missing: The line is omitted finished_goods.color (options from setting sds.color_options); SDSGenerator::section9 :2658; resale :477 reads raw_materials.color, which does not exist
Appearance:	FORMULA / RAW MATERIALS Per-product override > color plus state when the finished good has a color > the physical state in lower case. Resale sheets use the raw material's Appearance text before falling back to the state. <i>On a sheet: Black liquid</i> You maintain: Color on the finished good; Appearance on the raw material for resale sheets If missing: The lower-case physical state, e.g. 'liquid' SDSGenerator::section9 :2691-2713; text_overrides(9,'appearance') > section9.appearance_color_state > (resale) formula_props.appearance > section9.appearance_state
Odor:	FORMULA / RAW MATERIALS Per-product override > Odor of the raw material with the highest weight % > 'Not determined'. <i>On a sheet: Mild solvent</i> You maintain: Odor on the highest-weight raw material of each formula (Raw Materials form) If missing: 'Not determined' SDSGenerator::section9 :2719-2725; text_overrides(9,'odor') > formula_props.odor (raw_materials.odor of dominant RM) > labels.not_determined
Odor Threshold; pH; Melting Point/Freezing Point:	PRODUCT OVERRIDE Per-product Section 9 override only. The system holds no formula data for these properties. <i>On a sheet: Not determined</i> You maintain: Optionally a tested value in SDS > Edit Text, Section 9 (per language) If missing: 'Not determined' SDSGenerator::SECTION9_APPENDIX_D_FIELDS :2548, section9 :2740-2750; text_overrides(9,'odor_threshold' 'ph' 'melting_point')
Initial Boiling Point and Boiling Range:	FORMULA / RAW MATERIALS Per-product override, if it reads as a temperature and does not move the product between Flammable Liquids Category 1 and 2 > the lowest boiling point of any raw material in the formula, including sub-formulas, whatever its amount > 'Not determined'. <i>On a sheet: 78.4 °C (173.1 °F)</i> You maintain: Boiling Point (°C) on each volatile raw material (Raw Materials form) If missing: 'Not determined'. An edit that is not a temperature, or that contradicts the classification, is not printed (preview warning) SDSGenerator::resolveBoilingPointDisplay :1776, vetBoilingPointEdit :4118; TemperatureParser; FormulaCalcService :549-555 MIN(raw_materials.boiling_point_c)
Flash Point:	FORMULA / RAW MATERIALS Per-product override, if it reads as a temperature and gives the same flammable category and D001 outcome > the wt%-weighted average flash point of the raw materials that have one (recursive, rounded to 0.1 °C), prefixed '>' if any of them is flagged > 'Not determined'. Raw materials with no flash point are left out of the average. <i>On a sheet: 45.3 °C (113.5 °F)</i> You maintain: Flash Point (°C) and the '>' flag on every raw material, including a high value for water and other non-flammable raws (Raw Materials form) If missing: 'Not determined', and the product is treated as not flammable everywhere SDSGenerator::resolveFlashPointDisplay :1742, vetFlashPointEdit :4076; FormulaCalcService::weightedFlashPoint :690; raw_materials.flash_point_c, flash_point_greater_than

Evaporation Rate:	PRODUCT OVERRIDE Per-product Section 9 override only. There is no formula data for this property. <i>On a sheet: Not determined</i> You maintain: Optionally a tested value in SDS > Edit Text, Section 9 (per language) If missing: 'Not determined' SDSGenerator::SECTION9_APPENDIX_D_FIELDS :2548, section9 :2740-2750; text_overrides(9,'evaporation_rate')
Flammability (Solid, Gas):	PRODUCT OVERRIDE Per-product Section 9 override > 'Not applicable' when the resolved physical state is Liquid > 'Not determined'. <i>On a sheet: Not applicable</i> You maintain: Physical State; optionally a value in SDS > Edit Text, Section 9 If missing: 'Not determined' for every state other than Liquid (Paste and Powder included) SDSGenerator::section9 :2740-2750 (flammability_solid_gas; labels.not_applicable when state = Liquid)
Upper/Lower Flammability or Explosive Limits; Vapor Pressure; Vapor Density:	PRODUCT OVERRIDE Per-product Section 9 override only. There is no formula data for these properties. <i>On a sheet: Not determined</i> You maintain: Optionally a tested value in SDS > Edit Text, Section 9 (per language) If missing: 'Not determined' SDSGenerator::SECTION9_APPENDIX_D_FIELDS :2548, section9 :2740-2750; text_overrides(9,'flammability_limits' 'vapor_pressure' 'vapor_density')
Specific Gravity:	FORMULA / RAW MATERIALS A volume-additive blend of each raw material's specific gravity (total % divided by the sum of % / SG), rounded to 3 decimals. <i>On a sheet: 1.053</i> You maintain: Specific Gravity on every raw material (Raw Materials form) If missing: Each raw material with no value counts as 1.000, and a preview warning names it. The line never prints 'Not determined' VOCCalculator::getMixtureSG :248-281, applyDefaults :612-622 (else 1.0); FormulaCalcService::dataGapWarnings :658; SDSGenerator::section9 :2768
Solubility:	FORMULA / RAW MATERIALS Per-product override > a band from the weighted share of raw materials marked soluble (1), partially soluble (0.5) or negligible (0.03), among the raws that have any value: 90% or more soluble, 5% partial, 1% negligible, else not soluble. <i>On a sheet: Partially soluble in water</i> You maintain: Solubility on each raw material (Raw Materials form). Water, glycols and other water-soluble raws must read 'Soluble in water' If missing: 'Not determined' when no raw material has a value FormulaCalcService SOLUBILITY_WEIGHTS :30-35, thresholds :38-40, solubilityKeyForFraction :636; SDSGenerator::solubilityText :2559
Partition Coefficient: n-Octanol/Water; Auto-Ignition Temperature; Decomposition Temperature; Viscosity:	PRODUCT OVERRIDE Per-product Section 9 override only. There is no formula data for these properties. <i>On a sheet: Not determined</i> You maintain: Optionally a tested value in SDS > Edit Text, Section 9 (per language) If missing: 'Not determined' SDSGenerator::SECTION9_APPENDIX_D_FIELDS :2548, section9 :2740-2750; text_overrides(9,'partition_coefficient' 'auto_ignition_temp' 'decomposition_temp' 'viscosity')
VOC (lb/gal) (EPA Method 24):	FORMULA / RAW MATERIALS The VOC weight fraction (each raw's VOC % times its line %, divided by the formula's total line %) times the mixture specific gravity times 8.345. Exempt compounds and water are not subtracted. <i>On a sheet: 2.15</i> You maintain: VOC wt% (or the 'less than 1%' flag) and Specific Gravity on every raw material If missing: A missing VOC counts as 0 and a missing SG as 1.0. A preview warning names each raw material; nothing is marked on the sheet VOCCalculator::getVOCLbPerGal :289, getTotalVOCWeightPercent :181 (÷ totalLinePct :555); VOC_i = 0.99 if voc_less_than_one else voc_wt else 0
VOC (wt%):	FORMULA / RAW MATERIALS '<1' when every raw material is flagged less than 1% VOC > otherwise each raw's VOC % times its line %, divided by the formula's total line %. <i>On a sheet: 25.71 (or '<1')</i> You maintain: VOC wt% or the 'less than 1%' flag on every raw material If missing: A missing VOC counts as 0, with a preview warning SDSGenerator::section9 :2675-2679, :2775; FormulaCalcService all_voc_less_than_one; VOCCalculator::getTotalVOCWeightPercent :181

Solids (wt%):	FORMULA / RAW MATERIALS Each raw's solids % times its line %, divided by the formula's total line %. A raw with no solids % is estimated as 100 minus its VOC, exempt VOC and water. <i>On a sheet:</i> 74.3 You maintain: Solids wt% (or VOC, exempt VOC and water wt%) on each raw material; the Exempt VOC list (admin) If missing: Estimated. A raw material with no VOC or solids data counts as 100% solids VOC Calculator::getSolidsWeightPercent :390, applyDefaults :706-720; raw_materials.solids_wt else 100-voc-exempt-water; exempt_voc_list (FormulaCalcService :455-468)
Likely Routes of Exposure (Section 11):	HAZARD ENGINE Skin and eye contact always print. Inhalation is added for H330-H332 or H334-H336, or when any component has an exposure limit; Ingestion is added for H300-H302, H304 or H305. <i>On a sheet:</i> Inhalation, Skin Contact, Eye Contact, Ingestion. You maintain: Classifications per CAS (CAS determinations / CPD); exposure limits If missing: 'Skin Contact, Eye Contact.' SDSGenerator::deriveRoutesOfExposure :3221; labels.inhalation, skin_contact, eye_contact, ingestion
Most Important Symptoms/Effects, Acute and Delayed (Section 11):	HAZARD ENGINE Section 4 symptoms override > the sheet's H3xx statements, split into acute and delayed (H317, H334, H340-H362, H372, H373). The same text prints in Section 4. <i>On a sheet:</i> Acute: Causes skin irritation. Delayed: Suspected of causing cancer. You maintain: Classifications per CAS (CAS determinations / CPD) If missing: 'No significant symptoms or effects are known or expected under normal conditions of use.' SDSGenerator::section11 :3467, deriveSymptoms :1674, DELAYED_EFFECT_H_PREFIXES :98; text_overrides(4, 'symptoms')
Acute Toxicity (oral, dermal, inhalation lines):	HAZARD ENGINE Per-product override > for each route, the most severe Category 1-4 from the engine's classes or H-codes (cut-off or ATEmix), with its H-statement. <i>On a sheet:</i> Acute toxicity (oral): Category 4 — Harmful if swallowed (H302). You maintain: Acute toxicity categories per CAS (CAS determinations / CPD) If missing: 'Acute toxicity (oral): Not classified based on available data.' SDSGenerator::buildAcuteToxicity :3037-3213; text_overrides(11, 'acute_toxicity'); section11.acute_route_line(_no_category); h_statements
ATEmix value or 'Classified based on ingredient concentration.' (after each acute toxicity line):	HAZARD ENGINE The ATEmix (GHS additivity from each component's LD50/LC50, else its category's default point estimate) prints only when it produced the printed category. Otherwise the line ends 'Classified based on ingredient concentration.', unless the route came only from the finished-good override or a bare H-code. <i>On a sheet:</i> ATEmix = 1255 mg/kg bw. You maintain: Acute toxicity categories per CAS, and optionally the oral/dermal/inhalation LD50/LC50 values (CAS determinations). Solid and Powder products use the dust/mist route If missing: Category default ATEs are used silently. Nothing follows an override-only or H-code-only route HazardEngine::applyATECalculation :2303, CATEGORY_DEFAULT_ATES :207; SDSGenerator::buildAcuteToxicity :3138-3209, formatAte :3243
Section 11 Carcinogenicity:	FORMULA / RAW MATERIALS Per-product override > one line per component at 0.1% or more with an IARC 1/2A/2B, NTP or OSHA listing, with its band (plus an inhalable-dust note for bound carbon black or TiO2) > the 'none listed' sentence. IARC Group 3 is not reported, and listings of an ingredient supplied only by Solid or Powder raws are dropped when the formula has another state. <i>On a sheet:</i> No components present at or above 0.1% are listed as carcinogens by IARC, NTP, or OSHA. You maintain: Carcinogen list (admin); constituent %; raw material physical state (carbon black / TiO2) If missing: The 'none listed at or above 0.1%' sentence SDSGenerator::section11 :3431-3463; CarcinogenService::buildSummaryText :285, isReportableListing, LISTING_THRESHOLD_PCT 0.1; carcinogen_list; solid-in-liquid filter :5181
Section 11 component heading (name, CAS, range):	FORMULA / RAW MATERIALS Printed for each component at 0.1% or more that has an exposure limit or a reportable carcinogen listing, with the same band as Section 3. Trade-secret components print their description and 'TRADE SECRET'. <i>On a sheet:</i> Toluene (CAS 108-88-3) — 10 - 30% You maintain: As Section 3; exposure limits and the carcinogen list If missing: No band on legacy snapshots SDSGenerator::section11 :3379-3428; formatConcentration(compositionByCas); maskTradeSecretRow; PDFService :945-953

Section 12 component aquatic table (Conc. %, Aquatic Acute, Aquatic Chronic with M-factor):	HAZARD ENGINE Every component listed in Section 3 that has an aquatic classification, with its Section 3 band and, for Category 1, its M-factor; trade-secret identities are masked. The table is dropped under a replace-mode finished-good override. <i>On a sheet: Zinc oxide 1314-13-2 1 - 5% Category 1 (M = 10) Category 1 (M = 1)</i> You maintain: Aquatic categories and M-factors per CAS (CAS determinations / CPD) If missing: An empty cell for no class. A missing M-factor prints '(M = 1, default)' HazardEngine aquatic_components, resolveMFactor :1410; SDSGenerator::section12 :3509-3538, section3ListedCas :3705, formatAquaticCategory :3674
Persistence and Degradability / Bioaccumulative Potential:	REGULATORY LIST Per-product override > a line naming every SARA 313 PBT chemical present at any concentration, by its EPA list name > 'No data available.' Bioaccumulation then reads 'See Persistence and Degradability above.' <i>On a sheet: Contains component(s) identified as ... (PBT) under SARA 313 (40 CFR 372.28): Lead (CAS 7439-92-1).</i> You maintain: SARA 313 list with the PBT flag (admin) If missing: 'No data available.' on both lines SDSGenerator::section12 :3610-3642; SARA313Service reportable + below_threshold (sara313_list.is_pbt); section12.pbt_components, pbt_see_persistence
Section 13 RCRA ignitable (D001):	FORMULA / RAW MATERIALS Printed when the sheet carries H224-H226. The reason is the formula flash point when below 60 °C (never the Section 9 edit), else a hazard-based reason for an override class; oxidizer, pyrophoric, flammable gas and flammable solid H-codes add their own reasons. <i>On a sheet: ignitable (D001) — flash point below 60 °C (140 °F)</i> You maintain: Raw material flash and boiling points; Physical State; constituent classifications If missing: No D001 when the product has no flash point, is not a liquid, or is at exactly 60 °C, unless a finished-good hazard override sets Flammable Liquids 1-3 SDSGenerator::section13 :3806-3830, d001FlashReasonKey :4052, flammabilityOf :4025; section13.rcra_reason_flash_point(_gt)
Section 13 RCRA corrosive (D002) / reactive (D003):	HAZARD ENGINE D002 for H314 and/or H290, never on a Solid, Powder or Paste product. D003 from water-reactive, explosive, self-reactive or organic-peroxide H-codes. <i>On a sheet: corrosive (D002) — classified as corrosive to skin (H314)</i> You maintain: Constituent classifications (CAS determinations); Physical State If missing: Omitted SDSGenerator::section13 :3835-3862, rcraIsNonLiquidState :3970; H314/H290 -> D002; H260/H261, H200-H205, H240-H242 -> D003
Section 13 RCRA toxicity-characteristic and listed-waste components:	REGULATORY LIST Composition CAS numbers are matched to the admin RCRA list, and metal compounds to their element's D004-D011 codes. D codes print at any amount in the 'as sold' sentence; F/K/P/U codes print at 0.1% or more in a separate reference sentence. <i>On a sheet: contains Lead (D008 if the ... regulatory level of 5 mg/L (TCLP) is exceeded)</i> You maintain: RCRA waste code list (Admin > RCRA); TC metals per CAS (CAS determinations > TC metals) If missing: The 'No RCRA hazardous waste characteristic (40 CFR 261 Subpart C) has been identified...' sentence RCRAService::match :131-200 (rcra_waste_codes, cas_master.tc_metals), MIN_CONCENTRATION_PCT 0.1; SDSGenerator::section13 :3866-3925
UN Number / UN Proper Shipping Name:	TRANSPORT RULES Per-field override > classifier: Class 3 from H224-H226 on the sheet, 8 from H314 (or H290 on a liquid), 6.1 from acute Cat 1-3, 9 from H400/H410/H411, ranked by 49 CFR 173.2a. Class 3 alone uses UN1210/UN1263 (UN1993 for wash/cleaner types) by Transport Product Type, else description keywords; other entries are n.o.s. with up to two constituent names. <i>On a sheet: UN1210 / Printing ink, flammable; or 'Flammable liquids, n.o.s. (Toluene, Xylene)'</i> You maintain: Raw material flash and boiling points; constituent classifications; Transport Product Type on the finished good; Physical State If missing: 'Not regulated' when no class applies, including when there is no flash point. 'Not determined' (publishing blocked) for gas, aerosol, oxidizer-type codes, Gas state, or Class 3+8+6.1 where 8 or 6.1 outranks 3 SDSGenerator::section14 :4271-4390; TransportClassifier::classify :182-404, resolveProductType :414; override completeness :4332-4341
Transport Hazard Class(es):	TRANSPORT RULES Per-product override > primary class plus subsidiaries ranked by the 49 CFR 173.2a precedence table. <i>On a sheet: 3 (6.1, 8)</i> You maintain: As UN Number If missing: 'Not regulated' / 'Not determined' SDSGenerator::section14 :4359; text_overrides(14,'hazard_class') > TransportClassifier::classify :291-375

Packing Group:	TRANSPORT RULES Per-product override > the most stringent of: Class 3 PG I at IBP 35 °C or less, II for H224/H225, III for H226; Class 8 from the skin corrosion sub-category (H290 alone III); 6.1 from the acute category; Class 9 III. <i>On a sheet: //</i> You maintain: Raw material boiling and flash points; constituent classifications If missing: 'Not applicable' when not regulated; 'Not determined'. Class 8 with no sub-category defaults to II TransportClassifier::classify :258-287, corrosivePackingGroup :502, toxicPackingGroup :526; IBP from the engine flammability block (Section 9 edit not read)
Environmental Hazards (Marine pollutant):	TRANSPORT RULES Per-product override > 'Marine pollutant: Yes' when the sheet carries H400, H410 or H411, otherwise 'Marine pollutant: No'. <i>On a sheet: Marine pollutant: Yes</i> You maintain: Aquatic classifications per CAS If missing: 'Marine pollutant: No'; 'Not determined' while the classification is not determined and there is no override SDSGenerator::section14 :4302-4318, :4361; text_overrides(14,'environmental_hazards') > section14.marine_pollutant_yes _no
Transport in Bulk:	FIXED TEXT (TRANSLATED) A fixed sentence on every sheet; it cannot be overridden. <i>On a sheet: Not intended for transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code.</i> You maintain: Nothing If missing: Always printed (older snapshots without it skip the line) SDSGenerator::section14 :4363; section14.transport_in_bulk_text (en.php:456)
Special Precautions for User:	FIXED TEXT (TRANSLATED) A fixed sentence on every sheet; it cannot be overridden. <i>On a sheet: Read this safety data sheet before handling. Keep containers closed, upright and secured...</i> You maintain: Nothing If missing: Always printed (older snapshots without it skip the line) SDSGenerator::section14 :4364; section14.special_precautions_text (en.php:457)
Transport notes:	TRANSPORT RULES A combustible note for a non-regulated liquid with H227 and a flash point below 93 °C (or none), and a viscous-liquid note for Class 3 PG II with no subsidiary, then the fixed carrier sentence. Under a Section 14 override only the carrier sentence prints. <i>On a sheet: Combustible liquid; regulated in bulk packaging (> 450 L) only. ...verify with the carrier...</i> You maintain: Raw material flash points; Physical State If missing: Only the carrier sentence SDSGenerator::section14 :4347-4353; TransportClassifier :249 (combustible), :401 (viscous); section14.note_combustible, note_viscous, section14.note
SARA 313 reportable components:	REGULATORY LIST Composition CAS numbers on the SARA 313 list, or members of a TRI category (metal compounds, certain glycol ethers), at or above their de minimis; PBT chemicals and TRI PFAS are reported at any concentration. Each prints with the EPA list name, the Section 3 band, and its threshold or 'no de minimis exemption'. <i>On a sheet: • Ethylene glycol (CAS 107-21-1) — 1 - 5% (de minimis threshold: 1%)</i> You maintain: SARA 313 list with PBT and special-concern flags, and TRI categories (admin); constituent % If missing: The 'does not contain any toxic chemicals subject to ... Section 313' sentence SARA313Service::evaluate :77-140 (sara313_list de minimis_pct, is_pbt, is_special_concern; RegulatoryCategoryService); SDSGenerator::section15 :4431; PDFService :1107-1135
SARA 313 range note:	FIXED TEXT (TRANSLATED) A fixed sentence printed under the list whenever at least one SARA 313 chemical is reportable. <i>On a sheet: Concentrations are stated as the prescribed concentration ranges...; the upper end ... is the maximum</i> You maintain: Nothing If missing: Not printed when nothing is reportable labels.sara_313_range_note (en.php:724); PDFService :1129-1131
Hazardous Air Pollutants (rows and Total HAP Content):	REGULATORY LIST Composition CAS numbers on the HAP list or in a 112(b) category (metal compounds, glycol ethers) at 0.01% and up, plus HAPs entered by hand on raw materials, each with a band. The total is the summed % shown as a band. <i>On a sheet: Toluene 10 - 30% ... Total HAP Content: 10 - 30%</i> You maintain: HAP list and 112(b) categories (admin); manual HAP entries on raw materials If missing: 'This product does not contain any EPA HAPs listed under Clean Air Act Section 112(b).' HAPService::evaluate :102-160 (hap_list, RegulatoryCategoryService) + SDSGenerator::getManualHaps :5662 (raw_materials.haps_data); section15 :4431-4437

Preview Warnings box (operator only; never printed on the SDS): **FORMULA / RAW MATERIALS** Raised by missing or conflicting data: blank specific gravity or VOC, no physical state anywhere, flash point problems (edit dropped, IBP assumed, override disagrees, flammable ingredient on a raw with no flash point), non-temperature Section 9 edits, n.o.s. without technical names, and a Substance with no CAS. None of these blocks publishing.
On a sheet: Raw material R-123: no specific gravity entered; ... count it as 1.0.
You maintain: The raw-material or SDS editor field each warning names
If missing: No box when nothing is missing
 FormulaCalcService::dataGapWarnings :658; SDSGenerator::generate :127-341, flammabilityWarnings :4152, temperatureOverrideWarnings :1817

PRODUCT FAMILIES AND UV PRODUCTS

In this scope, text comes from the product family resolved for each item: its UV/LED flag, its active flag and its per-language Section 1 text. A product gets its family from a manual pick, a rule, or the content vote, where all UV/LED families count together. The UV/LED flag combines with the UV Acrylate Rule Pack switch, which drives the UV sentences folded into Sections 2, 4, 5, 6, 7 and 8 and the Section 11 note. The flag alone drives the Section 7 storage sentence and the Section 10 UV item. Acrylate detection now only supplies the names in the Section 4 sentence. Resale raw-material sheets ignore the family for Section 1. The family name no longer affects anything printed.

Recommended Use (Section 1, finished good / intermediate / alias / private-label sheet): **PRODUCT OVERRIDE** Per-product, per-language SDS editor text > the product's own Recommended Use field > the family's text in the sheet language > canned translation sentence. Inactive families are ignored, and a blank language never falls back to the family's English text.
On a sheet: Recommended Use: UV-curable offset ink for packaging.
You maintain: Family text in every language on an active family (Settings > Product Families), and the product must resolve to that family. Keep the product's Recommended Use field blank, because it has no language. Put per-language product wording in the SDS editor.
If missing: "Printing ink for commercial and industrial applications." (translated on ES/FR/DE). If the family has English text only, ES/FR/DE sheets print their canned translated sentence.
 SDSGenerator::resolveUseText (:1134): text_overrides(1,'recommended_use',fg,lang) > finished_goods.recommended_use > family_defaults[lang] (familyFields, active only) > section1.recommended_use

Recommended Use (Section 1, resale raw-material sheet): **FIXED TEXT (TRANSLATED)** Per-raw-material, per-language resale SDS editor text > canned resale sentence. The raw material's family is not used.
On a sheet: Recommended Use: Raw material for industrial formulation.
You maintain: Nothing normally. For product-specific wording, use the resale SDS editor (/sds/resale/{rm_id}/edit).
If missing: "Raw material for industrial formulation." (translated per language)
 resolveUseText on resale row (is_resale, recommended_use null; family tier skipped, Q15):
 text_overrides(raw_material_id,1,'recommended_use') via getOverrides :4628 > section1.recommended_use_resale

Restrictions on Use (Section 1, finished good / intermediate / alias / private-label sheet): **PRODUCT OVERRIDE** Per-product, per-language SDS editor text > the product's own Restrictions field > the family's restriction text in the sheet language > canned translation sentence. There is no English family fallback, and inactive families are ignored.
On a sheet: Restrictions on Use: For professional/industrial use only. Not for household consumer use.
You maintain: Family restriction text in every language (Settings > Product Families). Keep the product's Restrictions field blank: it has no language and masks the family text.
If missing: "For professional/industrial use only. Not for household consumer use." (translated per language)
 resolveUseText(field='restrictions'): text_overrides(1,'restrictions') > finished_goods.restrictions_on_use > family_defaults['restrictions'][lang] > section1.restrictions

Restrictions on Use (Section 1, resale raw-material sheet): **FIXED TEXT (TRANSLATED)** Per-raw-material, per-language resale SDS editor text > canned resale restriction sentence. The family is not used.
On a sheet: Restrictions on Use: For professional/industrial use only. Not for household consumer use.
You maintain: Nothing normally; the resale SDS editor is used for exceptions.
If missing: "For professional/industrial use only. Not for household consumer use." (resale key, translated)
 resolveUseText resale branch: text_overrides(raw_material_id,1,'restrictions') > section1.restrictions_resale (SDSGenerator.php:1145-1153)

Recommended Personal Protective Equipment (PPE): Respiratory / Hand Protection / Eye Protection / Skin/Body Protection (Section 2): **TEXT CHOSEN BY HAZARD** Only the fields whose tier is hazard-driven print, with the same text as Section 8: Section 8 override > hazard-tier sentence. On UV rule-pack products the field's UV sentence follows the tier sentence (never an override).
On a sheet: Hand Protection: Chemical-resistant gloves (nitrile...). For uncured acrylate product, wear...
You maintain: Nothing in this scope beyond the family UV/LED flag and the pack switch (the tiers come from H-codes)
If missing: Unclassified product: no PPE block in Section 2. Pack off or non-UV family: the tier sentence prints alone.
 SDSGenerator::section2 (:1290) → resolvePPE(\$uvPack) (:2483): text_overrides(8,field) > section8.ppe.<field>.<tier> + section8.uv_<field>

Type (Section 3: Substance / Mixture):	FORMULA / RAW MATERIALS Product's Substance/Mixture setting > Auto: 'Substance' only if the formula has exactly one line and it is a raw material marked Substance > 'Mixture'. Resale sheets use the raw material's setting; the product family has no influence. <i>On a sheet: Type: Mixture</i> You maintain: Mark single-substance raw materials as Substance (raw-material form). Set the finished-good setting only when the automatic rule is wrong. If missing: "Mixture" SubstanceMixtureResolver::resolveForFinishedGood / resolveForRawMaterial; labels.type / labels.substance / labels.mixture (SDSGenerator.php:1483-1485)
Skin Contact (Section 4, UV sentence):	PRODUCT FAMILY Added after the hazard-driven skin text when the rule pack is on and the product's active family is flagged UV/LED. Names acrylates at 0.1 % or more (14-CAS list or acrylate name), using the trade-secret description or 'Trade Secret' for secret ones. <i>On a sheet: ... UV/EB-curable product containing acrylate monomers/oligomers (HDDA): remove uncured ...</i> You maintain: UV/LED flag on the family (Settings > Product Families); pack switch (Admin Settings); constituent CAS numbers and names (raw-material form, CAS master); trade-secret descriptions on secret acrylates If missing: Pack off or family not UV: no UV sentence. No acrylate recognised: the sentence prints without the names. A Section 4 Skin override replaces the whole field. section4() :1580 + UVAcrylateRulePack::section4SkinFragment (section4.uv_skin / uv_skin_names); detectAcrylates :128; names COALESCE(prop65 name, cas_master, rmc)
Specific Hazards (Section 5, UV sentence):	PRODUCT FAMILY The same gate (pack on and a UV/LED family) adds a fixed translated sentence after the combustion sentence. Acrylate detection plays no part. <i>On a sheet: ... uncured UV/EB-curable product may polymerize uncontrollably, releasing heat ...</i> You maintain: As the Section 4 UV sentence (family flag + pack switch) If missing: Omitted. A Specific Hazards override replaces the whole field. section5() :1954 appends section5.uv_specific_hazards unless text_overrides(5,'specific_hazards')
Containment and Cleanup (Section 6, UV sentence):	PRODUCT FAMILY Same gate. A fixed translated sentence follows the physical-state containment sentence. <i>On a sheet: ... Keep collected uncured product in closed containers away from UV light ...</i> You maintain: As the Section 4 UV sentence If missing: Omitted. A Containment override replaces the whole field. section6() :2081 appends section6.uv_containment unless text_overrides(6,'containment')
Handling (Section 7, UV sentence):	PRODUCT FAMILY Same gate. Fixed translated sentence appended after the hazard-driven handling fragments. <i>On a sheet: ... Avoid prolonged or repeated skin contact with uncured product.</i> You maintain: As the Section 4 UV sentence If missing: Omitted. A Handling override replaces the whole field. section7() :2321 appends section7.uv_handling unless text_overrides(7,'handling')
Storage (Section 7, UV light sentence):	PRODUCT FAMILY Added, before the incompatible-materials sentence, whenever the product's active family is flagged UV/LED. Does not depend on the pack switch (same condition as the Section 10 UV item). <i>On a sheet: ... Store away from UV light sources and direct sunlight to prevent premature polymerization.</i> You maintain: UV/LED flag on the family (Settings > Product Families); product resolved to that family If missing: Omitted; the rest of the storage text prints section7() :2367 \$this->uvFamily (UVAcrylateRulePack::familyIsUv) → section7.uv_storage; skipped if text_overrides(7,'storage')
Respiratory Protection (Section 8, UV sentence):	PRODUCT FAMILY The hazard-tier sentence, plus the UV respirator sentence when the pack gate fires and the tier is hazard-driven or the mixture carries H317. A per-product override replaces the whole field. <i>On a sheet: ... Use local exhaust ventilation when handling uncured product; if mist or vapor is ...</i> You maintain: As the Section 4 UV sentence If missing: Only the tier sentence prints (an unclassified, non-sensitizing UV product keeps its 'No special ... required' sentence) resolvePPE (:2483-2504): section8.uv_respiratory appended if \$uvPack && (tier not in PPE_BASELINE_TIERS H317); text_overrides(8,'respiratory') wins
Hand Protection (Section 8, UV sentence):	PRODUCT FAMILY Tier sentence + UV glove sentence when the gate fires and the tier is hazard-driven or H317 is present; an override replaces the field. <i>On a sheet: ... For uncured acrylate product, wear chemical-resistant gloves (nitrile, minimum 0.4 mm...)</i> You maintain: As the Section 4 UV sentence If missing: Only the tier sentence prints resolvePPE: section8.uv_hand_protection under the same condition; text_overrides(8,'hand_protection') wins

Eye Protection (Section 8, UV sentence):	PRODUCT FAMILY Tier sentence + UV eye sentence when the gate fires and the tier is hazard-driven or H317 is present; an override replaces the field. <i>On a sheet: ... Chemical splash goggles or a face shield are required when handling uncured acrylate ...</i> You maintain: As the Section 4 UV sentence If missing: Only the tier sentence prints <code>resolvePPE: section8.uv_eye_protection</code> under the same condition; <code>text_overrides(8,'eye_protection')</code> wins
Skin Protection (Section 8, UV sentence):	PRODUCT FAMILY Tier sentence + UV skin sentence when the gate fires and the tier is hazard-driven or H317 is present; an override replaces the field. <i>On a sheet: ... Avoid prolonged or repeated skin contact with uncured product. A barrier cream is ...</i> You maintain: As the Section 4 UV sentence If missing: Only the tier sentence prints <code>resolvePPE: section8.uv_skin_protection</code> under the same condition; <code>text_overrides(8,'skin_protection')</code> wins
Conditions to Avoid (Section 10, UV light item):	PRODUCT FAMILY Added whenever the product's active family is flagged UV/LED; the pack switch and acrylate detection play no part. A per-product Conditions to Avoid override replaces the whole list. <i>On a sheet: Conditions to Avoid: Excessive heat, ..., exposure to UV light and direct sunlight (...)</i> You maintain: UV/LED flag on an active family (Settings > Product Families); product resolved to that family If missing: UV item omitted; the rest of the list (heat / ignition / oxidizers ...) prints <code>section10(\$isUv = UVAcrylateRulePack::familyIsUv) :2882</code> adds 'uv' → <code>section10.cond_uv</code> in <code>COND_ORDER</code>; skipped if <code>text_overrides(10,'conditions_avoid')</code>
UV Acrylate Information (Section 11, after the component table):	PRODUCT FAMILY Prints when the pack is on and the product's active family is flagged UV/LED, even if no acrylate is recognised. Sensitizer wording with H317, unclassified wording otherwise; a per-product override replaces it. <i>On a sheet: UV Acrylate Information: Contains acrylate monomers that are known skin sensitizers...</i> You maintain: As the Section 4 UV sentence; per-product wording in the SDS editor (Section 11) If missing: Note and label omitted. An override cannot add the note when the gate does not fire. <code>SDSGenerator :216/:262 section11Note</code> → <code>section11.uv_acrylate_note(unclassified)</code>; <code>text_overrides(11,'uv_acrylate_note')</code>; <code>PDFService :989</code>, preview :421
UV Acrylate Information (Sections 4-7, stored private-label snapshots):	FIXED TEXT The private-label preview renders the stored snapshot the customer received. Snapshots from before batch E hold separate UV notes in Sections 4-7 (English-only before batch C), which the renderers still print. <i>On a sheet: UV Acrylate Information: (old note text stored in the snapshot)</i> You maintain: Republish the private-label variant to replace the snapshot If missing: Nothing prints: new sheets never carry the key in Sections 4-7, and alias sends and reports now use the alias's own published PDF <code>PrivateLabelController::preview :958 snapshot_json</code> → <code>PDFService FIELD_LABEL_MAP / preview genericLabelMap 'uv_acrylate_note'</code> (Sections 4-7)
Abbreviations (Section 16, UV-related entries):	FIXED TEXT (TRANSLATED) A master-table entry prints when its term appears in printed text, including the UV sentences in Sections 4-8, the Section 11 note, the Section 10 UV item and Section 1 family text. Alias and private-label sheets are re-filtered against what they print. <i>On a sheet: UV: Ultraviolet (energy-curable); EB: Electron Beam (energy-curable); NIOSH: ...</i> You maintain: Nothing If missing: Entry omitted <code>AbbreviationService::build</code> over printed corpus (re-run in <code>createAliasVariant :835</code> and the private-label variant :977); <code>section16.abbreviation_table</code> (en.php: 'UV', 'EB', 'LED', NIOSH, CFR)

DATA YOU MUST KEEP POPULATED			
Grouped by the screen where the information is entered. Blank entries do not stop most sheets from publishing; they change what prints, as shown in the last column.			
Where	Keep populated	Feeds	If left blank
Raw materials	Constituent CAS numbers, chemical names and percentages (exact, or min and max) on every raw material	Section 2 classification and every hazard-driven sentence (Sections 4–15); Section 3 rows and bands; Sections 8, 11–13 and 15 list matches	No CAS or no % (counts as 0%): ingredient dropped, hazards under-reported, composition warning. No CAS also prints TSCA 'not verified'. A blank name prints blank unless a list name exists.
Raw materials	Hazard data for hidden ingredients: manual hazard entry on hazardous raw materials with no CAS; H-codes on trade-secret constituents	Section 2 classification and hazard sentences in Sections 4–7 and 10–12 (counted at the raw material's real share)	Those hazards are not counted. At or above the missing-data threshold with no manual hazard data saved, Publishing is blocked.

Where	Keep populated	Feeds	If left blank
Raw materials	Flash point (°C) with the '>' tick on every raw material (high value for water and similar), and initial boiling point on solvents	Product flash point and Flammable Liquids category: Sections 2, 5, 7, 8, 9, 10, 13 (D001) and 14 (Class 3, Packing Group)	Raw left out of the average. None on any raw: 'Not determined', treated as not flammable. No boiling point with flash below 23 °C: Category 2 / PG II assumed, warning.
Raw materials	Physical state on every raw material	Product state when the finished good has none; carbon black/TiO2 handling in Sections 2, 3, 8, 11 and 15; resale sheet state	Counts as not solid/powder, so carbon black/TiO2 are treated as bound (dropped from classification and the Prop 65 warning). No state anywhere: 'Liquid' with a preview warning.
Raw materials	Review and save each raw material after changes (any user save marks it reviewed)	Bulk and scheduled publish eligibility; readiness page verdict; resale Publish button	Unreviewed: the finished good, its aliases and its private-label items are left out of bulk publish with no reason given.
Raw materials	Trade-secret tick and description on constituents; manual Prop 65 and HAP entries	Masked name/CAS in Sections 3, 4 (UV names), 8, 11–13 and 15; Section 15 Prop 65 warning and HAP list	Blank description prints 'Trade Secret'; unticked prints the real identity. Ticked CAS on the Prop 65 list: Publishing is blocked. Without manual entries, only list matches print.
Raw materials	Specific gravity, VOC % (or '<1% VOC'), solids/exempt/water %, solubility, odor	Section 9 Specific Gravity, VOC (lb/gal and wt%), Solids (wt%), Solubility, Odor	SG 1.0 and VOC 0%, each with a preview warning naming the raw; solids back-calculated with no warning; 'Not determined' for solubility and odor
Raw materials	Resale items: supplier product name, appearance, and Substance/Mixture (also used on single-raw-material formulas)	Section 1 Product Identifier; Section 14 UN number keyword rule; Section 9 Appearance; Section 3 Type	Base code alone prints and shipping usually defaults to Printing ink (UN1210); appearance prints the state, e.g. 'liquid'; Type 'Mixture'
CAS determinations	Competent-person determination for every CAS without federal hazard data: classes, H/P codes, pictograms, signal word, LD50/LC50, M-factors, exposure limits	Section 2 and every hazard-driven sentence in Sections 3–15, including Section 8 limits, Section 11 acute toxicity, Section 12 aquatic	Publishing is blocked at or above the missing-data threshold (default 1%). Below it the CAS adds nothing, so hazards can be under-reported.
CAS determinations	Classification detail: skin-corrosion sub-category, acute-toxicity route, aquatic M-factors, and no stray physical-hazard codes (explosive, gas, aerosol, oxidizer, peroxide)	Section 14 class, packing group and marine pollutant; Section 12 aquatic; Sections 5, 7, 10, 13	Missing M-factor prints '(M = 1, default)'; no sub-category gives PG II. A stray physical code makes Section 14 'Not determined' and Publishing is blocked.
CAS determinations	CAS details: TSCA review status, preferred name, molecular formula, nitrogen/sulfur/halogen flags, RCRA metals (re-seed flags after adding CAS)	Section 15 TSCA; chemical names in Sections 3–15; Section 10 decomposition; Section 13 D004–D011; SARA 313, HAP and PBT metal matches	TSCA 'not verified' sentence and warning; constituent name used; base CO/CO2 sentence only; metal compounds missed unless listed directly
Finished goods	Active status and a current formula with correct line percentages (including every finished good used as a component)	Every section; concentration bands, cut-offs and classification	Inactive or no current formula: Publishing is blocked ('No current formula found'). A component with no formula adds no ingredients (readiness warning only).
Finished goods	Hazard override mode (add / replace) and contents, with rationale; needs the override permission	Section 2 and every hazard-driven sentence in Sections 4–15; a Flammable Liquids class or H224–H227 replaces the flash-point category	Calculated classification is used. Contents without a mode are ignored; an override class without its H-code prints the code with no phrase.
Finished goods	Physical state, colour and Substance/Mixture	Section 9 State, Color, Appearance; Sections 6, 8, 11, 13, 14; flammability (Solid, Powder, Paste never flammable liquids); Section 3 Type	State of the largest raw material that has one, else 'Liquid' with a warning; Color line omitted; Type 'Mixture' unless a single Substance raw
Finished goods	Product code and description, and alias customer code and description (from the CMS import)	Section 1 Product Identifier, footer code, PDF title, file names; description keywords also pick the Section 14 entry	Blank description prints the code alone (e.g. 'BK1080'); blank code leaves the footer empty and names the file 'unnamed_file'

Where	Keep populated	Feeds	If left blank
Finished goods	Transport product type (Automatic, Printing ink, Ink related, Paint, Paint related, Flammable liquid n.o.s.)	Section 14 UN number and proper shipping name for Class 3 products (UN1210 / UN1263 / UN1993)	Keyword rule on the description (COATING, VARNISH, WASH, THINNER...); no keyword defaults to Printing ink (UN1210)
Product families	UV/LED flag and Active flag on each family	UV wording in Sections 2 and 8 PPE, 4 Skin, 5, 6, 7 Handling and Storage, 10 'UV light', 11 note	No UV text anywhere, even on acrylate-rich inks. An inactive family is ignored everywhere, even when picked manually.
Product families	Recommended Use and Restrictions on Use text per language (EN, ES, FR, DE); keep products' own use fields blank	Section 1 use text on product, alias and private-label sheets (not resale)	That language's standard sentence prints, never the English family text. A filled product field prints as typed in every language.
Product families	Membership rules, Recompute, 'Reset legacy manual picks to Auto', and each item's Auto/manual family pick	Which family each product and raw material gets, and so all family-driven text	No family: standard Section 1 sentence and no UV text. Old family prints until Recompute; legacy manual picks stay until reset.
Manufacturers	Manufacturer name and emergency phone	Private-label Section 1 Company and Emergency lines, PDF Author, file names	The form refuses blanks; an older blank record means Publishing is blocked for that manufacturer's private-label items
Manufacturers	Address, phone, email, website, logo and disclaimer per language	Private-label Section 1 supplier lines, page 1 logo, disclaimer after Section 16	Line omitted with no fallback to your company (blank phone warns); no logo; disclaimer falls back to Admin settings, then built-in text
Manufacturers	Private-label item custom code and description (optional)	Private-label Product Identifier, footer code, PDF title, file names (printed as typed)	Falls back to the shared alias code/description, then the product's. An alias no longer on the product: publishing refused.
Admin settings	Company emergency phone	Section 1 Emergency line on standard, resale and alias sheets	Publishing is blocked on every path (the form refuses a blank; bulk publishes private-label items only and marks the job failed)
Admin settings	Missing-hazard-data block and threshold % (0.01–100)	Publish gate for any CAS with no hazard data and no competent-person determination (not printed)	Block on at 1.0%; a blank or out-of-range value is refused on save and read as 1.0%
Admin settings	Company name, address, phone, email, website, logo and legal disclaimer per language	Section 1 supplier block, PDF Author, page 1 logo, disclaimer after Section 16	Line omitted (name/phone warn; Author 'SDS System'); never saved: placeholder 'Your Ink Company, Inc.'; no logo; built-in translated disclaimer
Admin settings	Inhalation-only CAS list	Carbon black/TiO2 handling: Section 2 classification, Section 3 rows, Section 8 limits, Section 11 dust note, Section 15 Prop 65	Built-in carbon black 1333-86-4 and titanium dioxide 13463-67-7
Admin settings	UV acrylate rule pack, Sections 12–15 footnote, Prop 65 trace %, time zone, pick lists, pictogram artwork	UV sentences (Sections 2, 4–8 Handling, 11); footnote after Section 15; '(trace)' marker; effective date; Section 9 dropdowns; pictograms	UV on, footnote shown, 0.1%, America/Chicago, built-in lists, shipped images
Regulatory lists	Federal hazard data and exposure limits (NIOSH, ACGIH TLV, OSHA PEL), kept current; republish after imports (bulk won't pick them up)	Section 2 classification; Section 3 listing; Section 8 limits table; Sections 11 and 12	CAS counts as missing: Publishing is blocked at or above threshold without a determination. No limits: 'No occupational exposure limits apply...'
Regulatory lists	Carcinogen list (no admin screen; one correct row per CAS and agency)	Section 2 carcinogenicity, H350/H351, GHS08; Section 3 rows; Section 4 symptoms; Section 11 carcinogenicity	'No components ... listed' sentence prints; IARC Group 3 is never reported
Regulatory lists	TSCA inventory, Prop 65, SARA 313 (PBT, special concern), HAP, SNUR, RCRA waste codes and exempt VOC lists	Section 15 entries; Section 12 PBT; Section 13 RCRA; Section 9 Solids; chemical names in Sections 3 and 12	'None' or negative sentences; no TSCA import: every product prints 'not verified' with a warning

Where	Keep populated	Feeds	If left blank
SDS editor	Section 14 overrides: UN number, shipping name, class, packing group (+ environmental hazards), in every language	Section 14 lines; a complete set lifts the 'Not determined' block	Derived value prints. An incomplete core set, or a needed override missing in any language: Publishing is blocked.
SDS editor	Per-language text edits (SDS > Edit; resale: Readiness Check > Edit SDS text); 'Reset to automatic' on frozen old edits; then republish	Edited field in Sections 1–15, replaced whole (UV text included); Section 4 symptoms to 11, 10 incompatibles to 7, 8 PPE to 2	Automatic text prints; edits don't carry across languages. Frozen old wording prints until reset; published PDFs keep publish-time text until republished.

PUBLISH GATES

Conditions checked when an SDS is published. A block stops that publish; a warning lets it through and tells the operator.

Condition	Effect	Applies to	How to clear it
Missing federal hazard data: an ingredient CAS at or above the Missing Data Threshold (default 1%) has no hazard data and no active competent-person determination	Blocks publishing	Every publish path: single product, resale, SDS Updates republish, private label, alias (including on-demand publishes for sends), Bulk Publish and the scheduled run. Not shown in previews.	Add hazard data or an active determination on CAS Determinations for each listed CAS, or raise the threshold or turn the block off in Admin > Settings.
Section 14 transport 'Not determined' (unsupported hazard class or Gas physical state), or an incomplete Section 14 override	Blocks publishing	Every publish path, checked in every language. The reason also shows in every preview's Warnings box. A missing flash point never triggers it.	Correct the raw-material hazard data or physical state, or enter UN number, shipping name, hazard class and packing group in every language; clear an incomplete override.
Company emergency phone blank	Blocks publishing	Single product, resale, alias (including sends and auto-send) and SDS Updates republish. Bulk/scheduled runs publish only private-label items. Private-label publishing is not checked.	Enter the Emergency Phone in Admin > Settings (the form refuses a blank value).
Private-label manufacturer emergency phone or name blank	Blocks publishing	Every private-label publish: manual, republish, 'Republish Private Labels Only', the knock-on republish after a product publish, and bulk. Previews show a warning.	Enter the Emergency Phone and Name on the Manufacturers screen (the form refuses blanks).
Trade-secret ingredient whose CAS is on the Prop 65 list	Blocks publishing	Every publish path. Also shown in previews and on the readiness page.	Have the vendor disclose the chemical and clear its trade-secret box in Raw Materials > Edit, or correct the CAS number.
Inactive finished good	Blocks publishing	Single product publish, SDS Updates, alias publish and private-label republish. Bulk Publish leaves it out. Shown in the product preview.	Set the product's Status to Active (Finished Goods > Edit).
Private-label setup problems: shared alias no longer belongs to the product, no published base SDS, base missing a language, retired item, manufacturer missing	Blocks publishing	Private-label publish, the knock-on republish after a product publish, and bulk private-label items. The live preview also refuses a mismatched alias.	Re-point to a valid alias or enter a custom code, republish the base product in every language, restore the item or fix the manufacturer.
Routine controls: user permissions, Bulk Publish eligibility, scheduled-run blackout window, auto bulk publish switch, one runner at a time	Blocks publishing	Permissions: all manual publish and edit actions. Eligibility, blackout and runner lock: Bulk Publish and scheduled runs (ineligible products are left out, not failed).	Grant the edit permission; review raw materials, publish intermediates, activate products; wait for the window or other run, or change the settings.
TSCA inventory not verified, not listed, or INACTIVE entry	Warning only	Previews, readiness page and the message after single product and resale publish. Silent on SDS Updates, bulk, private-label and alias publishes.	Import the inventory or set a TSCA Review on CAS Determinations, and give every raw material its constituents with CAS. A text edit no longer silences it.

Condition	Effect	Applies to	How to clear it
Other data-quality warnings: flash/boiling point gaps or conflicting Section 9 edits, default 'Liquid' state, missing SG/VOC, incomplete constituents, trade-secret mismatch, shipping technical names, supplier name/phone	Warning only	Preview Warnings box; some also on the readiness page or formula Calculate page. Never printed on the sheet.	Enter the named data on the raw material, finished good, Admin > Settings or manufacturer record, or reset the named edit in the SDS editor.

WHAT CAN BE CHANGED PER PRODUCT

Text typed in the SDS editor for one product and one language replaces the automatic value for that product and language only.

Field	What a per-product edit does	Limits
Product Identifier, Manufacturer/Supplier details, Emergency phone, logo (Section 1)	None. The SDS editor has no override for these lines.	Change them on Finished Goods or Item Names, Admin > Settings > Manufacturer Information, or the Manufacturers screen for private-label sheets.
Recommended Use; Restrictions on Use (Section 1)	A text edit replaces the automatic text for that product and language. It takes priority over the product's own field, the family default and the standard sentence.	The finished good's own field holds one value for all languages. Leave it blank so the per-language Product Families defaults apply.
Signal Word, Pictograms, Hazard Classes, Hazard and Precautionary Statements (Section 2)	No text edit. The only per-product route is the Hazard Classification Override (add or replace) on Finished Goods.	Otherwise calculated from CAS Determinations data and the raw-material flash and boiling points. A Section 9 edit never changes the classification.
Other Hazards (Section 2)	A text edit replaces "None known." for that product and language.	No hazard data feeds this line, so it stays "None known." unless it is edited.
Composition table, Type (Substance/Mixture), trade-secret identities (Section 3)	No text edit. The table is built from formula and ingredient data.	Change constituents, CAS, %, and the trade-secret flag and description on Raw Materials > Edit. Set Substance/Mixture on Finished Goods.
Inhalation, Skin Contact, Eye Contact, Ingestion, Most Important Symptoms, Notes to Physician (Section 4)	A text edit replaces the whole automatic line, hazard-driven sentences included. The Symptoms edit also prints in Section 11.	A Skin Contact edit also removes the UV uncured-product sentence.
Suitable and Unsuitable Extinguishing Media, Specific Hazards, Advice for Firefighters (Section 5)	A text edit replaces the automatic text for that product and language.	Every time the sheet is generated, the flash point sentence inside a Specific Hazards edit is refreshed from Section 9. A Section 9 flash point edit does not change extinguishing media.
Personal Precautions, Environmental Precautions, Containment and Cleanup (Section 6)	A text edit replaces the automatic text, including the hazard-driven and UV sentences.	No limits beyond the edit. The automatic cleanup wording follows Physical State on Finished Goods or the raw materials.
Handling (Section 7)	A text edit replaces the standard sentence and all the hazard-driven and UV sentences.	None.
Storage (Section 7)	A text edit replaces the automatic storage text. While Storage stays automatic, a Section 10 Incompatible Materials edit also feeds this line.	An edited Storage line no longer names the incompatible materials, and the preview shows a warning.
Exposure Limits table (Section 8)	No text edit.	Built from exposure limits loaded by the hazard engine. Trade-secret rows are masked, and listed inhalation-only powders are left out of non-powder products.
Engineering Controls; Respiratory, Hand, Eye and Skin Protection (Section 8)	A text edit replaces the automatic line. Protection edits also change the Section 2 PPE line.	The Section 2 PPE line has no edit of its own. It prints only for hazard-driven tiers, even when a Section 8 edit is stored.

Field	What a per-product edit does	Limits
Odor, Appearance, Solubility, Flammability (Solid, Gas), and tested properties: pH, Odor Threshold, Melting Point, Evaporation Rate, explosive limits, vapor data, partition coefficient, ignition and decomposition temperatures, viscosity (Section 9)	A text edit replaces the derived value or "Not determined" for that product and language. Use it to enter tested values.	The system holds no formula data for the tested properties, so they print "Not determined" unless edited.
Flash Point (Section 9)	A text edit prints as typed. It also updates the flash point sentence in Section 5 Specific Hazards.	Must be a temperature that agrees with the Flammable Liquids category and the D001 outcome. Display only: it never changes classification, extinguishing media, Section 13 or Section 14.
Initial Boiling Point and Boiling Range (Section 9)	A text edit prints as typed instead of the lowest raw-material boiling point.	Saving refuses text that is not a temperature. An edit that contradicts the Category 1/2 split is dropped. It never changes classification or Section 14.
Physical State, Color, Specific Gravity, VOC (lb/gal and wt%), Solids (Section 9)	No text edit.	Set Physical State and Color on Finished Goods, and SG and VOC on every raw material. Missing SG counts as 1.0 and missing VOC as 0.
Reactivity, Chemical Stability, Conditions to Avoid, Incompatible Materials, Hazardous Decomposition Products (Section 10)	A text edit replaces the automatic text. While Storage is automatic, an Incompatible Materials edit also prints in Section 7 Storage.	None.
Acute Toxicity, Chronic Effects, Carcinogenicity, UV Acrylate Information (Section 11)	A text edit replaces the automatic text for that product and language.	Likely Routes, Component Toxicological Data and Symptoms have no edit here (edit Symptoms in Section 4). UV Acrylate prints only for UV/LED families with the rule pack on.
Ecotoxicity, Persistence and Degradability, Bioaccumulative Potential, Mobility in Soil (Section 12)	A text edit replaces the automatic text. When a PBT component is present, a Persistence edit puts the PBT sentence into Bioaccumulative Potential.	The Component Aquatic Hazard Data table has no edit.
Disposal Methods; RCRA Hazardous Waste Classification (Section 13)	A Disposal Methods text edit replaces the standard sentence.	The RCRA classification has no edit. It follows the final hazards, the RCRA list, CAS metal flags and the formula flash point.
UN Number, Proper Shipping Name, Hazard Class, Packing Group, Environmental Hazards (Section 14)	Edits replace the derived values. UN Number and Hazard Class count only as part of a complete four-field override on every language tab; otherwise publishing is blocked.	Shipping Name or Packing Group can be edited alone only when the product is regulated. Transport in Bulk, Special Precautions and Note cannot be edited.
OSHA Status; TSCA Status (Section 15)	No per-product override. Old stored OSHA edits are ignored.	OSHA follows the Section 2 classification test. TSCA follows per-CAS determinations and the EPA inventory; resolve gaps in CAS Determinations > TSCA Review.
SARA 313, HAPs, SNUR, California Proposition 65, State Regulations (Section 15)	Only State Regulations can be edited. Its text edit prints after the Prop 65 block; without an edit the line is left out.	The lists come from regulatory data plus manual HAP and Prop 65 entries on raw materials. A trade-secret Prop 65 match blocks publishing.
Version, Effective Date, Abbreviations, Disclaimer, GHS sections note (Section 16)	No per-product override.	Version and date are stamped on publish. The Disclaimer is set per language on Admin > Settings or on each manufacturer. Abbreviations follow the terms that print.

PRIVATE LABEL SHEETS

- A private-label sheet reuses the base product's content in each language, including the base product's text edits. There is no override at manufacturer or item level for Sections 2 to 16.
- Product code: the custom code, kept whole with its hyphens; else the shared alias code; else the base code. The description follows the same order. Both are frozen at publish, and a blank description prints the code alone.
- The supplier block, emergency phone, logo and disclaimer come only from the manufacturer record. The company's details are never used, and a manufacturer with no logo gets none.
- A blank manufacturer name or emergency phone refuses the item. The manufacturer form requires both.
- Each private-label item has its own version counter. All languages are published together or none are, including in bulk publishing.
- Private-label items still publish when the company emergency phone is blank: bulk publishing then runs only private-label items.
- File names carry a PL_<manufacturer> tag, and the PDF author is the manufacturer.
- If a manufacturer has no disclaimer for a language, the Admin disclaimer prints. If that text names your company, your name appears on the private-label sheet, so enter a manufacturer disclaimer per language.

LANGUAGES

- Sheets are produced in EN, ES, FR and DE. All printed wording comes from the four translation files: GHS phrases, hazard paragraphs, labels, PDF title and subject. A test checks that every key exists in all four.
- Data prints as entered, in English: chemical names, trade-secret descriptions, raw-material odor and appearance, regulatory list names, and CFR/ANSI/NFPA citations. DOT shipping names stay English, with a local gloss.
- Some text is maintained per language: family use text, the Admin disclaimer, manufacturer disclaimers and per-product edits. An edit in one language does not carry over to the others.
- A blank language falls back to the translated standard text, never to English.
- Every language uses '.' decimals, m/d/Y dates, US Letter and Helvetica. Only Latin-1 characters print; anything else prints as '?'.
• Section 14 is checked in every language before publishing, so a transport edit must be entered in each language.
- Accented capitals in section banners print correctly. French uses 'Page x sur y' and 'Rév.'. Older sheets that are re-rendered take the sheet language's strings instead of English.

TEXT THAT STAYS FIXED

Content that is the same on every sheet, and why.

Item	Why it is fixed
Section titles, field labels, table headings and page furniture (SAFETY DATA SHEET title, Page x of y, Rev., Draft text, DISCLAIMER bar)	Format constants required by the 16-section layout. They are translated per language and do not need to be data.
GHS hazard and precautionary statement wording, pictogram names, signal-word colours, and replacing withdrawn P281 with P280	Regulatory phrase tables under GHS and OSHA HazCom; the official wording must not vary.
Prescribed concentration bands (0.1-1% up to 80-100%), the never-understate band rule and the 0.1% disclosure cut-off	Required by HazCom 1910.1200(i) and Appendix A. Kept out of settings on purpose.
Flammability and flash-point rules: Flammable Liquids thresholds (23/60/93 °C, boiling point 35 °C), the weighted-average product flash point, Class 3/packing group limits, D001 below 60 °C and the combustible-liquid note below 93 °C	Set by GHS/HazCom Appendix B.6, 49 CFR 173 and 40 CFR 261.21. The averaging method is your decision; you accepted that it can understate flammability.

Item	Why it is fixed
Base paragraphs in Sections 4 to 8 and 10 (fresh air / wash / 15-minute eye flush, standard fire media, the drains sentence, base handling and storage, 'stable under recommended storage', CO/CO2 decomposition)	Standard SDS boilerplate. Hazard-specific sentences add to or replace it, and any field can be edited per product.
Mappings from hazard codes to sentences (first-aid fragments, PPE tiers, carcinogen-list categories, default ATE values, unsupported transport classes)	Regulatory interpretations held in code. They could become maintained tables but rarely change.
Fixed regulatory statements: the trade-secret withholding note, SARA range note, RCRA generator-responsibility sentence, Prop 65 warning templates, carrier verification note, Transport in Bulk and Special Precautions lines	Fixed regulatory citations; only whether each one prints depends on data. The Sections 12-15 footnote can be switched on or off.
No-data defaults: 'Not determined' for 12 Appendix D properties, 'No data available' for mobility in soil, 'None known' for Other Hazards, plus 'Liquid', SG 1.0 and VOC 0 when data is missing	There is no data field behind these, or they are your policy (print every Appendix D property). Missing data is flagged in the preview. A tested value can be typed per product.
UV/EB sentence wording and the built-in acrylate recognition list	Fixed rule-pack text, the same for every UV product. Whether it prints depends on the family's UV/LED flag and the Admin switch.
Number and date format ('.' decimals, m/d/Y) and English data text (chemical names, trade-secret descriptions, list names, CFR/ANSI citations) in every language	Your decision of 2026-10-09; the data is stored once and is not translated.
Page layout: US Letter, Helvetica (Latin-1 characters only), no header on pages 2+, PDF Creator 'SDS System'	Format and library constants. A4 for ES/FR/DE could become a setting but is fixed today.
Built-in standard disclaimer and standard Recommended Use / Restrictions sentences	Last-resort fallbacks only. The Admin, family and manufacturer text per language replaces them.

CHANGES SINCE THE FIRST AUDIT

Before this update roughly a third of the sheet came from product data. These are the main fields that moved from fixed text to data, or are new.

Field	Change	Note
Flash point and flammability (Sections 2, 5, 7, 8, 9, 10, 13, 14)	Changed	Was the lowest raw flash point plus ingredient flammable codes, so sections could disagree. Now one Flammable Liquids result, from the weight-averaged flash point of raws that have one, the lowest boiling point and the physical state. No flash point data means not flammable. A Section 9 edit only changes the displayed value.
Section 14 transport	Now from data	Was copied in English from the first ingredient with a DOT entry. Now a mixture classification (Class 3, 8, 6.1, 9) using the Transport Product Type, technical names, a marine-pollutant line and fixed bulk and special-precautions lines. 'Not determined' or an incomplete edit blocks publishing.
Section 2 classification	Changed	Each hazard class now brings only its own signal word, pictograms and precautionary statements. Classes use the standard GHS names. Acute toxicity Category 5 and IARC Group 3 no longer count, bound carbon black and TiO2 are excluded, and leftover hazard codes print as their own list.
Sections 4 to 8 and 10 paragraphs	Now from data	Were single canned or first-match paragraphs. Now built up from the final hazard codes and physical state, so combined hazards keep all their advice. New: the Section 4 symptoms line, hazard-driven notes to physician, storage that names the Section 10 incompatibles, and reactivity and decomposition built from hazards and element flags.
PPE (Sections 2 and 8)	Now from data	Was hard-coded English, and unclassified products got a respirator by default. Now translated tiers chosen from the hazard codes, with a 'no special PPE' tier. Sections 2 and 8 agree.
Section 3 composition	Changed	Type now follows the Substance/Mixture setting. Every row prints a prescribed band from the summed min/max. A Substance always lists its main constituent. The non-hazardous checkbox is gone, and the notes are reworded.
Concentrations outside Section 3	Changed	Exact percentages are replaced by the Section 3 band in Sections 8, 11, 12 and 15, including the HAP total.

Field	Change	Note
Trade-secret identity	Changed	Was masked only in Section 3. Now the description and 'TRADE SECRET' appear in Sections 3, 8, 11, 12, 13 and 15, with generic RCRA and SNUR wording. A trade secret on the Prop 65 list blocks publishing.
Section 9 calculated properties	Now from data	Boiling point, odor, appearance, physical state and solubility now come from finished-good and raw-material data. VOC and solids are now divided by the formula total instead of 100. VOC less water and Solids (vol%) are no longer printed.
Section 9 Appendix D properties	New	Twelve new lines (odor threshold to viscosity). Each prints a value typed per product, else 'Not determined'.
Section 11 toxicology	Now from data	Fixed sentences replaced by: one acute-toxicity line per exposure route (the ATEMix is shown only when it set the category), likely routes, symptoms, chronic effects from the hazard codes, and carcinogenicity limited to reportable listings. The UV note now prints.
Section 12 ecology	Now from data	Was canned paraphrases and 'No data available'. Now the actual aquatic statements with their source, a component table with M-factors and band, and a line naming SARA 313 PBT chemicals. Mobility in Soil is a new line.
Section 13 disposal	New	New calculated RCRA line: D001-D003 from the hazards and flash point, component D codes (metal compounds included) and F/K/P/U codes for reference. The disposal sentence is now one standard sentence that can be edited.
Section 15 regulatory	Now from data	OSHA status follows the classification and TSCA status the inventory check, and neither can be edited. SARA 313 now prints (before, it never did), with PBT and PFAS at any concentration and category matching. Prop 65 is translated and lists its chemicals. State Regulations has its own line.
Section 1 Recommended Use / Restrictions	Now from data	Per-language family text now sits between the product's own text and the standard sentence. Inactive families are ignored. Resale sheets keep the resale sentence and can take their own edits. The Default Product Use setting and the Product Family line were removed.
Section 1 supplier block and emergency phone	Changed	Values come from saved settings; the config file is used only for keys never saved, and never for the emergency phone. Country, email and website now print, and Fax is gone. Private-label sheets use only the manufacturer's details. A blank emergency number blocks publishing.
Private-label registry	New	Custom code and description, a version counter per item, all languages published together, a disclaimer per manufacturer and language, and no fallback to the company's details.
Alias sheets	Changed	Print the base code without the pack size. Each alias has its own published PDF, passed through the gates. Senders no longer re-render the base sheet.
Section 16 and footer revision	Changed	Shows the published version and effective date (admin time zone), or the Draft text, instead of the generation date. The Revision Note was removed. Abbreviations come from each language's table, filtered to the terms printed.
Disclaimer	Now from data	Was one English setting, blank on new installs. Now one Admin text per language, an optional text per manufacturer, and a translated built-in fallback.
ES/FR/DE wording	Changed	Hazard paragraphs, labels, PDF title and subject, banners and Prop 65 are now translated. Accented capitals print correctly. Old sheets that are re-rendered use the sheet language.
UV/LED products	Changed	Decided by the active family's UV/LED flag plus the Admin switch, no longer by the family name or by finding acrylates. UV sentences now sit inside the Section 4 to 8 fields, and only Section 11 keeps a separate note.
'Not required by OSHA HazCom' notes (Sections 12-15)	Removed	Replaced by one footnote after Section 15, which can be switched on or off in Admin Settings.
Per-product text edits	Changed	Only text that differs from the automatic text is saved, and there is a 'Reset to automatic' button. Blank rows are ignored, and two cleanup passes removed frozen old automatic text. Resale raw materials have their own editor. OSHA/TSCA status and the Revision Note can no longer be edited.
Publish gates and preview warnings	Changed	Gates now run on every publish path and in every language. The preview embeds the real PDF, with a Warnings box for operators only (data gaps, flash point conflicts, TSCA, incomplete composition, supplier details) that never prints. A settings change that affects printed text prompts a republish.